

The Curse of Seniority: Debt Priority Design and Inefficient Liquidation

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Abstract

We develop a dynamic theory of debt priority that trades off ex ante dilution protection against ex post efficiency in distress. Seniority protects legacy creditors from dilution but fragments control in default. When senior creditors are impaired, their capped upside biases them toward liquidation; when they are fully repaid and junior creditors are impaired, no offsetting continuation bias arises. Holding total face fixed, priority refinement weakly expands inefficient liquidation. Safe seniority disciplines dilution without distorting resolution, whereas beyond the safe region, additional dilution protection may come at the cost of distorted distress control. Efficient priority structures may remain coarse.

Keywords: debt priority; leverage ratchet effect; seniority fragmentation; inefficient liquidation; debt complexity

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1 Introduction

Corporate debt contracts assign priority over cash flows and shape who controls key decisions in distress. Priority can take many forms, including security interests, contractual subordination, and covenants restricting the issuer’s ability to create new claims of equal or higher rank. These arrangements are central to corporate debt markets because they protect existing creditors during ongoing financing while also allocating influence over resolution decisions when the firm approaches default. This paper studies the trade-off between these two roles of priority.

The force favoring stronger priority protection is debt dilution. By issuing new debt *pari passu* with outstanding claims, shareholders can dilute legacy creditors and capture the resulting value transfer. Priority can serve as a commitment device against this dilution by requiring subsequent borrowing to rank below protected claims. Taken to its logical extreme, this reasoning predicts a finely layered priority ladder that protects each issuance vintage from future borrowing. Yet firms often concentrate their borrowing in a small number of debt types, and observed priority structures remain much coarser than such an issue-by-issue ladder (Colla et al., 2013; Badoer et al., 2020).

What limits the use of priority protection? Our answer is that priority also shapes control in distress. Refining the priority ladder divides a previously unified creditor class into separately ranked claims that can become pivotal in different default states. When a senior class is pivotal, its payoff is capped at its own face value, so it captures less of the upside from continuation than a unified creditor class while remaining exposed to downside risk. Senior creditors may therefore prefer liquidation even when continuation is efficient, reflecting the familiar senior-creditor liquidation bias emphasized in the bankruptcy literature (Jackson, 1986). But this bias alone does not imply that priority refinement increases liquidation: if junior-pivotal states instead shifted decisions toward continuation relative to the unified-creditor benchmark, the two effects could offset.

We show that no such offset arises. When senior claims are fully repaid from contemporaneous cash and the junior class is pivotal, the junior class follows exactly the same liquidation rule as a unified creditor class with the same total face. Priority refinement therefore has a one-sided effect on distress resolution: it can introduce senior-pivotal states in which liquidation occurs even though a unified creditor class would continue, without creating junior-pivotal states in which the unified

class would liquidate but the junior class would continue. This non-offsetting asymmetry turns the familiar senior-creditor liquidation bias into a cost of ex ante priority design and generates the paper’s central trade-off between dilution protection and fragmented control in distress.

We study this trade-off in a four-date model of financing and distress resolution. At $t = 0$, the firm finances an initial investment and chooses its initial priority structure. At $t = 1$, it faces an additional financing need and can issue new debt *pari passu* with legacy junior claims but junior to protected senior claims. The firm may borrow beyond the financing need and distribute the residual to shareholders, generating the debt-dilution incentive. At $t = 2$, all outstanding debt matures, and the firm either repays, refinancing against date-3 resources when feasible, or defaults. Following default, cash-uncovered creditor classes choose between liquidation and continuation through a maturity extension that preserves priority. Liquidation occurs if any participating class objects to continuation. The initial priority structure therefore affects both interim dilution incentives and the allocation of control in distress.

Results preview. We first characterize distress resolution for a fixed terminal debt structure. Although every cash-uncovered creditor class can object to continuation, the outcome is determined by the first such class. If this pivotal class prefers continuation, all more junior cash-uncovered classes also weakly prefer continuation; if it objects, liquidation occurs. In the two-layer case, junior-pivotal states reproduce the liquidation rule of a unified creditor class, whereas senior-pivotal states can generate additional liquidation. It follows that, holding total promised repayment fixed, splitting a unified class into senior and junior claims weakly expands the set of forced-default states in which liquidation occurs. The increase is strict when the senior class liquidates in states in which a unified creditor class would continue.

This logic extends beyond the two-layer case. Holding total promised repayment fixed, adding priority layers weakly expands the set of forced-default states in which liquidation occurs by placing control in the hands of more tightly capped creditor classes. Under the baseline objection rule, as the priority ladder becomes arbitrarily fine, every forced default eventually results in liquidation. Thus, the issue-by-issue priority ladder suggested by the dilution problem can make inefficient liquidation pervasive in distress.

We next connect these fixed-face results to ex ante priority design. Under commitment, an optimal financing policy can avoid priority fragmentation and payout-driven issuance by placing

date-0 and date-1 debt in a single *pari passu* class. Without commitment, issuing additional *pari passu* debt can transfer value from legacy creditors to shareholders and induce borrowing beyond the interim financing need. Protecting legacy claims with seniority reduces this dilution incentive, but requires subsequent financing to rank below them and can therefore fragment control in distress. When no interim financing is required, an all-senior structure can, under the conditions of the model, implement the one-layer commitment benchmark. With positive interim financing needs, optimal priority design must balance dilution protection against the resulting resolution cost.

The trade-off has a sharp boundary. Priority layers contained entirely within the portion of debt always covered by contemporaneous cash never participate in distress resolution, so refining this safe portion leaves the liquidation rule and aggregate debt value unchanged at a fixed total face. At the same time, reallocating legacy debt from the junior class to safe seniority reduces the incentive for dilutive interim issuance without creating an ex post control cost. Under regularity conditions, an optimal structure can therefore be chosen to use all available safe seniority. Beyond the safe region, further seniority may continue to reduce dilution and equilibrium total face, but it can also create a direct control cost by making the senior class pivotal in distress. These results suggest a simple design principle: safe repayment capacity can support fine priority protection without fragmenting distress control, whereas extending priority protection into risky repayment capacity creates a force toward coarser structures. The numerical analysis illustrates that partial seniority can dominate both all-junior and all-senior debt. Over the reported financing-need domain, partial seniority implements the commitment benchmark outside a single interior interval. Within that interval, exact implementation fails and the optimal second-best policy switches from risky partial seniority to the safe-capacity boundary as financing needs increase.

We then extend the analysis from firm-wide priority to collateral. Collateral gives a creditor priority over a particular asset while leaving any deficiency claim to share *pari passu* with unsecured debt. Under proportional payoffs and exclusive secured-creditor enforcement of the pledged asset, introducing a secured layer weakly expands the set of states in which that asset is sold relative to a unified creditor class with the same total face and resolution technology. Collateral also creates an additional dilution channel: subsequent unsecured borrowing dilutes the secured creditor's deficiency recovery and reduces its payoff from continuation. When immediate priority recovery is sufficiently attractive relative to expected continuation recovery, a large unsecured pool can induce

the secured creditor to enforce against the pledged asset even though a unified creditor class would continue.

The model consequently yields empirical predictions for both priority design and distress outcomes. Holding leverage and firm fundamentals fixed, finer priority partitions within the cash-uncovered portion of the capital structure should be associated with more liquidation- and sale-oriented resolutions. Stronger cash coverage should support finer protected structures without distorting resolution, whereas weaker cash coverage should favor fewer or thicker priority classes and greater use of *pari passu* debt. Finally, pledged-asset sales should be especially likely when immediate collateral recovery is high relative to expected continuation recovery and a large unsecured pool dilutes the secured creditor’s deficiency claim.

Literature. Our paper first contributes to the literature on debt priority, security, and control-rights design. The classic anti-dilution rationale for priority is reflected in the “me-first” rule and in bond covenants restricting subsequent financing (Fama and Miller, 1972; Smith and Warner, 1979). Existing theories show how secured debt and collateral can support investment and borrowing capacity (Stulz and Johnson, 1985; Rampini and Viswanathan, 2010, 2013), while Hart and Moore (1995) study how debt and seniority constrain management’s future financing and investment choices. Related incomplete-contracting theories examine how the allocation of control and liquidation rights affects financing and investment (Aghion and Bolton, 1992; Hart and Moore, 1994, 1998).

More closely related, Gertner and Scharfstein (1991) study workouts with bank and public debt, while Gennaioli and Rossi (2013) show how allocating control between concentrated and dispersed debt classes can mitigate liquidation bias. Hackbarth and Mauer (2012) jointly characterize optimal capital and priority structures with multiple debt issues, and Donaldson et al. (2025) show how collateral and covenants interact to generate multilayered debt structures. Related recent theory studies creditor-on-creditor priority conflicts in distressed restructurings (Antill et al., 2025). Closest to our dynamic analysis, Zhong (2021) endogenizes creditor dispersion in a rollover environment, trading off stronger repayment incentives and pledgeability against coordination-driven rollover risk. We instead endogenize the vertical priority partition among common-maturity claims and isolate a governance cost of priority refinement. Our mechanism operates through the state-dependent identity and payoff cap of the pivotal creditor class: holding total promised repayment

fixed, dividing a unified class into priority-ranked claims creates additional liquidation in senior-pivotal states without changing the liquidation rule in junior-pivotal states. We extend this result to arbitrary priority refinements and to collateral as asset-specific priority.

Our theory also speaks to empirical work on creditor control, debt structure, and distress resolution. Creditor control rights activated through covenant violations or default affect investment, financing, and governance (Gilson, 1990; Chava and Roberts, 2008; Roberts and Sufi, 2009; Nini et al., 2012). Related work studies the role of liquidation values in renegotiation (Benmelech and Bergman, 2008), debt ownership and trading in Chapter 11 (Ivashina et al., 2016), and covenant complexity (Ivashina and Vallee, 2020). Evidence on debt structure documents variation in priority and security (Barclay and Smith, 1995; Rauh and Sufi, 2010), widespread debt specialization (Colla et al., 2013), and greater priority spreading as credit quality deteriorates (Badoer et al., 2020). Legal and empirical restructuring studies further document the growing influence of senior secured creditors and the increasing use of sale-oriented resolutions (Baird and Rasmussen, 2002; Ayotte and Morrison, 2009; Hotchkiss et al., 2023). We provide a mechanism linking ex ante priority design to the identity and incentives of the creditors who can block continuation. The collateral extension further links secured-creditor control to partial asset sales and identifies dilution of the secured creditor’s deficiency claim as an additional source of enforcement bias.

Finally, we build on the literature on dynamic leverage and debt dilution. Foundational work studies debt overhang and the distortions created by debt structure and sequential financing (Myers, 1977; Bizer and DeMarzo, 1992; Bolton and Scharfstein, 1996; Brunnermeier and Oehmke, 2013; Diamond and He, 2014). More recent work formalizes the leverage-ratchet force arising from shareholder–creditor conflicts in the absence of commitment (Admati et al., 2018; DeMarzo and He, 2021). This literature typically takes contractual protections such as maturity, covenants, or priority as given (e.g., Hu et al., 2025). We endogenize priority as a commitment device against dilution. Protecting legacy claims can restrain overborrowing, but placing subsequent debt below them fragments control in distress. Our analysis therefore connects dynamic dilution to distress governance: the priority protection that disciplines future issuance can also increase inefficient liquidation.

2 Baseline model

We consider a four-date environment $t = 0, 1, 2, 3$ designed to separate (i) *interim financing incentives* arising from the issuance of new claims at $t = 1$ from (ii) *creditor incentives in financial distress* governing continuation versus liquidation following default at $t = 2$. All debt issued at $t = 0$ and $t = 1$ matures at $t = 2$ and is structured into two priority layers: a senior tranche and a junior tranche. At $t = 2$, the firm either repays its outstanding debt in full, possibly by refinancing against $t = 3$ resources, or defaults. Upon default, cash-uncovered creditor classes decide whether to accept continuation through a maturity extension. Liquidation occurs if any cash-uncovered class objects to continuation.

2.1 Agents and cash flows

The firm is owned by an entrepreneur (the equity holder) and can issue securities to a competitive market of investors. All agents are risk neutral, financial claims are subject to limited liability, and the risk-free rate is normalized to zero. Competition implies break-even pricing: at each issuance date, the unit price of any claim equals its conditional expected payoff given public information at that date. The entrepreneur chooses issuance and payout policies to maximize contemporaneous payouts plus the expected continuation payoff to equity, net of expected privately borne real distress costs, taking competitive security prices as given. A private distress cost, introduced below, rules out a large-face escape in the issuance problem. Because it is borne privately by the entrepreneur and remains outside the bankruptcy estate, it does not enter creditor recoveries or resolution decisions.

Uncertainty is realized in two stages. At $t = 2$, a public state θ_2 is realized, and at $t = 3$, an additional public state θ_3 is realized. Let $(\mathcal{F}_t)_{t=0}^3$ denote the public filtration, with θ_2 being $\mathcal{F}_2$ -measurable and θ_3 being $\mathcal{F}_3$ -measurable. Let F_2 denote the distribution of θ_2 ; unconditional integrals over date-2 states are written with respect to $dF_2(\theta_2)$. No public information arrives between dates 0 and 1, so $\mathcal{F}_1 = \mathcal{F}_0$. Consequently, conditional on a given terminal face and issuance policy, claims with the same priority and maturity have the same competitive per-unit value at dates 0 and 1.

The firm generates operating cash flows at $t = 2$ and $t = 3$,

$$G(\theta_2) \geq 0 \quad \text{and} \quad G(\theta_3) \geq 0,$$

and yields liquidation proceeds $L(\theta_2, 2) > 0$ if liquidated at $t = 2$, or $L(\theta_3, 3) > 0$ if liquidated at $t = 3$. At $t = 2$, it is convenient to define the date-3 continuation value and total repayment capacity:

$$V_3(\theta_2) \equiv \mathbb{E}[G(\theta_3) + L(\theta_3, 3) \mid \theta_2], \quad (1)$$

$$V_C(\theta_2) \equiv G(\theta_2) + V_3(\theta_2). \quad (2)$$

The object $V_3(\theta_2)$ is the maximum market value that the firm can raise against its $t = 3$ resources, conditional on θ_2 . Because investors break even and payments on newly issued claims cannot exceed the firm's realized $t = 3$ resources, no refinancing can raise more than this amount. Accordingly, $V_C(\theta_2)$ is the maximum amount available to satisfy date-2 claims while continuing the firm, combining contemporaneous cash with refinancing capacity. We assume that $V_C(\theta_2) < \infty$ for every θ_2 and, in addition, that

$$\mathbb{E}[V_C(\theta_2)] = \mathbb{E}[G(\theta_2) + V_3(\theta_2)] < \infty, \quad (3)$$

so that every expected payoff considered below is real-valued.

Our analysis focuses on default states in which creditors' private liquidation incentives may conflict with total value maximization. In particular, we restrict attention to environments in which liquidation destroys going-concern value.

Assumption 1. *For every θ_2 in the support of the $t = 2$ state, liquidation destroys going-concern value:*

$$L(\theta_2, 2) < V_3(\theta_2). \quad (4)$$

2.2 Timeline

Figure 1 illustrates the timeline of the model. At $t = 0$, the firm must finance an exogenous outlay $I_0 > 0$ by issuing two-period zero-coupon debt that matures at $t = 2$. The capital structure has two layers:

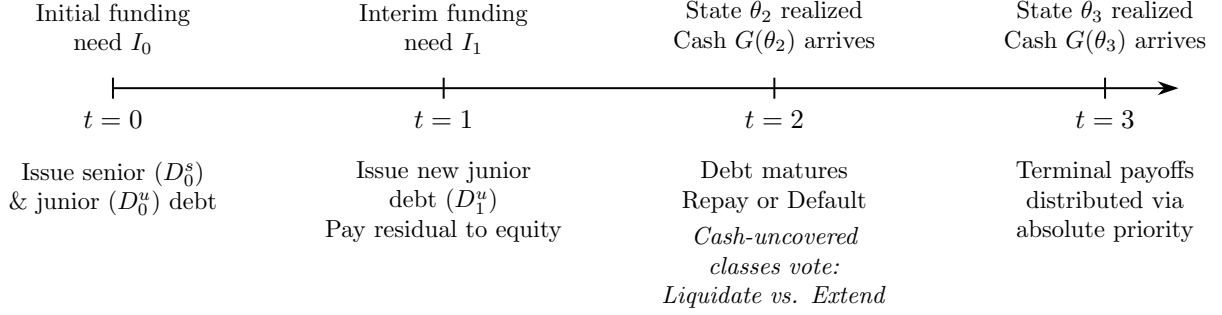


Figure 1: **Timeline of the model.** This figure summarizes the sequence of financing, default, and distress resolution.

- Senior debt with face value $D_0^s \geq 0$ and unit price p_0^s .
- Junior debt with face value $D_0^u \geq 0$ and unit price p_0^u .

The date-0 budget constraint is

$$I_0 = p_0^s D_0^s + p_0^u D_0^u. \quad (5)$$

Date-0 proceeds fund the initial project, so the model permits no date-0 payout. Payout-driven issuance is available only at $t = 1$ through equation (6).

At $t = 1$, the firm faces an additional funding need $I_1 \geq 0$. The date-0 senior tranche is protected by a covenant prohibiting the subsequent issuance of claims that are senior to or pari passu with it. New borrowing must therefore be junior to the legacy senior tranche. In the baseline model, the firm issues new debt pari passu with the legacy junior tranche.¹ Let $D_1^u \geq 0$ be the face value of one-period junior debt issued at $t = 1$ and let p_1^u be its unit price. The firm may issue more than is required to finance I_1 and distribute the residual to equity as a payout $P_1 \geq 0$:

$$p_1^u D_1^u = I_1 + P_1, \quad P_1 \geq 0. \quad (6)$$

Define total junior face and total face due at $t = 2$ as

$$D^u \equiv D_0^u + D_1^u, \quad D_1 \equiv D_0^s + D^u = D_0^s + D_0^u + D_1^u. \quad (7)$$

¹This restriction gives the minimal environment in which priority protection and dilution coexist. Allowing date-1 debt to be strictly junior to all existing claims would add another priority layer and attenuate the dilution of legacy junior debt, while precommitted future funding would amount to additional date-0 commitment. We abstract from these margins to focus on the trade-off between protecting legacy senior claims and fragmenting control in distress.

At $t = 2$, the state θ_2 is realized, the firm receives cash $G(\theta_2)$, and all outstanding debt matures. The firm can use its contemporaneous cash together with funds raised against its $t = 3$ resources to repay this debt. Because the legacy debt is retired upon full repayment, the maximum market value the firm can raise against its $t = 3$ resources is $V_3(\theta_2)$. Full repayment is therefore feasible if and only if

$$V_C(\theta_2) = G(\theta_2) + V_3(\theta_2) \geq D_1. \quad (8)$$

Condition (8) partitions date-2 states into repayment-feasible states, in which $V_C(\theta_2) \geq D_1$, and forced-default states, in which $V_C(\theta_2) < D_1$. Upon forced default, and before creditors choose between liquidation and extension, the entrepreneur incurs the private real distress cost introduced in Assumption 3. The cost is independent of the subsequent resolution outcome and remains outside the bankruptcy estate. Because it represents the severity of insolvency rather than the use of the default procedure itself, it is not incurred following strategic default in a repayment-feasible state.

When repayment is feasible, the entrepreneur chooses whether to repay D_1 in full, refinancing as necessary, or to strategically default and enter the creditor-controlled distress-resolution stage described below.²

Assumption 2. *In default at $t = 2$, the firm cannot issue new debt or equity. Existing claimants choose between liquidation at $t = 2$ and continuation through a maturity extension of the unpaid portions of outstanding claims to $t = 3$. The extension preserves their relative priority.*

Only cash-uncovered creditor classes participate in the decision. A class is cash-uncovered if, after more senior claims have been paid to the extent possible from contemporaneous cash $G(\theta_2)$ and without liquidating the firm, its promised claim cannot be fully satisfied. Each participating class objects to continuation if and only if its liquidation payoff is strictly positive and weakly exceeds its expected payoff under extension.³ The firm is liquidated if at least one participating class objects to continuation; otherwise, the unpaid claims are extended to $t = 3$.

Assumption 2 captures an incomplete market for distress resolution. In practice, reorganizations

²In the main analysis, the entrepreneur cannot commit ex ante to future issuance or to the date-2 repayment/default decision. Thus, interim dilution may arise at $t = 1$, and strategic default may occur at $t = 2$ even when full repayment is feasible. The commitment benchmark in Section 3.1.3 allows the firm to commit ex ante to both margins.

³A participating class with zero liquidation payoff is assumed not to block continuation. If such a class also receives zero expected payoff under extension, it is indifferent; allowing arbitrarily small consent transfers would select continuation.

that reallocate priority or introduce new claims are costly and subject to legal, informational, and coordination frictions. We therefore model continuation in reduced form as a maturity extension that preserves the relative priority of existing claims and requires the consent of cash-uncovered creditor classes. This incompleteness is essential for our mechanism: under frictionless reorganization with complete contracting, continuation and liquidation would be efficiently renegotiated, and inefficient liquidation would not arise. Appendix B provides a bargaining interpretation and shows that the same resolution outcome survives bounded consent transfers whenever the available pledgeable bargaining slack is no larger than the liquidation advantage of a class that objects to continuation. Our “liquidation” outcome should be interpreted as a cash-out resolution of the pre-petition capital structure, not necessarily as a piecemeal shutdown of the firm’s assets. Modern Chapter 11 cases often involve going-concern sales, including sales under Section 363 of the U.S. Bankruptcy Code, in which assets are sold free and clear of existing liens and proceeds are distributed according to priority.⁴ Such a sale may preserve the firm’s assets in operation, but it terminates pre-petition creditors’ exposure to the firm’s future cash flows. In this sense, it corresponds to liquidation in our model: creditors receive a priority-based cash payoff rather than claims on the continued firm. By contrast, continuation in the model represents a resolution that preserves creditors’ exposure to future firm value, captured here by maturity extension.

The final primitive is a private real cost of forced financial distress borne by the entrepreneur. For total promised face D , the insolvency shortfall in state θ_2 is the amount by which promised repayment exceeds the firm’s continuation value, $(D - V_C(\theta_2))_+$. The cost is incurred when this shortfall is positive, at the onset of forced default and before creditors choose between liquidation and extension, but remains outside the bankruptcy estate. It therefore affects the entrepreneur’s issuance incentives without entering creditor recoveries or resolution decisions.

Assumption 3 (Private cost of forced distress). *For total promised face D due at $t = 2$, the entrepreneur bears the private real distress cost*

$$\phi\left((D - V_C(\theta_2))_+\right)$$

⁴Ayotte and Morrison (2009) document that secured creditors play a central role in Chapter 11: in their sample, 75% of firms enter bankruptcy with a prepetition credit facility, 97% of those facilities are secured, 76% of firms use either DIP financing or cash collateral, and 66% of cases end in a sale of the entire firm. Hotchkiss et al. (2023) show that large Chapter 11 cases have become increasingly prearranged and sale-oriented, with significant Section 363 sales rising from 12% of large cases filed before 2000 to 34% during 2000–2021.

in state θ_2 , where $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is continuous and weakly increasing,

$$\phi(0) = 0 \quad \text{and} \quad \lim_{z \rightarrow \infty} \phi(z) = \infty.$$

The associated expected distress cost is

$$\Psi(D) \equiv \mathbb{E} \left[\phi \left((D - V_C(\theta_2))_+ \right) \right], \quad (9)$$

and is finite for every finite D .

Holding total face D fixed, the cost is invariant to the number, size, and ordering of priority classes. It is incurred at the onset of forced default and is identical whether creditors subsequently liquidate or extend. The cost is borne by the entrepreneur outside the bankruptcy estate and therefore does not enter creditor recoveries or resolution decisions.

Assumption 3 captures reduced-form, privately borne consequences of severe financial distress, such as losses of entrepreneur-specific human and reputational capital. Consistent with this interpretation, managers of distressed firms experience substantial turnover and adverse subsequent employment outcomes, while bankruptcy can impose large career and compensation losses on incumbent CEOs (Gilson, 1989; Eckbo et al., 2016). Because these losses are not transferable to creditors, they remain outside the bankruptcy estate.

The conditions in the assumption imply that Ψ is finite, continuous, and weakly increasing, with $\Psi(D) \rightarrow \infty$ as $D \rightarrow \infty$. Moreover, because the cost is invariant to the subsequent resolution outcome and to how a given total face is divided across priority classes, it leaves the default-stage resolution decision unchanged and cancels from comparisons across priority structures holding total face fixed.

Under liquidation at $t = 2$, total resources available for distribution are

$$V_L(\theta_2) \equiv G(\theta_2) + L(\theta_2, 2). \quad (10)$$

Under continuation, total resources available for distribution across dates $t = 2$ and $t = 3$ are

$$Z(\theta_2, \theta_3) \equiv G(\theta_2) + G(\theta_3) + L(\theta_3, 3), \quad (11)$$

viewed as a random variable conditional on θ_2 .

2.3 Priority and tranche payoffs

Payoffs follow absolute priority. Senior creditors are paid first up to D_0^s , then junior creditors share pro rata up to D^u , and equity receives any residual.

In a default state θ_2 , liquidation at $t = 2$ delivers the pool $V_L(\theta_2)$ in equation (10). Senior and junior payoffs are

$$R_2^s(\theta_2) = \min\{D_0^s, V_L(\theta_2)\}, \quad (12)$$

$$R_2^u(\theta_2) = \min\{D^u, V_L(\theta_2) - R_2^s(\theta_2)\}. \quad (13)$$

If creditors extend to $t = 3$, total resources are $Z(\theta_2, \theta_3)$ in equation (11) and payoffs are

$$R_3^s(\theta_2, \theta_3) = \min\{D_0^s, Z(\theta_2, \theta_3)\}, \quad (14)$$

$$R_3^u(\theta_2, \theta_3) = \min\{D^u, Z(\theta_2, \theta_3) - R_3^s(\theta_2, \theta_3)\}. \quad (15)$$

2.4 Equilibrium and pricing

A competitive equilibrium specifies issuance and payout choices by the entrepreneur, a refinancing/default rule, and security prices consistent with investor break-even.

Definition 1. *A competitive equilibrium consists of (i) a date-0 issuance policy (D_0^s, D_0^u) satisfying equation (5), (ii) a date-1 issuance and payout policy (D_1^u, P_1) satisfying equation (6) and the no-new-senior restriction, (iii) a date-2 refinancing/default policy specifying, for each θ_2 , whether the entrepreneur repays in full or defaults, where full repayment is feasible only when equation (8) holds, (iv) a default-state liquidation rule consistent with Assumption 2, so that liquidation occurs if and only if at least one participating creditor class objects to continuation, and (v) price functions (p_0^s, p_0^u, p_1^u) such that*⁵

- (i) *Given prices, the entrepreneur's policies in (i)–(iii) are sequentially optimal. At each issuance date, her objective is the sum of contemporaneous payouts and the expected continuation payoff*

⁵Security prices depend on the full promised-face vector and, where relevant, the current state. Formally, date-0 prices are $p_0^s(D_0^s, D_0^u)$ and $p_0^u(D_0^s, D_0^u)$, while the date-1 junior price is $p_1^u(y; D_0^s, D_0^u)$ for an issuance of junior face value y . We suppress these and the corresponding arguments when no confusion arises.

- to equity, net of expected privately borne real distress costs, subject to feasibility.*
- (ii) *Each unit price equals the conditional expected per-unit payoff of the corresponding claim under the equilibrium continuation, default, and liquidation rules.*

When repayment is feasible and the entrepreneur is indifferent between repayment and default, indifference is resolved toward repayment if default would trigger liquidation and toward default if default would lead to extension. This convention selects among payoff-equivalent actions and pins down the repayment/default policy used below. In the latter case, repayment and extension generate the same claim values, total resources, and real loss.

Risk neutrality implies that security prices equal the appropriate conditional expected repayment rates. In particular, because the date-1 issue is *pari passu* with legacy junior debt, both vintages receive the same pro rata payoff. Consequently, conditional on legacy senior face D_0^s , the unit price p_1^u depends on the two junior vintages only through total junior face, $D^u = D_0^u + D_1^u$. Additional junior issuance can therefore reduce the per-unit payoff of both new and legacy junior claims. This effect on the value of legacy claims generates the dilution incentives studied in the next section.

3 Analysis

This section analyzes how debt priority affects liquidation decisions and protects creditors against dilution. We first characterize default and liquidation for a fixed priority structure, showing how priority refinement can expand inefficient liquidation. We then formulate the date-0 issuance problem and study a commitment benchmark, which isolates the cost of priority fragmentation absent future dilution. Finally, we relax commitment and characterize interim dilution incentives and their implications for priority design. Sections 3.3.2–3.3.3 illustrate the resulting trade-off and derive comparative statics in a tractable specification.

3.1 The Seniority Fragmentation Effect

3.1.1 The Default and Liquidation Decisions at Date-2

We analyze the equilibrium backwards starting from the entrepreneur’s strategic default decision and creditors’ liquidation decision at $t = 2$.

Given Assumptions 1 and 2, strategic default is attractive only when it leads to creditor extension rather than liquidation. The next lemma characterizes the entrepreneur's repayment/default choice in repayment-feasible states and implies that liquidation can occur only in forced-default states.

Lemma 1. *In any state θ_2 with $V_C(\theta_2) \geq D_1$ (full repayment feasible), the entrepreneur refinances and repays D_1 in full if and only if default would trigger liquidation under the default-stage rule in Assumption 2; otherwise she strategically defaults and creditors extend. Consequently, liquidation at $t = 2$ occurs if and only if $V_C(\theta_2) < D_1$ and at least one cash-uncovered creditor class objects to continuation in default.*

Proof. See Appendix A.1.

Lemma 1 implies that repayment-feasible states never lead to liquidation. We therefore characterize creditors' liquidation incentives only in forced-default states with $V_C(\theta_2) < D_1$. Fix promised faces (D_0^s, D_0^u, D_1^u) and such a state θ_2 .

We first consider states in which contemporaneous cash covers the senior tranche. When $G(\theta_2) \geq D_0^s$, senior creditors are fully repaid from cash and do not participate in the default-resolution decision. The junior class is cash-uncovered and decides whether to object to continuation. Junior creditors prefer liquidation if and only if $R_2^u(\theta_2) \geq \mathbb{E}[R_3^u(\theta_2, \theta_3) \mid \theta_2]$, or

$$V_L(\theta_2) - D_0^s \geq \mathbb{E}\left[\min(D^u, (Z(\theta_2, \theta_3) - D_0^s)^+) \mid \theta_2\right]. \quad (16)$$

Because $G(\theta_2) \geq D_0^s$ implies $Z(\theta_2, \theta_3) \geq D_0^s$ almost surely, condition (16) is equivalent to

$$V_L(\theta_2) \geq \mathbb{E}\left[\min(D_1, Z(\theta_2, \theta_3)) \mid \theta_2\right]. \quad (17)$$

This is exactly the liquidation rule of a unified creditor class with total face value D_1 .

Proposition 1 (Junior-pivotal neutrality). *Fix promised faces (D_0^s, D^u) with total face $D_1 = D_0^s + D^u$, and fix any state θ_2 in which the firm has entered default and the resolution stage is triggered. Suppose $G(\theta_2) \geq D_0^s$, so the senior claim is cash-covered and the junior class is pivotal. Then the junior class chooses liquidation if and only if a unified creditor class with total face D_1 chooses liquidation; equivalently, condition (17) holds.*

Proof. Since $G(\theta_2) \geq D_0^s$ and $Z(\theta_2, \theta_3) = G(\theta_2) + G(\theta_3) + L(\theta_3, 3)$, we have $Z(\theta_2, \theta_3) \geq D_0^s$ almost surely. Hence the junior class's continuation payoff is

$$\mathbb{E}\left[\min(D^u, Z(\theta_2, \theta_3) - D_0^s) \mid \theta_2\right].$$

The junior class chooses liquidation if and only if condition (16) holds. Adding D_0^s to both sides and using

$$D_0^s + \min(D^u, Z(\theta_2, \theta_3) - D_0^s) = \min(D_0^s + D^u, Z(\theta_2, \theta_3)) = \min(D_1, Z(\theta_2, \theta_3)),$$

gives condition (17). A unified creditor class with total face D_1 compares liquidation payoff $V_L(\theta_2)$ with continuation payoff $\mathbb{E}[\min(D_1, Z(\theta_2, \theta_3)) \mid \theta_2]$, so the two liquidation rules coincide. $\square$

Proposition 1 shows that junior pivotality reproduces the unified-creditor benchmark. This neutrality distinguishes the effect of priority fragmentation from the familiar liquidation bias of senior creditors: senior bias alone would not imply that priority splitting expands inefficient liquidation if junior control generated an offsetting continuation bias. Proposition 1 rules out such an offset. Thus, any additional liquidation created by priority splitting must arise in states in which senior creditors are cash-uncovered.

We next consider states in which contemporaneous cash does not cover the senior tranche. When $G(\theta_2) < D_0^s$, the senior class is cash-uncovered and participates in the default-resolution decision. Since contemporaneous cash is exhausted before junior claims are paid, the junior class is also cash-uncovered whenever $D^u > 0$. Senior creditors prefer liquidation if and only if $R_2^s(\theta_2) \geq \mathbb{E}[R_3^s(\theta_2, \theta_3) \mid \theta_2]$, or

$$\min(D_0^s, V_L(\theta_2)) \geq \mathbb{E}\left[\min(D_0^s, Z(\theta_2, \theta_3)) \mid \theta_2\right]. \quad (18)$$

The next lemma shows that, in these states, the senior class is sufficient to determine whether continuation is blocked.

Lemma 2 (Senior pivotality). *When $G(\theta_2) < D_0^s$, if senior creditors do not object to continuation in state θ_2 , then no junior cash-uncovered creditor class objects to continuation.*

Proof. If $D_0^s \leq V_L(\theta_2)$, then the senior class is fully repaid under liquidation, so $\min\{D_0^s, V_L(\theta_2)\} =$

D_0^s . Moreover, $\min\{D_0^s, Z(\theta_2, \theta_3)\} \leq D_0^s$ pointwise, so $\mathbb{E}[\min\{D_0^s, Z(\theta_2, \theta_3)\} \mid \theta_2] \leq D_0^s$. By the default-stage rule in Assumption 2, the senior class objects to continuation.

Hence, whenever the senior class does not object to continuation, it must be that $D_0^s > V_L(\theta_2)$. In this case junior creditors receive zero under liquidation and, by Assumption 2, do not block continuation. $\square$

Lemma 2 implies that, when senior claims are cash-uncovered, the senior class is pivotal: if it objects to continuation, the firm is liquidated; if it does not object, no junior cash-uncovered class objects. Thus, in senior cash-uncovered states, liquidation is governed by condition (18). Together with Proposition 1, this establishes the two relevant resolution rules: junior-pivotal states follow the unified-creditor condition (17), while senior-pivotal states follow the senior condition (18).

We now compare a unified debt class with a two-layer priority structure, holding total promised repayment fixed.

Proposition 2 (Seniority fragmentation effect). *Fix total face D_1 and compare a one-layer debt structure, in which all debt belongs to a single priority class, with a two-layer debt structure with senior face $D_0^s \in (0, D_1)$ and junior face $D^u = D_1 - D_0^s$. In any forced-default state θ_2 , if liquidation occurs under the one-layer structure, then liquidation also occurs under the two-layer structure. Hence introducing a strict senior layer weakly expands the set of forced-default states in which liquidation occurs.*

The increase is strict whenever there is positive probability of forced-default states with $G(\theta_2) < D_0^s$ in which the senior liquidation condition (18) holds but the unified liquidation condition (17) fails.

Proof. Under one-layer debt, the single creditor class has face value D_1 and liquidates if and only if condition (17) holds.

Fix a forced-default state θ_2 in which the one-layer structure liquidates, so condition (17) holds. Under the two-layer structure, if $G(\theta_2) \geq D_0^s$, Proposition 1 implies that liquidation is again governed by condition (17), so the firm liquidates. If instead $G(\theta_2) < D_0^s$, liquidation is governed by the senior condition (18). Since $D_1 \geq D_0^s$ and $\min(D_1, x) \geq \min(D_0^s, x)$ pointwise, condition (17) implies

$$V_L(\theta_2) \geq \mathbb{E}\left[\min(D_0^s, Z(\theta_2, \theta_3)) \mid \theta_2\right].$$

If $V_L(\theta_2) \leq D_0^s$, this is exactly condition (18). If $V_L(\theta_2) > D_0^s$, senior creditors are fully repaid under liquidation and therefore object to continuation under Assumption 2. Thus the firm also liquidates under the two-layer structure. For strictness, suppose there is positive probability of forced-default states with $G(\theta_2) < D_0^s$ in which condition (18) holds but condition (17) fails. In these states, the one-layer structure continues while the two-layer structure liquidates. $\square$

Proposition 2 formalizes the seniority fragmentation effect. Priority splitting can change the liquidation decision only in senior-pivotal states. In those states, senior creditors do not internalize continuation payoffs accruing to junior claims and may therefore liquidate when a unified creditor class would continue. Priority refinement thus adds senior-pivotal liquidation distortions without creating symmetric junior-pivotal continuation distortions.

3.1.2 The Date-0 Objective

Next, we analyze the entrepreneur's optimal debt issuance problem at $t = 0$. Under competitive pricing, creditors break even in expectation. The entrepreneur's expected payoff equals contemporaneous payouts plus continuation equity value net of the privately borne real distress cost. As a result, maximizing her expected payoff is equivalent to maximizing total resources net of liquidation loss and the distress cost.

Define the per-state deadweight loss from liquidation at $t = 2$ as

$$\Delta(\theta_2) \equiv V_3(\theta_2) - L(\theta_2, 2) > 0. \quad (19)$$

For any outstanding face values (D_0^s, D_0^u, D_1^u) and the induced equilibrium date-2 liquidation rule, let $\mathbf{1}_{\text{liq}}(\theta_2; D_0^s, D^u) \in \{0, 1\}$ indicate liquidation at $t = 2$.

Then the expected liquidation deadweight loss can be written as

$$\Lambda(D_0^s, D^u) \equiv \mathbb{E}[\Delta(\theta_2) \mathbf{1}_{\text{liq}}(\theta_2; D_0^s, D^u)]. \quad (20)$$

Using the general cost in Assumption 3, define total expected real loss as

$$\mathcal{T}(D_0^s, D^u) \equiv \Lambda(D_0^s, D^u) + \Psi(D_0^s + D^u).$$

Because the private distress cost enters no creditor payoff and depends only on total face, it does not affect the default-stage resolution rule or any comparison across priority partitions holding total face fixed. Its role is to discipline total promised repayment, not priority granularity.

The following proposition shows that the entrepreneur's date-0 problem can be written as minimizing total expected loss subject to the date-0 financing constraint.

Proposition 3. *For each inherited date-0 pair (D_0^s, D_0^u) , let $\text{BR}_1(D_0^s, D_0^u)$ denote the set of sequentially optimal date-1 issuance choices. Define the date-0-feasible best-response graph by*

$$\mathcal{E}^{\text{nc}} \equiv \{(D_0^s, D_0^u, D_1^u) \in \mathbb{R}_+^3 : D_1^u \in \text{BR}_1(D_0^s, D_0^u), \quad I_0 = p_0^s D_0^s + p_0^u D_0^u\},$$

where the date-0 prices are the competitive prices induced by the specified date-1 continuation. A feasible no-commitment competitive equilibrium is optimal if and only if its issuance triple satisfies

$$(D_0^{s,*}, D_0^{u,*}, D_1^{u,*}) \in \arg \min_{(D_0^s, D_0^u, D_1^u) \in \mathcal{E}^{\text{nc}}} \mathcal{T}(D_0^s, D_0^u + D_1^u).$$

Proof. See Appendix A.

A useful way to see the reduction is to write expected terminal resources before subtracting the privately borne distress cost as

$$\mathbb{E}[\mathcal{R}] = \mathbb{E}[G(\theta_2) + V_3(\theta_2)] - \Lambda(D_0^s, D^u).$$

Because Ψ is a privately borne real resource cost outside the bankruptcy estate, total surplus net of all real losses is $\mathbb{E}[\mathcal{R}] - \Psi(D_0^s + D^u)$. The first term $\mathbb{E}[G(\theta_2) + V_3(\theta_2)]$ is finite by equation (3) and depends only on primitives. Because investors break even at each issuance date, changes in security design affect the entrepreneur's date-0 expected payoff only through their effect on total surplus net of the privately borne cost. Thus, conditional on satisfying the financing constraints, maximizing the entrepreneur's expected payoff is equivalent to minimizing $\mathcal{T}$.

Higher debt face can increase liquidation loss by expanding the set of default states, but it can also decrease liquidation loss by strengthening creditors' incentives to continue the firm in default. We therefore impose monotonicity on total loss rather than on liquidation loss alone.

Assumption 4 (Monotone total loss on the relevant financing domain). *Let $\mathcal{D}$ denote the policy-relevant domain of outstanding face vectors described in Appendix A.2A. For any two vectors*

$$(D_0^s, D^u), (D_0^{s'}, D^{u'}) \in \mathcal{D}$$

satisfying $D_0^{s'} \geq D_0^s$ and $D^{u'} \geq D^u$,

$$\mathcal{T}(D_0^{s'}, D^{u'}) \geq \mathcal{T}(D_0^s, D^u).$$

Assumption 4 keeps the main analysis focused on the dilution–fragmentation trade-off. In the absence of the private distress cost, expected liquidation loss can be hump-shaped in total face because sufficiently high face makes a unified creditor class behave like a residual claimant over continuation resources. Assumption 3 supplies a coercive cost of excessive promised repayment while leaving the fixed-face creditor-control mechanism unchanged.

Assumption 4 holds whenever any reduction in expected liquidation loss caused by higher face is dominated by the associated increase in private distress cost. Appendix A.2A formalizes this dominance condition in marginal terms. Assumption 4 is maintained throughout the remainder of the analysis unless stated otherwise.

3.1.3 Optimal Date-0 Debt Issuance with Commitment

We begin with a benchmark that isolates the ex post distress-governance consequences of priority from interim debt dilution. At $t = 0$, the entrepreneur chooses the date-0 structure (D_0^s, D_0^u) and commits to a date-1 junior-issuance plan D_1^u (and hence the payout P_1 via equation (6)). The firm also commits to its date-2 repayment/default strategy. Under competitive pricing, strategic default cannot create ex ante surplus and may induce inefficient liquidation. Hence an optimal committed policy can be chosen to repay whenever full repayment is feasible, i.e., whenever $V_C(\theta_2) \geq D_1$. This benchmark therefore provides a clean comparison of priority structures based on their liquidation decisions within forced default.

One-layer pricing regularity. For total face $D \geq 0$, let $\Pi^{\text{one}}(D)$ denote aggregate creditor value under one-layer debt when the firm repays whenever repayment is feasible and the one-layer rule

applies in forced default. Throughout Sections 3.1.3–3.3.3, we maintain continuity of Π^{one} on the compact face domains used in the analysis and nonemptiness of each funding set invoked below. These conditions hold in the affine specification and are verified on the reported numerical domain in Appendix A.11F.

When Assumption 4 holds, the same surplus-accounting identity used in Proposition 3 implies that excess date-1 issuance cannot improve the committed policy: date-1 borrowing is used only to finance investment, rather than an interim payout.

For any fixed total face value due at $t = 2$, a one-layer debt structure delivers weakly higher expected creditor value than a fragmented two-layer structure when default occurs only when repayment is infeasible.⁶ Outside default, creditors are repaid the same promised face value under either structure. In forced-default states, however, a one-layer structure assigns the liquidation decision to a single creditor class that internalizes total creditor value, whereas a two-layer structure may place control in the hands of a senior class that liquidates even when continuation yields a higher total payoff to creditors as a whole. Under competitive pricing, one-layer debt therefore has weakly greater aggregate market value at any fixed total face.

When $I_1 = 0$, this fixed-face dominance implies that any feasible fragmented issuance can be replaced by a weakly smaller one-layer issuance that raises the required date-0 funds. When $I_1 > 0$, the same statewise dominance must be combined with the separate date-0 and date-1 financing constraints. The following proposition collects both cases.

Proposition 4 (Optimal debt structure under commitment). *Suppose the firm can commit at $t = 0$ to its date-1 issuance policy and date-2 repayment/default strategy. There exists an optimal committed policy satisfying*

$$p_1^{u,\text{commit}} D_1^{u,\text{commit}} = I_1 \quad \text{and} \quad P_1 = 0.$$

Moreover:

- (i) *If $I_1 = 0$, an optimal policy can be chosen with $D_1^{u,\text{commit}} = 0$ and a single date-0 priority*

⁶Without commitment to the repayment/default decision, repayment-feasible states introduce an additional threat channel: a more liquidation-biased creditor class can deter strategic default by making default more costly for equity. The commitment benchmark shuts down this channel by imposing repayment whenever $V_C(\theta_2) \geq D_1$, so that the comparison across priority structures is driven solely by liquidation choices within forced default. The state-by-state liquidation comparisons within forced default are unaffected by this modeling choice.

class. The resulting one-layer policy may be represented either by $D_0^u = 0$ (all senior) or by $D_0^s = 0$ (all junior).

- (ii) If $I_1 > 0$, an optimal policy can be chosen with $D_0^s = 0$, so that the date-0 and date-1 debt belong to a single *pari passu* junior class.

Proof. See Appendix A.3.

Proposition 4 shows that commitment allows the firm to avoid seniority fragmentation without sacrificing financing feasibility. The next subsection relaxes commitment to both date-1 issuance and the date-2 repayment/default decision, focusing on seniority's role in protecting date-0 creditors against dilutive *pari passu* borrowing.

3.2 Seniority as Protection against Debt Dilution

Under commitment, an optimal debt policy can avoid seniority fragmentation. Once the entrepreneur cannot commit to future issuance, however, date-1 borrowing can transfer value away from legacy creditors. When debt is outstanding from $t = 0$, issuing additional junior debt at $t = 1$ may dilute legacy claims, creating an incentive to borrow beyond the interim financing need and pay the excess proceeds to equity.

Let $V_1^{\text{leg}}(D_1^u; D_0^s, D_0^u)$ denote the date-1 market value of all legacy debt issued at $t = 0$ under a date-1 issuance choice D_1^u . Let $U_1(D_1^u; D_0^s, D_0^u)$ denote the entrepreneur's date-1 expected payoff, equal to the immediate payout P_1 plus continuation equity value net of the expected private real distress cost. A standard surplus decomposition implies

$$U_1(D_1^u; D_0^s, D_0^u) = \mathbb{E}[V_C(\theta_2)] - I_1 - V_1^{\text{leg}}(D_1^u; D_0^s, D_0^u) - \Lambda(D_0^s, D_0^u + D_1^u) - \Psi(D_0^s + D_0^u + D_1^u). \quad (21)$$

The term I_1 is the required interim investment; any proceeds beyond I_1 are paid out immediately to equity, as in the budget identity $p_1^u D_1^u = I_1 + P_1$, with $P_1 \geq 0$. Because the new issue is competitively priced, changing D_1^u does not create surplus from new investors. Instead, it affects the entrepreneur's payoff through the value transferred from legacy creditors and the resulting total real loss. The entrepreneur's date-1 problem is therefore

$$\max_{\substack{D_1^u \geq 0 \\ p_1^u D_1^u \geq I_1}} U_1(D_1^u; D_0^s, D_0^u) \quad \Longleftrightarrow \quad \min_{\substack{D_1^u \geq 0 \\ p_1^u D_1^u \geq I_1}} \left\{ V_1^{\text{leg}}(D_1^u; D_0^s, D_0^u) + \mathcal{T}(D_0^s, D_0^u + D_1^u) \right\}. \quad (22)$$

Thus, additional issuance is privately attractive when the resulting reduction in legacy debt value outweighs the increase in expected real loss. Although this dilution benefit accrues to equity at $t = 1$, it is anticipated and incorporated into debt prices at $t = 0$. Interim dilution therefore creates no free ex ante gain and, when it induces excessive promised face and thereby increases real loss, reduces the entrepreneur's ex ante payoff.

It is useful to distinguish two channels through which date-1 issuance can reduce legacy debt value. *Pari-passu dilution* arises when new claims are issued into the same priority class as legacy claims, reducing the legacy claims' pro-rata share of the class payoff. *Default-set dilution* arises when higher total promised repayment pushes additional states into default and creditor control, reducing legacy debt value even without pari passu sharing. Designating date-0 debt senior to future borrowing eliminates pari passu dilution of that debt, but does not mechanically eliminate default-set dilution.

Suppose that $I_1 = 0$. Designating date-0 debt senior to any debt subsequently issued at $t = 1$ eliminates pari passu dilution without constraining required interim financing. A remaining question is whether the entrepreneur might nevertheless issue junior debt solely to reduce the value of outstanding senior debt through default-set expansion. Under endogenous refinancing, this channel cannot generate a private transfer to equity. Any reduction in legacy senior value caused by additional junior issuance arises from states that newly end in liquidation, in which equity is wiped out. Such issuance creates real loss rather than transferring value from senior creditors to equity.

Recall Π^{one} and the one-layer pricing regularity from paragraph 3.1.3. Let $\Pi^{\text{strat}}(D)$ denote the corresponding one-layer value under the endogenous repayment/default rule in Lemma 1. Both functions equal zero at $D = 0$ and can differ only because of strategic default in repayment-feasible states. To ensure that strategic default does not reduce debt value, we use the following condition where it is required.

Condition 1 (No value-relevant strategic default). *Fix a total face $D > 0$. The condition holds at D if, in every repayment-feasible state θ_2 with $V_C(\theta_2) \geq D$, at least one of the following holds:*

- (i) *aggregate creditor value under liquidation weakly exceeds that under extension,*

$$\mathbb{E}[\min\{D, Z(\theta_2, \theta_3)\} \mid \theta_2] \leq \min\{D, V_L(\theta_2)\}; \quad (23)$$

(ii) *extension delivers the full promised face in expectation,*

$$\mathbb{E}[\min\{D, Z(\theta_2, \theta_3)\} \mid \theta_2] = D. \quad (24)$$

Under condition (23), the unified creditor class weakly prefers liquidation and has a strictly positive liquidation payoff because $D > 0$ and $V_L(\theta_2) > 0$. It therefore objects under the baseline rule, triggering liquidation and, by Lemma 1, refinancing and repayment. Under condition (24), the capped creditor payoff equals D almost surely. If a default would trigger liquidation, Lemma 1 selects repayment; otherwise default leads to extension, which delivers the full promised face without liquidation or forced-distress loss and is payoff-equivalent to repayment. Thus, in either case, endogenous strategic default does not reduce the value of one-layer debt.

Consequently, whenever Condition 1 holds at D ,

$$\Pi^{\text{strat}}(D) = \Pi^{\text{one}}(D). \quad (25)$$

For any interim funding need $I \geq 0$ for which the funding set below is nonempty, define the minimum one-layer commitment funding face by

$$D^{\text{com}}(I) \equiv \min \{D \geq 0 : \Pi^{\text{one}}(D) \geq I_0 + I\}. \quad (26)$$

Continuity and $\Pi^{\text{one}}(0) = 0 < I_0 + I$ imply

$$\Pi^{\text{one}}(D^{\text{com}}(I)) = I_0 + I. \quad (27)$$

Proposition 5 (Optimality of all-senior debt with no interim funding). *Suppose $I_1 = 0$.*

- (i) *Under any all-senior date-0 structure, $D_1^u = P_1 = 0$ constitutes a date-1 weak best response. Any positive junior issuance that induces liquidation on a positive-probability set of states satisfying $V_C(\theta_2) > D_0^s$ that would otherwise continue strictly reduces the entrepreneur's date-1 expected payoff.*

(ii) If Condition 1 holds at $D^{\text{com}}(0)$, then there exists an optimal no-commitment equilibrium with

$$D_0^s = D^{\text{com}}(0), \quad D_0^u = 0, \quad D_1^u = P_1 = 0.$$

This equilibrium coincides with the one-layer commitment benchmark in total face, liquidation rule, total loss, and the entrepreneur's expected payoff.

Proof. See Appendices A.4–A.5.

Part (i) isolates the date-1 anti-dilution effect: an all-senior structure contains no inherited *pari passu* junior claim, and issuing subordinated debt solely to expand the default set cannot transfer value to equity. For part (ii), Condition 1 preserves the funding capacity of the commitment debt at $D^{\text{com}}(0)$. Designating that debt senior therefore eliminates dilution without changing the commitment allocation. The conclusion is conditional because fragmentation may otherwise deter strategic default and increase debt value at a given total face.

3.3 The Trade-off Between Seniority Fragmentation and Debt Dilution

When interim borrowing is required, the firm must issue debt at $t = 1$ even if all date-0 debt is protected by priority. Seniority then has two opposing effects: it protects legacy debt against *pari passu* dilution, but the resulting fragmentation across senior and junior classes can induce inefficient liquidation. We first show that an optimal structure can always be chosen to provide priority protection to date-0 debt that can be repaid from contemporaneous cash in every state. We then study the trade-off created by seniority beyond this safe cash-flow capacity.

For the issuance-design results below, restrict total face to a compact choice-relevant interval

$$\mathcal{I} \equiv [D_{\mathcal{I}}^-, D_{\mathcal{I}}^+] \subset \mathbb{R}_+$$

containing the relevant commitment funding faces and date-1 upper-contour sets. Continuity of the relevant pricing and payoff mappings, together with the coercivity of the private distress cost in Assumption 3, permits $\mathcal{I}$ to be chosen so that, for every inherited structure considered below, any total face outside $\mathcal{I}$ is strictly dominated at date 1 by a feasible minimal-financing policy.

Let

$$\mathcal{J} \equiv [0, D_{\mathcal{I}}^+].$$

The one-layer pricing and secant-slope conditions below are imposed on $\mathcal{J}$, because relevant safe senior faces may lie below the lower endpoint of $\mathcal{I}$.

Throughout Sections 3.3.1–3.3.3, we maintain Condition 1 at every total face $D \in \mathcal{I}$. Remark 1 shows that, under every two-layer structure with total face in $\mathcal{I}$, strategic default in a repayment-feasible state either induces refinancing and repayment or is payoff-equivalent to repayment. The benchmark repayment rule is therefore valid for the pricing and loss calculations below. This restriction is used only for the issuance-design results and is not needed for the fixed-face pivotality or priority-refinement results.

3.3.1 Safe Seniority

Let $\underline{G} \equiv \inf_{\theta_2} G(\theta_2)$ denote the firm's lower bound on contemporaneous cash flow at date 2, and define $\bar{S} \equiv \min\{I_0, \underline{G}\}$. If $D_0^s \leq \underline{G}$, senior debt is fully repaid from contemporaneous cash in every state and therefore never participates in default resolution. The junior class is pivotal whenever default occurs, so Proposition 1 implies

$$\Lambda(D_0^s, D_0^u + D_1^u) = \Lambda(0, D_0^s + D_0^u + D_1^u), \quad \mathcal{T}(D_0^s, D_0^u + D_1^u) = \mathcal{T}(0, D_0^s + D_0^u + D_1^u). \quad (28)$$

Thus, at any given total face, reallocating date-0 debt from the junior to the senior class while maintaining $D_0^s \leq \underline{G}$ changes neither the liquidation rule nor total real loss.

Together with Condition 1, junior-pivotal neutrality also makes safe seniority neutral for aggregate pricing. Appendix A.6 shows that, for any feasible safe senior face $S \in [0, \bar{S}]$ and total face $D \in \mathcal{I}$, aggregate no-commitment debt value equals $\Pi^{\text{one}}(D)$. Because the safe senior claim has market value S , the junior pool of face $D - S$ has aggregate value $\Pi^{\text{one}}(D) - S$.

Safe seniority nevertheless reduces pari passu dilution of legacy debt. Write $D_0 \equiv D_0^s + D_0^u$. For any $S \in [0, \bar{S}]$ and $D \in \mathcal{I}$ with $D > S$, define the average value per unit of junior face by

$$v(D; S) \equiv \frac{\Pi^{\text{one}}(D) - S}{D - S}. \quad (29)$$

Because one-layer debt with face S is fully repaid, $\Pi^{\text{one}}(S) = S$, so $v(D; S)$ is the secant slope of Π^{one} between S and D . Whenever Π^{one} is concave on $\mathcal{J}$, as assumed in Proposition 6 below, $v(D; S)$ is weakly decreasing in D for each fixed S .

After the entrepreneur issues new junior face D_1^u , legacy debt has market value

$$V_1^{\text{leg}}(D_1^u; S, D_0 - S) = S + (D_0 - S)v(D_1; S). \quad (30)$$

The new creditors receive the complementary value $D_1^u v(D_1; S)$. Define the dilution benefit relative to no date-1 issuance by

$$B(D_1^u; S, D_0) \equiv \Pi^{\text{one}}(D_0) - V_1^{\text{leg}}(D_1^u; S, D_0 - S). \quad (31)$$

For intuition, suppose additionally that Π^{one} is differentiable. Holding D_0 fixed,

$$\frac{\partial B}{\partial D_1^u} = \frac{D_0 - S}{D_1 - S} [v(D_1; S) - \Pi^{\text{one}'}(D_1)] \geq 0, \quad (32)$$

and

$$\frac{\partial^2 B}{\partial D_1^u \partial S} = -\frac{(D_0 - S)[1 - v(D_1; S)] + D_1^u [v(D_1; S) - \Pi^{\text{one}'}(D_1)]}{(D_1 - S)^2} \leq 0. \quad (33)$$

The inequalities follow because $v(D_1; S) \leq 1$, as junior debt value cannot exceed junior face, and $v(D_1; S) \geq \Pi^{\text{one}'}(D_1)$ by concavity. Thus, holding total legacy face fixed, replacing legacy junior debt with safe senior debt reduces the marginal benefit of additional issuance.

Let $\mathcal{A}_1(S, D_0 - S)$ denote the feasible date-1 issuance set given the inherited senior and junior faces. Within the safe region, the date-1 problem can therefore be written, up to terms independent of D_1^u , as

$$\max_{D_1^u \in \mathcal{A}_1(S, D_0 - S)} \{B(D_1^u; S, D_0) - \mathcal{T}(0, D_1)\}. \quad (34)$$

Because $B(0; S, D_0) = 0$ and equation (33) holds, increasing safe seniority weakly reduces the payoff advantage of any larger issuance choice over a smaller one. Equation (28) shows that the real-loss schedule is unchanged at each total face.

This fixed- D_0 comparison isolates the dilution channel but does not by itself compare competitively priced equilibria: increasing safe seniority changes both the legacy junior face required

to raise I_0 and the mapping from new junior face to date-1 issuance proceeds. The proposition shows that the ordering survives competitive repricing at both dates and the resulting equilibrium adjustment in total face. We impose standard continuity, compactness, and feasibility conditions; Appendix A.6 states them formally.

Proposition 6 (Optimal use of safe seniority). *Suppose $I_1 > 0$, Condition 1 holds at every $D \in \mathcal{I}$, and Π^{one} is continuous, weakly increasing, and concave on $\mathcal{J}$, with $\Pi^{\text{one}}(D) > 0$ for every $D \in \mathcal{J}$ with $D > 0$. Under regularity Condition 2 stated in Appendix A.6, there exists an optimal no-commitment debt structure satisfying*

$$D_0^s \geq \min\{I_0, \underline{G}\}.$$

In particular, if $I_0 \leq \underline{G}$, an optimal structure can be chosen with

$$D_0^s = I_0, \quad D_0^u = 0.$$

Proof. See Appendix A.6.

The compactness and continuity conditions first ensure that the no-commitment problem attains an optimum. Starting from such an optimum, the proof raises the safe senior face while competitively repricing the remaining legacy junior claim. At a fixed total face, safe seniority leaves the resolution rule and real-loss schedule unchanged and weakly reduces the dilution benefit of upward date-1 issuance deviations. A monotone best-response construction then yields an equilibrium with weakly lower total face and no larger total loss.

Corollary 1 (Commitment implementation under safe seniority). *Suppose $I_1 > 0$, $D^{\text{com}}(I_1)$ is well defined and belongs to $\mathcal{I}$, and Condition 1 holds at every $D \in \mathcal{I}$. Fix*

$$S \in [0, \min\{\underline{G}, I_0\}].$$

If, for the corresponding date-0 structure financing I_0 , the zero-payout date-1 issuance that brings total face to $D^{\text{com}}(I_1)$ is a date-1 best response, then that structure supports an optimal no-commitment equilibrium that matches the commitment benchmark in total face, payout, liquidation rule, creditor value, total loss, and the entrepreneur's expected payoff. Under clause (ii) of

Condition 1, the repayment/default action may differ while remaining payoff-equivalent.

Proof. See Appendix A.7.

Proposition 6 establishes a lower bound on optimal priority protection rather than a complete characterization of the optimal priority structure. Seniority beyond $\underline{G}$ may provide further protection against dilution, but senior creditors are then cash-uncovered in some states and may become pivotal in distress. Risky seniority therefore introduces a trade-off between dilution protection and distorted liquidation, which we study next.

3.3.2 Risky Seniority: Extensive and Intensive Margins

The preceding result leaves open the liquidation consequences of extending priority protection beyond safe cash-flow capacity. To isolate these consequences, fix total face D_1 due at $t = 2$ and write $S \equiv D_0^s \in [0, D_1)$ for the senior face. Once $S > \underline{G}$, increasing seniority operates through two opposing margins: it creates new states in which the senior class is pivotal, but weakly raises its continuation payoff in states in which it was already pivotal.

For a claim with face D that has first priority over the firm's continuation resources, define its expected continuation payoff by

$$\mathcal{C}(D, \theta_2) \equiv \mathbb{E}[\min\{D, Z(\theta_2, \theta_3)\} \mid \theta_2]. \quad (35)$$

Using Proposition 1 and Lemma 2, the liquidation set at senior face S is

$$\begin{aligned} \mathcal{L}_{D_1}(S) \equiv & \{\theta_2 : V_C(\theta_2) < D_1, G(\theta_2) < S, V_L(\theta_2) \geq \mathcal{C}(S, \theta_2)\} \\ & \cup \{\theta_2 : V_C(\theta_2) < D_1, G(\theta_2) \geq S, V_L(\theta_2) \geq \mathcal{C}(D_1, \theta_2)\}. \end{aligned} \quad (36)$$

The first set contains senior-pivotal states, while the second contains junior-pivotal states and therefore applies the unified-creditor liquidation rule. The senior condition in equation (36) is equivalent to condition (18): because $\mathcal{C}(S, \theta_2) \leq S$, senior creditors liquidate if and only if $V_L(\theta_2) \geq \mathcal{C}(S, \theta_2)$.

Write

$$\Lambda_{D_1}(S) \equiv \int_{\mathcal{L}_{D_1}(S)} \Delta(\theta_2) dF_2(\theta_2) = \Lambda(S, D_1 - S) \quad (37)$$

for expected liquidation loss. Because total face is fixed, the private distress cost $\Psi(D_1)$ is also fixed, so a change in $\Lambda_{D_1}(S)$ is exactly the corresponding change in total real loss. Proposition 2 implies that $\mathcal{L}_{D_1}(0) \subseteq \mathcal{L}_{D_1}(S)$. The fragmentation loss relative to one-layer debt with the same total face is therefore

$$\text{Frag}(S; D_1) \equiv \Lambda_{D_1}(S) - \Lambda_{D_1}(0) \geq 0. \quad (38)$$

For the remainder of the fixed-face analysis, we suppress the subscript D_1 on $\mathcal{L}$ and Λ .

Lemma 3 (Extensive and intensive margins of risky seniority). *Fix total face D_1 and let $0 \leq S < S' < D_1$. Define*

$$\mathcal{L}^+(S, S') \equiv \mathcal{L}(S') \setminus \mathcal{L}(S), \quad \mathcal{L}^-(S, S') \equiv \mathcal{L}(S) \setminus \mathcal{L}(S').$$

States entering the liquidation set can occur only where raising the senior face shifts the pivotal class from junior to senior, whereas states leaving it can occur only where the senior class is pivotal under both structures. Consequently,

$$\mathcal{L}^+(S, S') \subseteq \{\theta_2 : S \leq G(\theta_2) < S'\}, \quad \mathcal{L}^-(S, S') \subseteq \{\theta_2 : G(\theta_2) < S\}. \quad (39)$$

Moreover,

$$\Lambda(S') - \Lambda(S) = \int_{\mathcal{L}^+(S, S')} \Delta(\theta_2) dF_2(\theta_2) - \int_{\mathcal{L}^-(S, S')} \Delta(\theta_2) dF_2(\theta_2). \quad (40)$$

Proof. Because total face D_1 is fixed, the forced-default set $\{\theta_2 : V_C(\theta_2) < D_1\}$ is unchanged. If $G(\theta_2) \geq S'$, the junior class is pivotal under both senior faces and applies the same unified-creditor liquidation rule. If $G(\theta_2) < S$, the senior class is pivotal under both faces. Since $\mathcal{C}(S', \theta_2) \geq \mathcal{C}(S, \theta_2)$, increasing senior face makes the senior liquidation condition weakly harder to satisfy, so states can leave but cannot enter the liquidation set. Finally, if $S \leq G(\theta_2) < S'$, control shifts from the junior to the senior class. Because $\mathcal{C}(S', \theta_2) \leq \mathcal{C}(D_1, \theta_2)$, states can enter but cannot leave the liquidation set. This proves equation (39). Equation (40) follows by partitioning the two liquidation sets into their intersection and their set differences. $\square$

The first integral in equation (40) is the *extensive margin*. Raising senior face makes the senior class cash-uncovered in additional states, and its tighter continuation-payoff cap can convert continuation into liquidation. The second integral is the *intensive margin*. In states in which the

senior class was already pivotal, raising its face relaxes the cap on its continuation payoff and can eliminate liquidation. Thus, Proposition 2 implies that every strict priority structure weakly increases liquidation relative to one-layer debt at the same total face, but it does not imply that fragmentation loss is monotone in the size of the senior tranche.

Equation (40) gives the general condition for monotonicity. On any interval $\mathcal{S}$ of senior faces, fragmentation loss is weakly increasing if and only if, for every $S < S'$ in $\mathcal{S}$,

$$\int_{\mathcal{L}^+(S, S')} \Delta(\theta_2) dF_2(\theta_2) \geq \int_{\mathcal{L}^-(S, S')} \Delta(\theta_2) dF_2(\theta_2). \quad (41)$$

Condition (41) requires the expected real loss in newly senior-pivotal liquidation states to dominate the expected real loss removed by the relaxation of senior creditors' liquidation bias. The general model does not impose this ordering. We next give primitive conditions under which it holds.

Assumption 5 (Affine cash flows and two-point continuation risk). *Let $\theta_2 \sim U[0, 1]$. Conditional on θ_2 ,*

$$\theta_3 = \theta_2 + \varepsilon, \quad \Pr(\varepsilon = +x \mid \theta_2) = \Pr(\varepsilon = -x \mid \theta_2) = \frac{1}{2},$$

where $x > 0$. Cash flows and liquidation values are affine:

$$G(\theta) = g_0 + g\theta, \quad L(\theta, t) = \ell_t + \lambda_t\theta,$$

with $g > 0$, $g_0 - gx \geq 0$, $\lambda_t > 0$, and $\ell_t - \lambda_t x > 0$. Liquidation-value depreciation is captured by $\ell_2 > \ell_3$ and $\lambda_2 \geq \lambda_3$. Moreover, continuation is socially efficient in every date-2 state, as ensured by

$$\ell_2 < g_0 + \ell_3, \quad \ell_2 + \lambda_2 < g_0 + \ell_3 + g + \lambda_3.$$

Under Assumption 5, $\underline{G} = g_0$ and

$$V_C(\theta_2) = a_C + b_C\theta_2, \quad a_C \equiv 2g_0 + \ell_3, \quad b_C \equiv 2g + \lambda_3, \quad (42)$$

$$V_L(\theta_2) = a_L + b_L\theta_2, \quad a_L \equiv g_0 + \ell_2, \quad b_L \equiv g + \lambda_2, \quad (43)$$

$$Z(\theta_2, \theta_3) = V_C(\theta_2) + c(\theta_3 - \theta_2), \quad c \equiv g + \lambda_3, \quad (44)$$

$$\Delta(\theta_2) = \delta_0 + \delta_1\theta_2, \quad \delta_0 \equiv g_0 + \ell_3 - \ell_2, \quad \delta_1 \equiv g + \lambda_3 - \lambda_2. \quad (45)$$

The two endpoint inequalities in Assumption 5 ensure that $\Delta(\theta_2) > 0$ throughout $[0, 1]$.

Fix total face D_1 on the capped-upside branch

$$D_1 \leq a_C + cx. \quad (46)$$

The relevant cutoffs are

$$\bar{\theta}(D_1) \equiv \frac{D_1 - a_C}{b_C}, \quad (47)$$

$$\theta_S(S) \equiv \frac{S - g_0}{g}, \quad (48)$$

$$\theta_L(D) \equiv \frac{D + \ell_3 - 2\ell_2 - cx}{2d}, \quad d \equiv \lambda_2 - \frac{\lambda_3}{2} > 0, \quad (49)$$

$$\theta_L^s(S) \equiv \theta_L(S), \quad \theta_L^u(D_1) \equiv \theta_L(D_1), \quad (50)$$

$$\theta_F^s(S) \equiv \frac{S - a_L}{b_L}, \quad (51)$$

$$\underline{\theta}^s(S) \equiv \min\{\theta_L^s(S), \theta_F^s(S)\}. \quad (52)$$

The cutoff $\bar{\theta}(D_1)$ solves $V_C(\theta_2) = D_1$ and therefore separates forced-default states, while $\theta_S(S)$ solves $G(\theta_2) = S$ and separates senior-pivotal from junior-pivotal states. The function $\theta_L(D)$ gives the liquidation-indifference cutoff for a first-priority claim with face D when its upside continuation payoff is capped at face but its downside continuation payoff is not. On this branch, only the downside continuation payoff, which occurs with probability one-half, remains sensitive to θ_2 , so the expected continuation payoff has slope $b_C/2$. Before liquidation proceeds repay the claim in full, its liquidation payoff has slope b_L . Since $b_L - b_C/2 = \lambda_2 - \lambda_3/2 = d > 0$, liquidation becomes relatively more attractive as θ_2 rises and therefore occurs above, rather than below, $\theta_L(D)$. Accordingly, $\theta_L^s(S)$ governs liquidation when the senior class is pivotal on this branch, whereas $\theta_L^u(D_1)$ governs liquidation when the junior class is pivotal and the unified-creditor rule applies. The cutoff $\theta_F^s(S)$ solves $V_L(\theta_2) = S$ and marks the point above which liquidation proceeds repay the senior claim in full. Hence the effective senior liquidation cutoff is $\underline{\theta}^s(S) = \min\{\theta_L^s(S), \theta_F^s(S)\}$.

Thus, repayment of total face D_1 is infeasible for $\theta_2 < \bar{\theta}(D_1)$. Conditional on default, senior creditors are pivotal when $\theta_2 < \theta_S(S)$; within those states, they liquidate when $\theta_2 \geq \underline{\theta}^s(S)$. In

junior-pivotal default states, liquidation occurs when $\theta_2 \geq \theta_L^u(D_1)$. Hence the fragmentation loss is

$$\text{Frag}(S; D_1) = \int_{\max\{0, \underline{\theta}^s(S)\}}^{\min\{1, \theta_S(S), \theta_L^u(D_1), \bar{\theta}(D_1)\}} \Delta(\theta_2) d\theta_2, \quad (53)$$

where the integral is zero whenever its lower endpoint is weakly above its upper endpoint.

Consider an interval $\mathcal{S}$ of senior faces satisfying the interior-branch ordering

$$0 < \theta_L^s(S) < \theta_S(S) < \min\{\theta_L^u(D_1), \bar{\theta}(D_1), 1\}, \quad \theta_L^s(S) \leq \theta_F^s(S), \quad S \in \mathcal{S}. \quad (54)$$

On this branch, fragmentation loss reduces to

$$\text{Frag}(S; D_1) = \int_{\theta_L^s(S)}^{\theta_S(S)} \Delta(\theta_2) d\theta_2.$$

Differentiating gives

$$\frac{\partial \text{Frag}(S; D_1)}{\partial S} = \frac{\Delta(\theta_S(S))}{g} - \frac{\Delta(\theta_L^s(S))}{2d}. \quad (55)$$

Corollary 2 (When risky seniority increases fragmentation loss). *Fix total face D_1 and consider an interval $\mathcal{S}$ satisfying condition (54). If*

$$\lambda_2 - \lambda_3 \leq g \leq 2\lambda_2 - \lambda_3, \quad (56)$$

then fragmentation loss is weakly increasing in senior face throughout $\mathcal{S}$. The increase is strict if either inequality in condition (56) is strict.

Proof. Condition (56) implies $\delta_1 = g + \lambda_3 - \lambda_2 \geq 0$ and $2d = 2\lambda_2 - \lambda_3 \geq g$. Because $\theta_S(S) > \theta_L^s(S)$ on the interior branch and Δ is positive and weakly increasing,

$$\frac{\Delta(\theta_S(S))}{g} \geq \frac{\Delta(\theta_L^s(S))}{g} \geq \frac{\Delta(\theta_L^s(S))}{2d}.$$

Equation (55) therefore implies the result. The first inequality is strict if the lower bound in condition (56) is strict, and the second is strict if the upper bound is strict. $\square$

The two terms in equation (55) are the local counterparts of the extensive and intensive margins in Lemma 3. The control-switch cutoff $\theta_S(S)$ moves at rate $1/g$, adding newly senior-pivotal

liquidation states, while the senior liquidation cutoff $\theta_L^s(S)$ moves at rate $1/(2d)$, removing states as the higher senior face induces continuation. Condition (56) requires the state sensitivity of contemporaneous cash flow to be intermediate. The lower bound makes liquidation loss weakly increasing in the state, so the higher states added at $\theta_S(S)$ carry at least as much loss as the lower states removed at $\theta_L^s(S)$. The upper bound ensures that the control-switch cutoff moves at least as rapidly as the senior liquidation cutoff, since $1/g \geq 1/(2d)$. Together, these restrictions make the extensive margin dominate.

The parameter restriction does not ensure that the interior branch exists, because intercepts such as g_0 and the total face D_1 determine the cutoff ordering in condition (54). Nor does the corollary imply global monotonicity. Once $\theta_S(S)$ reaches $\min\{\theta_L^u(D_1), \bar{\theta}(D_1), 1\}$, additional seniority no longer adds fragmentation states, and the intensive margin may cause fragmentation loss to fall. If the lower endpoint is instead determined by $\theta_F^s(S)$ or the boundary of the state space, the local derivative also changes.

A second branch is particularly relevant for the baseline parameterization. Fix D_1 and consider an interval $\mathcal{S}$ on which

$$\underline{\theta}^s(S) \leq 0 < \theta_S(S) < \min\{\theta_L^u(D_1), \bar{\theta}(D_1), 1\}, \quad S \in \mathcal{S}.$$

Because the lower boundary of the fragmentation region is then truncated at zero,

$$\text{Frag}(S; D_1) = \int_0^{\theta_S(S)} \Delta(\theta_2) d\theta_2, \quad \frac{\partial \text{Frag}(S; D_1)}{\partial S} = \frac{\Delta(\theta_S(S))}{g} > 0.$$

On this branch, increasing seniority cannot remove additional states through the lower liquidation cutoff because that cutoff already lies outside the state space. It only raises the control-switch cutoff $\theta_S(S)$ and therefore expands the fragmentation region.

The baseline parameterization does not satisfy condition (56), because

$$\lambda_2 - \lambda_3 = 0.52 > g = 0.50.$$

Its initial rising segment instead satisfies the boundary ordering above. For the fixed all-senior face reported in Table 1, Appendix A reports the corresponding senior-face interval.

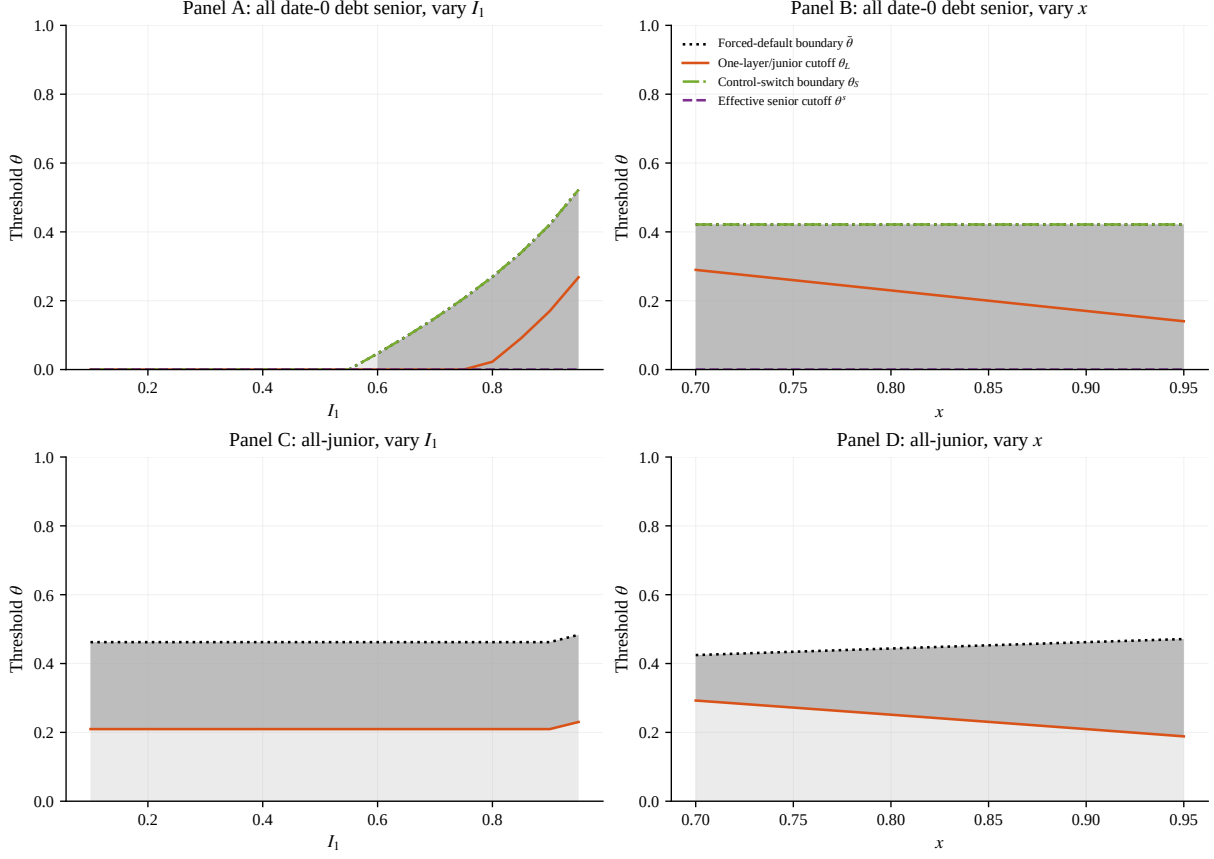


Figure 2: Liquidation thresholds and equilibrium regions. Under the baseline weak-objection convention in Assumption 2, light shading denotes forced-default states in which the firm continues, while dark shading denotes equilibrium liquidation. The black dotted curve is the forced-default boundary $\bar{\theta}(D_1)$; the red solid curve is the junior-control cutoff $\theta_L^u(D_1)$; the green dash-dotted curve is the control-switch boundary $\theta_S(S)$; and the purple dashed curve is the effective senior-control cutoff $\theta^s(S) = \min\{\theta_F^s(S), \theta_L^s(S)\}$. The left panels vary the interim financing need I_1 , holding $x = 0.90$ fixed, while the right panels vary $x \in [0.70, 0.95]$, holding $I_1 = 0.90$ fixed. The baseline parameterization is reported in Table 1 below.

Figure 2 illustrates the cutoff geometry behind the extensive and intensive margins. Under senior control, liquidation can extend below the unified-creditor cutoff $\theta_L^u(D_1)$, creating the fragmentation region. The shaded regions also show why the branch ordering matters: depending on which cutoff is active, a parameter change may move only one boundary of the fragmentation region or may move both.

3.3.3 The Optimal Extent of Priority Protection

The preceding results imply a simple design trade-off. At a fixed total face, safe seniority protects legacy debt without changing liquidation, whereas risky seniority may create fragmentation loss. Endogenously, priority protection can also reduce total face by weakening the date-1 dilution incentive. Because dilution is a transfer between legacy creditors and equity, its ex ante welfare effect operates through this induced change in borrowing and real loss.

For each feasible senior face S in the design range considered below, select a loss-minimizing equilibrium from the nonempty compact set of conditional no-commitment equilibria, and let $D(S)$ denote its total face. The regularity conditions in Appendix A.6 ensure that such a selection exists; differentiability of the selected branch is a separate local condition.

Suppose that $D(S)$, $\Lambda_D(S)$, and $\Psi(D)$ are differentiable on the branch under consideration. Expected total loss along the design locus is

$$\hat{\mathcal{T}}(S) = \Lambda_{D(S)}(S) + \Psi(D(S)), \quad (57)$$

and

$$\frac{d\hat{\mathcal{T}}(S)}{dS} = \left. \frac{\partial \Lambda_D(S)}{\partial S} \right|_{D=D(S)} + \left[\frac{\partial \Lambda_D(S)}{\partial D} + \Psi'(D) \right]_{D=D(S)} D'(S). \quad (58)$$

The first term is the direct priority-control effect at a fixed total face; the second is the induced-face effect. For $S \leq \underline{G}$, equation (28) implies that the direct priority-control effect is zero. Under Assumption 4, the bracketed marginal-loss term is nonnegative, so the induced-face effect is non-positive whenever additional priority protection lowers equilibrium total face.

Continuation risk affects the direct priority-control effect through the liquidation cutoffs. The affine specification yields a concise comparison across the two principal cutoff orderings.

Lemma 4 (Continuation risk and liquidation losses). *Fix total face D and consider interior branches on which the effective senior liquidation cutoff is $\theta_L^s(S)$. Under one-layer debt,*

$$\frac{\partial \Lambda_D(0)}{\partial x} = \frac{c}{2d} \Delta(\theta_L^u(D)) > 0. \quad (59)$$

Under two-layer debt, if

$$\theta_L^s(S) < \theta_S(S) < \min\{\theta_L^u(D), \bar{\theta}(D)\},$$

then

$$\frac{\partial \text{Frag}(S; D)}{\partial x} = \frac{c}{2d} \Delta(\theta_L^s(S)) > 0. \quad (60)$$

If instead

$$\theta_L^s(S) < \theta_L^u(D) < \min\{\theta_S(S), \bar{\theta}(D)\},$$

then

$$\frac{\partial \text{Frag}(S; D)}{\partial x} = \frac{c}{2d} [\Delta(\theta_L^s(S)) - \Delta(\theta_L^u(D))], \quad (61)$$

whose sign is $-\text{sign}(\delta_1)$.

Proof. See Appendix A.

Greater continuation risk lowers the liquidation cutoffs. This expands liquidation under one-layer debt, but its effect on fragmentation depends on the active upper endpoint. If the control-switch cutoff binds, only the lower endpoint moves and fragmentation expands. If the junior liquidation cutoff binds, both endpoints move and the sign depends on the state profile of liquidation loss. These are fixed-face comparisons and do not by themselves rank priority structures once issuance is endogenous.

Numerical illustration. We next illustrate the overall design trade-off under endogenous issuance. The example shows that making all date-0 debt senior can produce a worse outcome even though it eliminates pari passu dilution and payout-driven overborrowing in the reported equilibrium. It also shows how partial seniority can provide sufficient dilution protection without incurring the full priority-control cost of the all-senior policy.

Table 1: **Commitment benchmark and no-commitment polar financing policies**

	Commitment (one layer)	All-junior	All-senior
Total face D_1	2.079	2.151	2.098
Closed-form branch slack $D_{\text{cap}} - D_1$	0.191	0.119	0.172
D_0^s	0.000	0.000	1.000
D_0^u	1.094	1.111	0.000
D_1^u	0.985	1.040	1.098
Payout P_1	0.000	0.036	0.000
Forced-default probability	0.407	0.462	0.422
Liquidation probability	0.251	0.253	0.422
Strict share among liquidated states	1.000	1.000	0.310
Liquidation loss Λ	0.0488	0.0488	0.0825
Quadratic refinancing-shortfall cost $\Psi(D_1)$	0.0285	0.0417	0.0317
Total loss $\mathcal{T} = \Lambda + \Psi$	0.0773	0.0905	0.1142

Notes: Forced-default probability is $\Pr(V_C(\theta_2) < D_1)$. Liquidation probability is $\Pr(\mathbf{1}_{\text{liq}}(\theta_2; \cdot) = 1)$. Liquidation loss is $\Lambda = \mathbb{E}[\Delta(\theta_2)\mathbf{1}_{\text{liq}}(\theta_2; \cdot)]$. The quadratic refinancing-shortfall cost is $\Psi(D_1) = \frac{\chi}{2}\mathbb{E}[(D_1 - V_C(\theta_2))_+^2]$. D_{cap} is a branch diagnostic and is not imposed as an economic or numerical cap. Baseline parameters are $(g_0, g, \ell_2, \lambda_2, \ell_3, \lambda_3) = (0.55, 0.50, 0.80, 0.82, 0.45, 0.30)$, $(I_0, I_1) = (1.00, 0.90)$, $x = 0.90$, and $\chi = 1.50$. The implied constants are $a_C = 1.55$, $b_C = 1.30$, $c = 0.80$, $cx = 0.72$, $d = 0.67$, and $D_{\text{cap}} = 2.27$.

For the closed-form and numerical analysis, we specialize Assumption 3 to

$$\phi(z) = \frac{\chi}{2}z^2, \quad z \geq 0, \quad \chi > 0.$$

Consequently,

$$\Psi(D) = \frac{\chi}{2}\mathbb{E}\left[(D - V_C(\theta_2))_+^2\right]. \quad (62)$$

This specification makes the date-1 issuance problem well posed without changing creditor payoffs or liquidation decisions. The baseline parameterization is

$$(g_0, g, \ell_2, \lambda_2, \ell_3, \lambda_3) = (0.55, 0.50, 0.80, 0.82, 0.45, 0.30),$$

$$(I_0, I_1) = (1.00, 0.90), \quad x = 0.90, \quad \chi = 1.50.$$

It satisfies Assumption 5: $g_0 - gx = 0.100$, $\ell_2 - \lambda_2x = 0.062$, $\ell_3 - \lambda_3x = 0.180$, and $\Delta(\theta_2) = 0.20 - 0.02\theta_2 \in [0.18, 0.20]$.

Table 1 compares the commitment benchmark with the two polar no-commitment policies, each evaluated at its own equilibrium face. All-junior debt preserves unified creditor control but selects a higher total face and a positive payout, raising the refinancing-shortfall cost. All-senior date-0 debt eliminates the payout but creates a senior-controlled liquidation region; in the baseline, its total

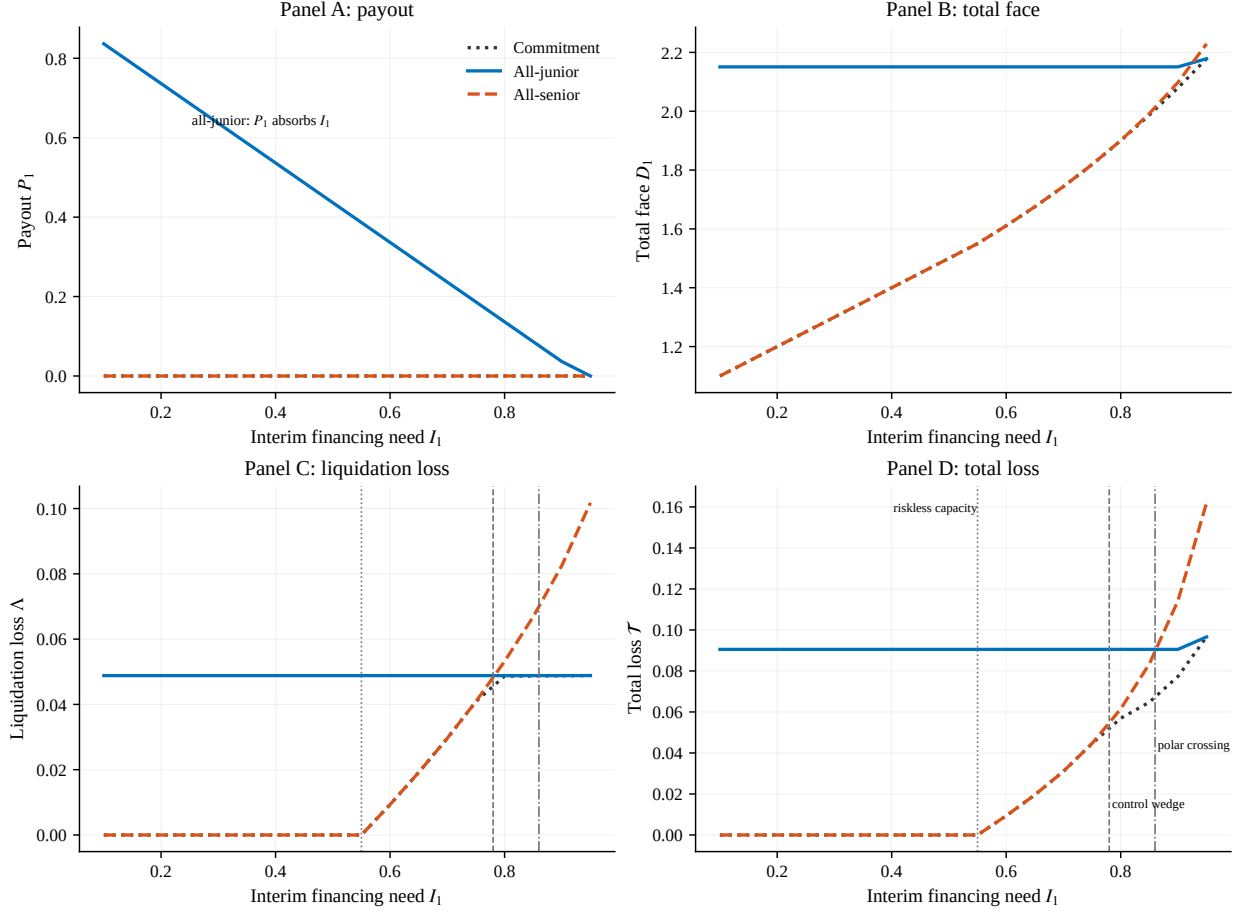


Figure 3: **Financing-need comparative statics.** Panels A–D report date-1 payout, total face, liquidation loss Λ , and total loss $\mathcal{T}$ as I_1 varies. The commitment, all-junior, and all-senior policies are solved separately at every point. The vertical lines mark exhaustion of riskless capacity, onset of the additional senior-control wedge, and the crossing of total loss under the two polar no-commitment policies.

loss exceeds that under all-junior debt. Thus, eliminating pari passu dilution need not improve the outcome. Because the policies select different total faces, the comparison combines the induced-face and priority-control effects in equation (58); it is not a fixed-face fragmentation comparison. The positive strict-share statistic shows that the date-0 all-senior liquidation outcome is not driven entirely by creditor indifference, while the reported values of $D_{\text{cap}} - D_1$ verify the capped-upside condition for each policy.

Figure 3 contrasts payout-driven borrowing with financing-driven issuance. Under all-junior debt, total face initially remains at its unconstrained date-1 optimum, so a larger financing need crowds out payout almost one-for-one until the financing constraint binds. Under all-senior debt,

payout is zero and issuance tracks the financing need. Thus, a larger I_1 weakens the benefit of priority protection by reducing the amount of payout-driven overborrowing it prevents.

The lower panels show the corresponding cost. Once protected debt becomes risky, senior control can liquidate states that unified creditor control would continue, opening the fragmentation-loss gap. The final crossing compares only the two polar policies; it does not characterize the globally optimal mixed structure.

The polar comparisons conceal an important feature of the design problem: partial seniority can dominate both. Figure 4 therefore reports the lowest-loss equilibrium candidates identified by the numerical search over the full range of date-0 senior faces at the baseline parameterization.

Starting from the all-junior policy at $S = 0$, additional seniority reduces the legacy debt exposed to pari passu dilution. Equilibrium total face and the refinancing-shortfall cost consequently fall. At approximately $S = 0.313$, the payout reaches zero. For $S \in [0.313, g_0]$, the baseline solution reproduces the commitment total face, liquidation rule, and total loss. Importantly, all policies are evaluated at the same distress-cost parameter $\chi = 1.50$. Under this common cost, all-junior debt still selects a positive payout and a higher total face than the commitment benchmark (Table 1), whereas payout is zero and total face coincides with the commitment benchmark throughout the partial-seniority interval. Thus, the change in issuance behavior is generated by seniority rather than by the distress-cost level alone. Once $S > g_0 = 0.55$, senior debt is no longer fully covered by contemporaneous cash flow in every state, and the direct priority-control cost becomes active on the reported branch. Partial seniority therefore dominates both polar policies by providing enough protection to eliminate payout-driven overborrowing without exposing a risky senior class to distorted control.

The right endpoint of the plateau depends on how creditor indifference is resolved. Under a tie rule favoring continuation, it extends to $a_C - cx = 0.83$ in the baseline calculation. To see why, for $S \leq a_C - cx$ the downside continuation payoff satisfies

$$Z^-(\theta_2) = V_C(\theta_2) - cx \geq a_C - cx \geq S,$$

while $V_L(\theta_2) \geq a_L > S$. Whenever the senior class is cash-uncovered, it therefore receives S under both liquidation and extension and is exactly indifferent. Resolving this indifference toward con-

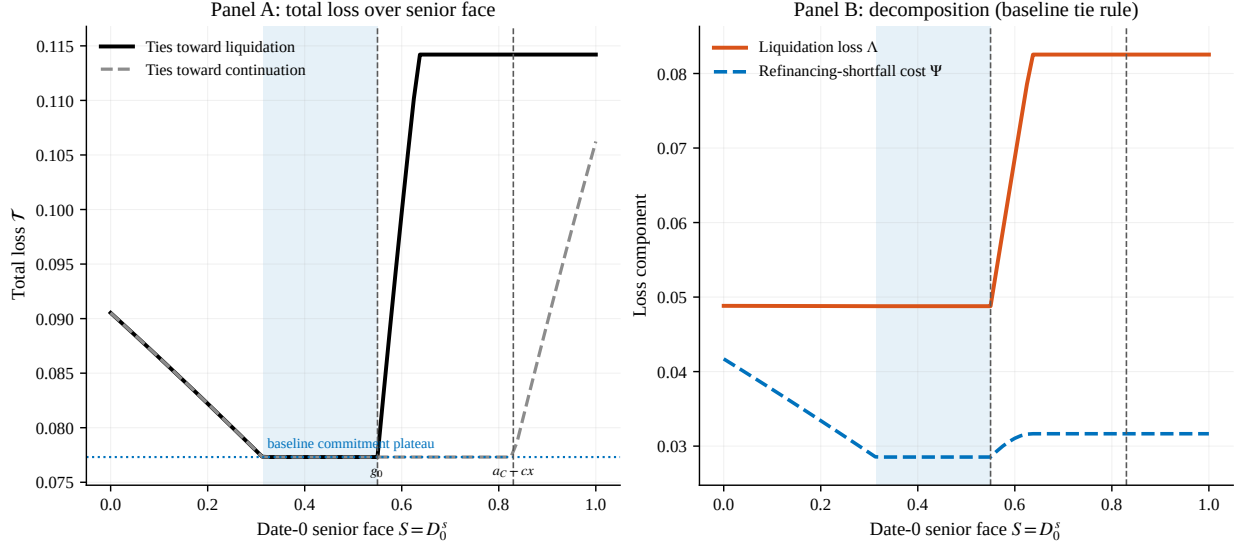


Figure 4: **Priority-design locus.** Panel A reports total expected loss over date-0 senior face S , with the date-1 issuance best response and date-0 break-even condition imposed at each point. The black solid line applies the baseline tie rule favoring liquidation, while the gray dashed line resolves creditor indifference toward continuation. The shaded interval marks the commitment plateau under the baseline rule. Panel B decomposes total loss under the baseline rule into liquidation loss and the refinancing-shortfall cost.

tinuation leaves the residual junior comparison equivalent to the unified-creditor rule and preserves the commitment allocation. The alternative locus is a default-stage resolution robustness exercise: the tie convention changes the right endpoint of the implementation plateau, but not the central result.

Figure 5 completes the priority-design comparison over the full senior-face domain. At low financing needs, the commitment face lies below the fragmentation kink, so the fragmentation region is empty. Every $S \in [S_{\text{lower}}(I_1), I_0]$ implements commitment, including risky senior faces. The lower endpoint rises with I_1 because the commitment face and the dilution incentive rise with the interim funding need.

Implementation fails on the certified window $I_1 \in (0.781991, 0.859514)$. Entry satisfies

$$I_1^{\text{enter}} = \Pi^{\text{one}}(2\ell_2 - \ell_3 + cx) - I_0 = 0.781991.$$

At $I_1 = 0.80$ and $S = I_0$, commitment-face financing slack is -1.69×10^{-4} , so the commitment target cannot finance the interim need. At $I_1 = 0.85$ and $S = g_0$, financing is feasible and the date-1

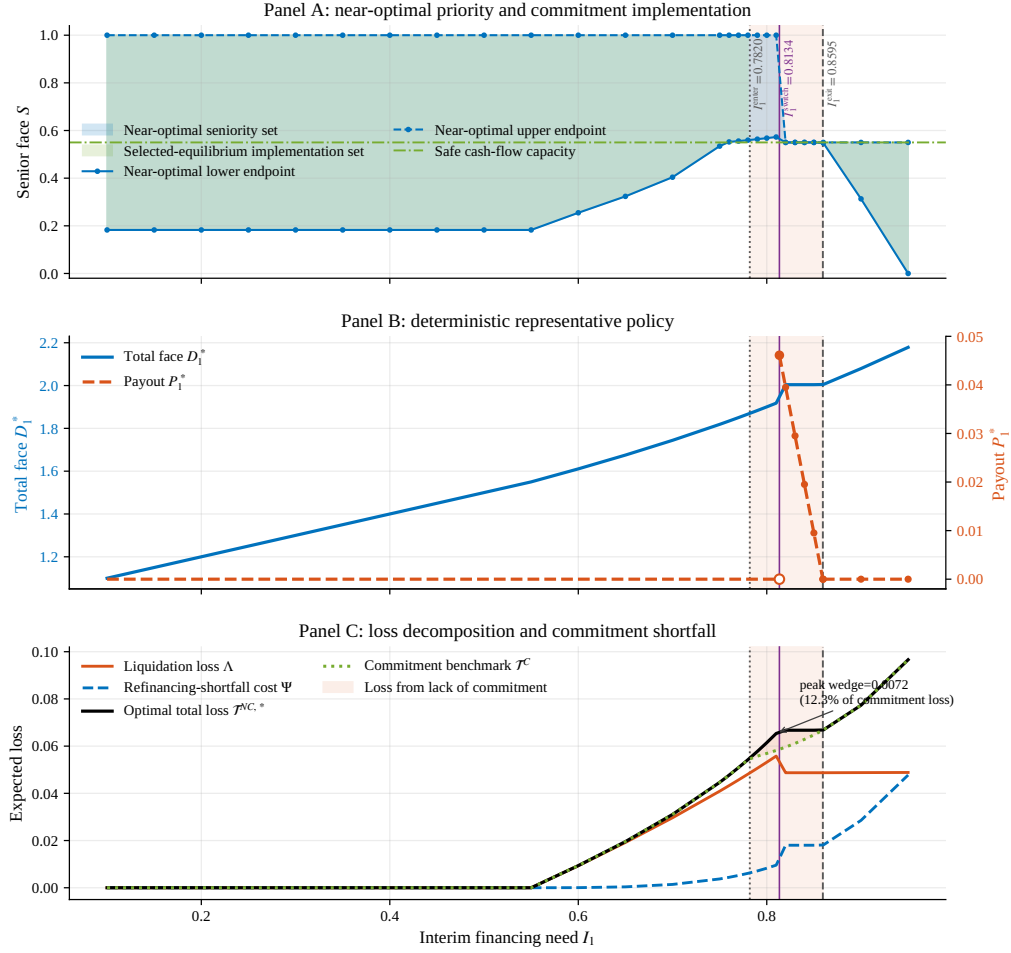


Figure 5: **Near-optimal priority protection and commitment implementation.** Panel A reports the seniority correspondence retained within an absolute objective tolerance of 10^{-7} , with a relative safeguard of 10^{-9} . Component interiors are sampled, and every reported seniority endpoint is certified by a symmetric bracket of width at most 9.16×10^{-7} . The green band uses the stronger selected-equilibrium implementation criterion; weak-best-response membership is reported separately in the replication diagnostics. The shaded window is the certified implementation gap. The vertical markers give entry $I_1^{\text{enter}} = 0.781991$, the risky-partial-to-safe-partial policy switch $I_1^{\text{switch}} = 0.813407$, and exit $I_1^{\text{exit}} = 0.859514$; all three bracket widths are at most 7.63×10^{-7} . Panel B reports the deterministic representative policy, defined as the midpoint of the widest certified near-optimal component, and renders the payout discontinuity at the certified interior switch explicitly rather than interpolating across the switch. On the safe-partial branch, $S^* = g_0 = 0.55$ up to endpoint-certification tolerance, $D_1^* = 2.004126$, and total loss is invariant in I_1 up to numerical tolerance; payout absorbs the financing need one-for-one and vanishes at the certified gap exit up to numerical tolerance. Panel C reports the loss decomposition and commitment benchmark. The peak shortfall is 0.0072 at $I_1 = 0.81$, equal to 0.72% of normalized date-0 investment and 12.3% of commitment-benchmark loss. The selected policy correspondence jumps at the interior switch, while the minimized-loss envelope is continuous because the two branch losses coincide at the certified crossing.

deviation gain is 7.16×10^{-5} ; the best-response face is 2.0031, compared with $D^{\text{com}} = 1.9870$, and payout is 0.0109. The implementation gap therefore identifies where the corollary’s conditional hypothesis fails; it does not contradict the corollary.

Inside the gap, priority protection is second best. The risky-partial branch is selected from entry through the certified branch crossing $I_1^{\text{switch}} = 0.813407$. The selected priority correspondence then jumps to the safe-capacity branch. The minimized-loss envelope remains continuous because the two branch losses are equal at the crossing. On $I_1 \in \{0.82, 0.83, 0.84, 0.85\}$, the safe-partial policy is invariant at the safe-capacity point $S^* = g_0 = 0.55$ (numerically 0.549999), $D_1^* = 2.004126$, and total loss 0.066737. Issue proceeds on the frozen safe-partial branch equal 0.859518; hence $P_1^* = 0.859518 - I_1$. The implied zero crossing is 3.48×10^{-6} above the independently certified exit $I_1^{\text{exit}} = 0.859514$; this small discrepancy is numerical drift in the frozen-branch proceeds rather than uncertainty about the certified transition. The excess-loss wedge peaks at 0.0072 at $I_1 = 0.81$.

Implementation resumes at $I_1 = 0.859514$. At $I_1 = 0.90$, the baseline implementation plateau is $S \in [0.313, 0.550]$. At $I_1 = 0.95$, the financing constraint disciplines issuance directly: all-junior payout is 0.0000, and the near-optimal lower endpoint falls to 0.0000, so all-junior debt belongs to the near-optimal set.

4 Extensions and Empirical Implications

The baseline model has two priority layers and a binary firm-wide liquidation decision. We first extend the analysis to arbitrary seniority partitions and characterize how priority refinement affects liquidation. We then examine collateral as asset-specific priority and show how the same mechanism can generate inefficient partial liquidation, before discussing the model’s empirical implications.

4.1 Multiple layers of seniority

We consider the liquidation–continuation problem at $t = 2$ when the firm has multiple layers of debt, holding all other aspects of the baseline setup fixed. We show that a finer seniority partition weakly expands liquidation and characterize the statewise liquidation limit as the partition mesh converges to zero.

4.1.1 Seniority partitions and the pivotal class

Fix total promised debt $D_1 > 0$. A K -layer seniority structure is represented by the partition

$$\mathcal{P} = \{0 = C_0 < C_1 < \dots < C_K = D_1\} \subset [0, D_1]. \quad (63)$$

The interval $(C_{i-1}, C_i]$ defines class i . Claims in this class are junior to all claims below C_{i-1} and senior to all claims above C_i . Thus, the cutoffs C_i record cumulative face value from the most senior to the most junior claims.

Under Assumption 2, participation in a default state θ_2 is determined by cash coverage. Class i participates in the liquidation–continuation decision if and only if

$$C_i > G(\theta_2).$$

Classes satisfying $C_i \leq G(\theta_2)$ are fully covered by contemporaneous cash, and do not participate. Although every cash-uncovered class has an objection right, only the first such class can affect the outcome.

Lemma 5. *Fix a default state θ_2 with $G(\theta_2) < D_1$ and a seniority partition*

$$\mathcal{P} = \{0 = C_0 < C_1 < \dots < C_K = D_1\}.$$

Let

$$j = \min\{i : C_i > G(\theta_2)\},$$

so that

$$C_{j-1} \leq G(\theta_2) < C_j.$$

Under Assumption 2, the liquidation–continuation outcome is determined solely by class j . Equivalently, the outcome is the same as if class j were the unique decisive class.

Proof. See Appendix A.

Lemma 5 reduces the creditor-objection protocol to a pivotal-class representation. For a parti-

tion $\mathcal{P}$ and a default state θ_2 , define the pivot cutoff by

$$C_{\text{pivot}}(\mathcal{P}; \theta_2) \equiv \min\{C_i \in \mathcal{P} : C_i > G(\theta_2)\}. \quad (64)$$

Because $G(\theta_2) < D_1$ in default, this cutoff is well defined. If $G(\theta_2)$ coincides with a cutoff, the strict inequality in equation (64) makes the next higher class pivotal.

Let j denote the pivotal class. Since $C_{j-1} \leq G(\theta_2)$, the cumulative face C_{j-1} of all claims senior to class j is repaid in full under either liquidation or continuation and therefore cancels from class j 's payoff comparison. Under Assumption 2, the pivotal class objects to continuation if and only if its own liquidation payoff is strictly positive and

$$\min\{C_{\text{pivot}}(\mathcal{P}; \theta_2), V_L(\theta_2)\} \geq \mathbb{E}[\min\{C_{\text{pivot}}(\mathcal{P}; \theta_2), Z(\theta_2, \theta_3)\} \mid \theta_2]. \quad (65)$$

Thus, despite the presence of multiple creditor classes, the resolution decision depends on a single state-dependent cutoff.

4.1.2 Priority refinement and coarsening

Let

$$\mathcal{P} = \{0 = C_0 < C_1 < \dots < C_K = D_1\}$$

and

$$\mathcal{P}' = \{0 = \tilde{C}_0 < \tilde{C}_1 < \dots < \tilde{C}_M = D_1\}$$

be seniority partitions with the same total promised debt. We say that $\mathcal{P}'$ is a *refinement* of $\mathcal{P}$ if

$$\mathcal{P} \subseteq \mathcal{P}'.$$

Equivalently, $\mathcal{P}'$ is obtained by inserting one or more cutoffs into $\mathcal{P}$, thereby splitting existing classes without changing total face or their relative priority.

For any state satisfying $G(\theta_2) < D_1$, refinement can only weakly lower the pivot cutoff:

$$C_{\text{pivot}}(\mathcal{P}'; \theta_2) \leq C_{\text{pivot}}(\mathcal{P}; \theta_2). \quad (66)$$

The smaller cutoff caps the pivotal class's payoff more tightly and therefore weakly reduces its exposure to continuation value.

To characterize increasingly fine priority structures, define the mesh of a partition by

$$\text{mesh}(\mathcal{P}) \equiv \max_{i=1,\dots,K} (C_i - C_{i-1}). \quad (67)$$

For every state satisfying $G(\theta_2) < D_1$,

$$0 < C_{\text{pivot}}(\mathcal{P}; \theta_2) - G(\theta_2) \leq \text{mesh}(\mathcal{P}). \quad (68)$$

Proposition 7 (Priority refinement and its limit). *Fix total promised debt D_1 .*

- (i) *Let $\mathcal{P}'$ be a refinement of $\mathcal{P}$. In every forced-default state, liquidation under $\mathcal{P}$ implies liquidation under $\mathcal{P}'$. Thus, liquidation is weakly increasing in priority refinement.*
- (ii) *Fix a forced-default state θ_2 satisfying $V_C(\theta_2) < D_1$. If*

$$\text{mesh}(\mathcal{P}) \leq V_L(\theta_2) - G(\theta_2) = L(\theta_2, 2),$$

then the firm liquidates at $t = 2$ in state θ_2 . Consequently, for any sequence of partitions $(\mathcal{P}_n)_{n \geq 1}$ satisfying $\text{mesh}(\mathcal{P}_n) \rightarrow 0$, there exists a state-dependent threshold $N(\theta_2)$ such that the firm liquidates in state θ_2 under every $\mathcal{P}_n$ with $n \geq N(\theta_2)$.

Proof. See Appendix A.

Lemma 6. *If liquidation is chosen when the decisive cutoff is C , then it is also chosen when the decisive cutoff is any $C' \in (0, C]$.*

Figure 6 illustrates the mechanism behind Proposition 7. Under the coarse partition, the pivotal junior class is not fully repaid in liquidation and therefore retains exposure to continuation value. Refinement can instead make a smaller intermediate class pivotal. When that class is fully repaid in liquidation, its payoff is capped at face value and it weakly favors liquidation. As the partition mesh converges to zero, the pivot cutoff approaches contemporaneous cash and the pivotal class's incremental exposure to continuation value vanishes.

Let $\mathbf{1}_{\text{liq}}(\theta_2; \mathcal{P}, D_1)$ denote the equilibrium liquidation indicator under partition $\mathcal{P}$, setting the

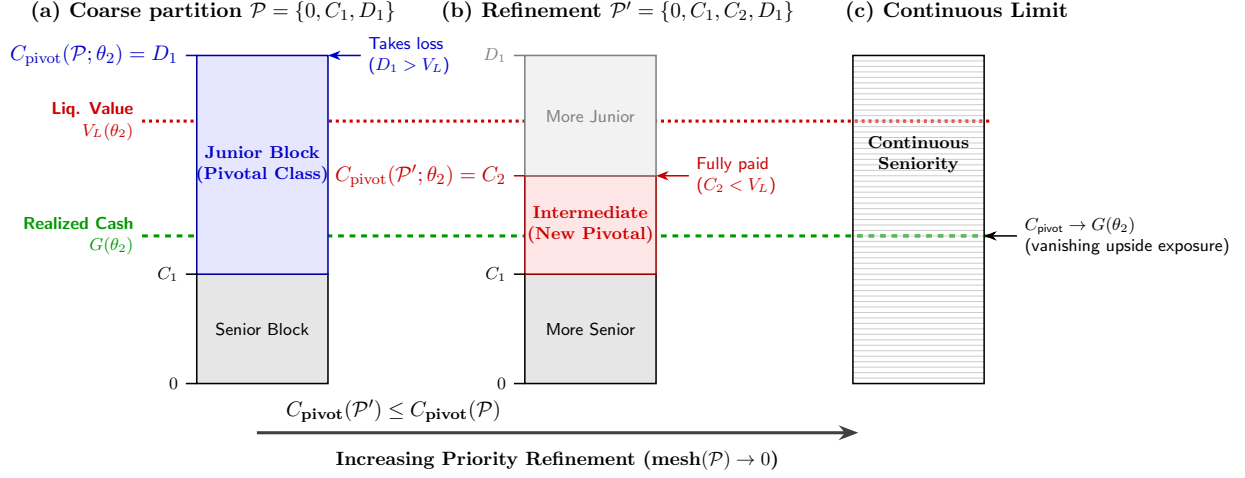


Figure 6: **Priority refinement and the pivotal class.** For a fixed state θ_2 , the pivot cutoff is the first partition cutoff above contemporaneous cash $G(\theta_2)$. Panel (a) shows a coarse partition in which the junior class is pivotal. Panel (b) inserts an intermediate cutoff, lowering the pivot cutoff and making the new pivotal class fully covered by liquidation value. Panel (c) illustrates the limit in which the partition mesh converges to zero and the pivot cutoff approaches $G(\theta_2)$.

indicator equal to zero whenever $V_C(\theta_2) \geq D_1$. Part (ii) of Proposition 7 implies that, along any sequence satisfying $\text{mesh}(\mathcal{P}_n) \rightarrow 0$,

$$\mathbf{1}_{\text{liq}}(\theta_2; \mathcal{P}_n, D_1) \longrightarrow \mathbf{1}\{V_C(\theta_2) < D_1\} \quad \text{for every } \theta_2.$$

Since

$$0 < \Delta(\theta_2) < V_3(\theta_2) \leq V_C(\theta_2),$$

equation (3) and dominated convergence imply

$$\lim_{n \rightarrow \infty} \mathbb{E}[\Delta(\theta_2) \mathbf{1}_{\text{liq}}(\theta_2; \mathcal{P}_n, D_1)] = \mathbb{E}[\Delta(\theta_2) \mathbf{1}\{V_C(\theta_2) < D_1\}]. \quad (69)$$

Thus, in the limit of continuous seniority, every forced default results in liquidation.

Proposition 7 uses the baseline weak-objection rule. Appendix Section B separately provides a sufficient condition for the same statewise conclusion when continuation proposals may include bounded consent transfers. Online Appendix Section OA.1 shows that refinement monotonicity in part (i) also holds under the strict-objection convention; see Proposition 11(i). The fine-partition liquidation limit in part (ii) survives under the additional downside-continuation condition in Propo-

sition 11(ii).

The number of priority layers alone, however, does not determine the fragmentation cost. Refinement has no effect when all newly created cutoffs remain covered by contemporaneous cash.

Corollary 3 (Safe priority refinement). *Fix total promised debt $D > 0$ and a seniority partition $\mathcal{P}$ of $[0, D]$. Let $\mathcal{P}'$ be a finite refinement of $\mathcal{P}$ satisfying*

$$\mathcal{P}' \setminus \mathcal{P} \subseteq [0, \underline{G}], \quad \underline{G} \equiv \inf_{\theta_2} G(\theta_2).$$

Then:

- (i) *Whenever a cash-uncovered class exists, $\mathcal{P}$ and $\mathcal{P}'$ have the same pivotal cutoff. They therefore induce the same repayment, default, and liquidation decisions.*
- (ii) *The two partitions induce the same liquidation set, expected liquidation deadweight loss, state-wise aggregate creditor payoff, and aggregate debt value. Every tranche whose upper cutoff is weakly below $\underline{G}$ is repaid in full from contemporaneous cash and has unit price.*

Proof. See Appendix A.

Corollary 3 shows that the firm can create any finite refinement within the always cash-covered portion of its capital structure without generating a fragmentation distortion. A newly inserted cutoff affects resolution only if it can become cash-uncovered and hence pivotal. This is an equality case of Proposition 7, which otherwise establishes only weak monotonicity.

The fixed-face refinement result also yields a conditional coarsening principle for ex ante design. Holding total protected face S , legacy junior face U , and date-1 junior issuance y fixed, subdividing the protected block provides no additional protection against dilution by the junior pool but can place control in the hands of a more tightly capped intermediate class. Corollary 4, stated and proved in Appendix A.13, shows that coarsening the protected block weakly contracts the liquidation set and weakly lowers expected liquidation deadweight loss, with a strict reduction whenever the two liquidation sets differ on a set of positive probability.

If $S \leq \underline{G}$, the refined and coarsened structures differ only through safe cutoffs. Corollary 3 then implies equality for every candidate junior issue y : date-1 junior pricing and best responses, as well as aggregate date-0 issuance proceeds, are unchanged. This is a fixed-architecture result and does not determine the optimal protected face or the optimal response to future financing needs.

Online Appendix Section OA.2 extends this logic to n financing dates and compares two polar financing policies. Repeated protected issuance generates an increasingly fine maturity priority partition, whereas repeated unprotected issuance accumulates in a common *pari passu* pool. The appendix decomposes the difference in expected loss between these policies into a fixed-face fragmentation component and a dilution-induced leverage component, without characterizing the unrestricted sequential financing problem.

4.2 Collateral as Asset-Specific Priority

The baseline model represents priority as a claim on a common pool of value and treats liquidation as a binary, firm-wide decision. Collateral is economically distinct: it gives a creditor priority over a subset of the firm's assets and a *pari passu* deficiency claim on the remainder. This subsection formalizes that distinction, extending our priority-fragmentation mechanism to environments with partial liquidation and asset lumpiness.

Fix a forced-default state θ_2 and suppress the state variable below. Write $G_2 = G(\theta_2)$ and $L_2 = L(\theta_2, 2)$, while G_3 and L_3 denote the date-3 cash flow and liquidation proceeds. All expectations are conditional on θ_2 , and G_2 remains available under either immediate enforcement or continuation.

Let A denote the pledged component of the firm and B the unpledged component. Suppose that the pledged component accounts for a constant share $\alpha \in (0, 1)$ of asset payoffs:

$$(L_2^A, G_3^A, L_3^A) = \alpha(L_2, G_3, L_3), \quad (L_2^B, G_3^B, L_3^B) = (1 - \alpha)(L_2, G_3, L_3).$$

We maintain the going-concern condition $\mathbb{E}[G_3 + L_3] > L_2$, which, by proportionality, implies $\mathbb{E}[G_3^A + L_3^A] > L_2^A$. The proportionality restriction is a tractability condition: the results below concern pledged components whose payoffs scale proportionally with aggregate asset payoffs.

A creditor holds a secured claim with face $D^{\text{sec}} > 0$, while the remaining unsecured claims have total face $D^{\text{unsec}} \geq 0$. The secured creditor has first priority over L_2^A under immediate enforcement and over L_3^A under continuation. Operating cash flows are unencumbered, and any unpaid portion of the secured claim becomes a general unsecured deficiency claim that shares the residual pool *pari passu* with D^{unsec} . Any collateral proceeds in excess of the secured claim return to the residual estate. At forced default, contemporaneous cash G_2 remains in the unencumbered estate and is

not distributed before the enforcement decision. The secured creditor's priority therefore extends only to pledged collateral proceeds; any recovery from operating cash arises through its *pari passu* deficiency claim.

The pledged component A is fixed contractually, and the secured creditor has the exclusive right to enforce against it. Enforcement produces partial liquidation, denoted PL : A is liquidated at $t = 2$ while B continues to $t = 3$. Non-enforcement produces full continuation, denoted C : both components continue to $t = 3$. We retain the weak-objection convention from Assumption 2: the secured creditor enforces when its partial-liquidation payoff is strictly positive and weakly exceeds its continuation payoff. The baseline cash-coverage participation rule is replaced in this subsection by the secured creditor's exclusive contractual enforcement right over component A . Unsecured creditors cannot separately force the sale of A , and we abstract from any separate resolution decision concerning B to isolate the collateral-enforcement channel. Online Appendix Section OA.2.2 studies a broader divisible-asset environment under payoff conservation, common incremental objection rights, and path-independent class payoffs. The proportional-payoff environment and enforcement technology described above are maintained throughout this subsection. The total and unencumbered residual pools under these outcomes are

$$\begin{aligned} \text{PL:} \quad X^{PL} &= L_2^A + G_2 + G_3^B + L_3^B, & Y^{PL} &= G_2 + G_3^B + L_3^B, \\ \text{C:} \quad X^C &= G_2 + G_3 + L_3, & Y^C &= G_2 + G_3 + L_3^B. \end{aligned}$$

For a deficiency claim of size $h \geq 0$ against an unencumbered residual pool $Y \geq 0$, define its *pari passu* recovery by

$$\mathcal{R}_{\text{def}}(h, Y; D^{\text{unsec}}) \equiv \begin{cases} \frac{h}{h + D^{\text{unsec}}} \min\{Y, h + D^{\text{unsec}}\}, & h + D^{\text{unsec}} > 0, \\ 0, & h + D^{\text{unsec}} = 0. \end{cases}$$

The secured creditor's expected payoffs from partial liquidation and continuation are therefore

$$U_{\text{sec}}^{PL} = \min\{D^{\text{sec}}, L_2^A\} + \mathbb{E}[\mathcal{R}_{\text{def}}((D^{\text{sec}} - L_2^A)^+, Y^{PL}; D^{\text{unsec}})], \quad (70)$$

$$U_{\text{sec}}^C = \mathbb{E}[\min\{D^{\text{sec}}, L_3^A\} + \mathcal{R}_{\text{def}}((D^{\text{sec}} - L_3^A)^+, Y^C; D^{\text{unsec}})]. \quad (71)$$

To isolate the effect of priority fragmentation, consider a one-layer benchmark with the same total face

$$D = D^{\text{sec}} + D^{\text{unsec}},$$

in which all debt belongs to a single *pari passu* class. Hold the resolution technology fixed by giving the unified class the same choice between partial liquidation of A and full continuation, governed by the same objection rule. Its expected payoffs are

$$U_{\text{one}}^{PL}(D) = \mathbb{E}[\min\{D, X^{PL}\}], \quad U_{\text{one}}^C(D) = \mathbb{E}[\min\{D, X^C\}].$$

Proposition 8 (Collateral fragmentation effect). *Fix total face D and compare the one-layer structure with a collateralized structure having secured face $D^{\text{sec}} \in (0, D)$ and unsecured face $D^{\text{unsec}} = D - D^{\text{sec}}$. In any forced-default state, if the one-layer class objects to full continuation in favor of partial liquidation of A , then the secured creditor also objects and, by its exclusive enforcement right, liquidates A . Hence introducing a secured layer weakly expands the set of forced-default states in which A is liquidated.*

Proof. See Appendix A.15.

Proposition 8 gives a direct collateral counterpart to the seniority-fragmentation effect. Under proportional asset payoffs, the secured creditor inherits any preference of the unified class for partial liquidation, so introducing a secured layer cannot eliminate liquidation of A . Because $\mathbb{E}[G_3^A + L_3^A] > L_2^A$, any such enforcement is socially inefficient. Thus, when collateral can be enforced separately, the fragmentation distortion can take the form of an inefficient sale of the pledged asset. Beyond this weak fragmentation effect, collateral creates an additional channel through the dilution of the secured creditor's deficiency claim. To characterize this channel, define

$$\Gamma_{\text{sec}} \equiv \min\{D^{\text{sec}}, L_2^A\} - \mathbb{E}[\min\{D^{\text{sec}}, L_3^A\}].$$

Proposition 9 (Deficiency dilution and secured-creditor liquidation bias). *Fix D^{sec} and the asset-payoff primitives. As $D^{\text{unsec}} \rightarrow \infty$,*

$$U_{\text{sec}}^{PL} - U_{\text{sec}}^C \longrightarrow \Gamma_{\text{sec}},$$

whereas

$$\begin{aligned} & U_{\text{one}}^{PL}(D^{\text{sec}} + D^{\text{unsec}}) - U_{\text{one}}^C(D^{\text{sec}} + D^{\text{unsec}}) \\ & \longrightarrow \mathbb{E}[X^{PL} - X^C] = -\alpha(\mathbb{E}[G_3 + L_3] - L_2) < 0. \end{aligned}$$

Consequently, if $\Gamma_{\text{sec}} > 0$, there exists $\bar{D}^{\text{unsec}} < \infty$ such that, for every $D^{\text{unsec}} > \bar{D}^{\text{unsec}}$, the secured creditor objects to full continuation in favor of partial liquidation, whereas the corresponding one-layer class strictly prefers continuation.

Proof. See Appendix A.16.

The proposition identifies the source of secured creditors' liquidation bias. As the unsecured pool becomes large, deficiency recoveries under both outcomes vanish, leaving the secured creditor focused on the priority-recovery comparison Γ_{sec} . The unified class instead internalizes the difference in total resources and therefore preserves the pledged component's going-concern surplus.

When immediate collateral proceeds fully cover the secured claim, $L_2^A \geq D^{\text{sec}}$,

$$\Gamma_{\text{sec}} > 0 \iff \Pr(L_3^A < D^{\text{sec}}) > 0.$$

Thus, if continuation collateral is insufficient with positive probability, sufficiently extensive unsecured issuance makes the secured creditor strictly prefer immediate enforcement. At the same time, the fully covered enforcement payoff remains protected from additional unsecured claims.

Collateral also introduces a payoff-allocation channel absent from ordinary all-asset seniority. For comparison, an all-asset senior unsecured creditor with face D^{sec} has expected payoffs

$$U_{\text{sen}}^{PL} = \mathbb{E}[\min\{D^{\text{sec}}, X^{PL}\}], \quad U_{\text{sen}}^C = \mathbb{E}[\min\{D^{\text{sec}}, X^C\}].$$

Lemma 7 (Secured versus all-asset senior debt). *For either stipulated resolution outcome,*

$$U_{\text{sen}}^{PL} \geq U_{\text{sec}}^{PL}, \quad U_{\text{sen}}^C \geq U_{\text{sec}}^C.$$

If immediate collateral proceeds fully cover the secured claim, $L_2^A \geq D^{\text{sec}}$, then

$$U_{\text{sen}}^{PL} = U_{\text{sec}}^{PL} = D^{\text{sec}}$$

and

$$U_{\text{sec}}^{PL} - U_{\text{sec}}^C \geq U_{\text{sen}}^{PL} - U_{\text{sen}}^C \geq 0.$$

Proof. See Appendix A.17.

The lemma distinguishes asset-specific collateral from all-asset seniority. An all-asset senior creditor has priority over the entire firm resource pool, whereas a secured creditor has priority only over A and shares any deficiency claim with the unsecured pool. When immediate collateral proceeds fully cover the common face value, both creditors receive the same enforcement payoff, but dilution of the secured creditor’s deficiency recovery can lower its continuation payoff. Collateral can therefore generate a stronger enforcement incentive than all-asset seniority.

The baseline and collateral models describe two resolution technologies. When only firm-wide resolution is available, fragmentation can produce inefficient liquidation of the firm. When a fixed collateral package can be enforced separately, the same capped-payoff mechanism can instead produce an inefficient sale of the pledged asset. Collateral additionally exposes the secured creditor’s continuation payoff to dilution of its deficiency claim, a channel absent from all-asset senior debt. The model does not derive which resolution technology is available or a comparative static in asset redeployability; those are empirical interpretations of the two environments.

4.3 Empirical Relevance and Implications

An important empirical question is why many firms concentrate borrowing in a small number of debt types, as documented by Colla et al. (2013). Debt specialization is broader than priority concentration, but our mechanism speaks specifically to its priority and security dimensions. At the same time, corporate liabilities display substantial heterogeneity in priority and security (Barclay and Smith, 1995; Rauh and Sufi, 2010). Badoer et al. (2020) further show that firms spread priority as their credit quality deteriorates, increasing their reliance on secured and subordinated debt, and argue that security protects creditors against dilution more effectively than payment-priority covenants. This evidence speaks to the dilution-protection benefit of priority. Our framework offers a priority-based explanation for why concentration along these dimensions may nevertheless be valuable: creating protected layers can place resolution control in the hands of a more tightly capped creditor and thereby expand inefficient liquidation. The resulting trade-off can favor priority

concentration when resolution distortions dominate and priority dispersion when protection against future dilution is especially valuable. This interpretation is closely aligned with John et al. (2021), who find that stronger creditor protection is associated with more concentrated debt structures and attribute this pattern to the greater value of concentration in mitigating creditor liquidation bias and facilitating distressed renegotiation.

Institutional and empirical work on Chapter 11 underscores the relevance of the model’s control channel. Baird and Rasmussen (2002) emphasize the shift from traditional negotiated reorganizations toward creditor control and going-concern sales. More directly, Ayotte and Morrison (2009) find pervasive creditor control and show that Chapter 11 cases are more likely to result in a sale when secured lenders are oversecured, consistent with secured-creditor fire-sale bias. Hotchkiss et al. (2023) review broader evidence that the growing influence of senior secured creditors is associated with greater use of sales and asset dispositions, while noting that such sales need not be inefficient. This evidence supports the link between secured-creditor control and sale outcomes, although it does not by itself establish the model’s inefficiency prediction. Complementary evidence from covenant violations shows that creditor control affects investment, financing, and governance decisions outside bankruptcy (Chava and Roberts, 2008; Roberts and Sufi, 2009; Nini et al., 2012). In the baseline model, the relevant outcome is firm-wide liquidation; in the collateral extension, it is enforcement against the pledged component.

Against this background, the model yields three empirical implications that distinguish cash-covered from cash-uncovered priority and firm-wide from collateral-specific enforcement.

Prediction 1: Cash-uncovered priority refinement weakly increases liquidation.

Proposition 7(i) shows that, holding total face fixed, refining the priority partition weakly expands the set of forced-default states in which liquidation occurs. Part (ii) provides the limiting benchmark: as the partition mesh converges to zero, every forced default eventually results in liquidation under the baseline objection rule. The relevant empirical measure is not simply the number of priority classes. A newly created cutoff affects resolution only when it can become cash-uncovered and hence pivotal. Conditional on total leverage and firm fundamentals, finer priority partitions within the cash-uncovered portion of the capital structure should therefore be associated with more liquidation or sale-oriented resolutions and less continuation through traditional reorganization. Among structures in which a priority class controls resolution, a lower cumulative face at

the first cash-uncovered cutoff should likewise be associated with a stronger liquidation incentive.

Prediction 2: Safe cash coverage supports finer priority structures.

Corollary 3 shows that priority layers contained within the portion of debt always covered by contemporaneous cash do not affect resolution. A firm can therefore preserve issue-by-issue priority within this cash-covered portion, potentially creating many layers without generating a fragmentation cost. Once a protected layer can become cash-uncovered, however, the same protection can distort resolution. Conditional on financing needs and the value of dilution protection, stronger cash coverage should therefore be associated with more and finer protected layers, whereas weaker cash coverage should tilt debt design toward fewer or thicker priority classes, greater use of *pari passu* issuance, and greater exposure to subsequent dilution. This prediction concerns cash coverage rather than collateral coverage: a claim fully covered by collateral can still control enforcement against the pledged asset.

Prediction 3: Collateral fragmentation weakly increases pledged-asset sales, while deficiency dilution can make the increase strict.

In environments in which the secured creditor controls enforcement against the pledged asset, Proposition 8 compares a collateralized structure with a one-layer structure having the same total face and the same partial-liquidation-versus-continuation technology. If the unified class prefers liquidation of the pledged component, the secured creditor does as well. Introducing a secured layer therefore weakly expands the set of states in which the pledged asset is sold. Proposition 9 identifies an additional channel specific to collateral. Holding the secured face and asset-payoff primitives fixed, a larger unsecured pool dilutes the secured creditor's deficiency recovery. When immediate priority recovery exceeds expected continuation priority recovery, sufficiently extensive unsecured issuance induces the secured creditor to enforce even though the corresponding unified class strictly prefers continuation. Thus, holding the secured face and asset fundamentals fixed, pledged-asset sales should be especially likely when immediate collateral recovery is high relative to expected continuation collateral recovery and the unsecured pool is sufficiently large to dilute the secured creditor's deficiency claim.

Together, these predictions emphasize two distinctions: priority layers that remain cash-covered differ from layers that can become pivotal, and firm-wide resolution differs from collateral-specific enforcement. Debt priority is therefore not merely distributional; it changes who controls distress

resolution and whether the firm or a pledged component continues.

5 Conclusion

This paper develops a theory of debt priority design that trades off ex ante dilution protection against ex post resolution efficiency. Priority protects legacy creditors from dilution by future borrowing and can thereby restrain payout-driven overborrowing. Yet it also divides creditors into separately ranked classes and changes who controls the firm in distress. Priority therefore allocates not only cash flows, but also decision-making power.

Holding total promised repayment fixed, priority splitting weakly expands inefficient liquidation. When more-senior claims are covered by contemporaneous cash, the pivotal junior class follows the same liquidation rule as a unified creditor class. When a senior class becomes cash-uncovered, however, its capped continuation payoff can make liquidation privately attractive even when a unified class would continue. This asymmetry extends to arbitrary priority structures: refining the priority partition weakly expands liquidation, and sufficiently fine partitions lead every forced-default state to liquidation under the baseline rule. Refinements within the always cash-covered portion of the capital structure are neutral. Holding total protected face, legacy junior face, and subsequent junior issuance fixed, subdividing a cash-uncovered protected block can increase liquidation loss without providing additional dilution protection.

The ex ante design implication is a trade-off between dilution and fragmentation. Safe seniority protects legacy creditors without distorting resolution and, under the conditions of the model, an optimal structure can be chosen to use all available safe seniority. When this protection is sufficient to discipline interim issuance, it can implement the commitment allocation. Beyond the safe region, further priority protection may reduce dilution and equilibrium total face but also creates a direct priority-control cost. The numerical illustration shows that partial seniority can dominate both all-junior and all-senior debt by providing sufficient dilution protection while limiting distorted senior control. Over the reported financing-need domain, partial seniority implements the commitment benchmark outside a single interior interval; within that interval, the second-best policy switches from risky partial seniority to the safe-capacity boundary.

The collateral extension shows how the mechanism operates under asset-specific priority in the

maintained collateral environment. Introducing a secured layer weakly expands inefficient sales of the pledged asset, while subsequent unsecured borrowing can further encourage enforcement by diluting the secured creditor's deficiency claim. More broadly, the analysis provides a rationale for coarse priority structures despite the anti-dilution appeal of issue-by-issue priority. It predicts that cash-uncovered priority fragmentation should be associated with more liquidation- and sale-oriented resolutions. The relation between priority granularity and liquidation should be weak while the relevant cumulative cutoffs remain cash-covered and stronger once they become cash-uncovered.

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A Proofs

This appendix collects omitted proofs. Section A.0 gives a normalization lemma used in minimal-face funding arguments. Sections A.1–A.2A prove the refinancing/default rule, the date-0 reduction, and sufficient conditions for monotone total loss. Section A.3 gives the consolidated proof of Proposition 4. Sections A.4–A.5 prove the two parts of Proposition 5. Section A.6 establishes the two-layer strategic-default transmission result, aggregate pricing under safe seniority, existence, and Proposition 6; Section A.7 proves Corollary 1. Sections A.8–A.11 derive the tractable liquidation cutoffs, continuation-risk comparative statics, closed-form pricing objects, and numerical implementation. Sections A.12–A.14 establish pivot-cutoff monotonicity, priority-refinement monotonicity, safe-refinement neutrality, conditional dominance of coarsening, and the continuous-seniority limit; Sections A.15–A.17 prove the collateral results. The remaining multi-date and general-fragmentation proofs are collected in the Online Appendix.

A.0 A normalization lemma used repeatedly

Lemma 8. *Fix some $\bar{I} \geq 0$. Suppose $\Pi : [0, \infty) \rightarrow \mathbb{R}$ is continuous with $\Pi(0) = 0$. Suppose there exists $D \geq 0$ such that $\Pi(D) \geq \bar{I}$. Then there exists $\underline{D}(\bar{I}) \in [0, D]$ such that*

$$\Pi(\underline{D}(\bar{I})) = \bar{I}.$$

Proof. If $\bar{I} = 0$, take $\underline{D}(0) = 0$. Now suppose $\bar{I} > 0$. Because Π is continuous, the set $\{d \in [0, D] : \Pi(d) \geq \bar{I}\}$ is closed and nonempty, hence the minimum

$$\underline{D}(\bar{I}) \equiv \min\{d \in [0, D] : \Pi(d) \geq \bar{I}\}$$

is attained. Note that $\underline{D}(\bar{I}) > 0$ since $\Pi(0) = 0 < \bar{I}$. If $\Pi(\underline{D}(\bar{I})) > \bar{I}$, by continuity there exists $\varepsilon \in (0, \underline{D}(\bar{I}))$ such that $\Pi(\underline{D}(\bar{I}) - \varepsilon) > \bar{I}$, contradicting minimality. Thus $\Pi(\underline{D}(\bar{I})) = \bar{I}$. $\square$

A.1 Proof of Lemma 1

Proof. When full repayment is feasible, $V_C(\theta_2) \geq D_1$, and repayment and default give the entrepreneur identical expected payoffs, the selection follows the creditors' resolution in default: if

default would trigger liquidation under Assumption 2, the entrepreneur repays; if default would lead to extension, she defaults. The convention applies only to payoff-equivalent actions and pins down the repayment/default selection used below.

Fix θ_2 with $V_C(\theta_2) \geq D_1$. Full repayment gives equity $V_C(\theta_2) - D_1$.

If default would trigger liquidation, equity receives $(V_L(\theta_2) - D_1)^+$. Assumption 1 implies $V_L(\theta_2) < V_C(\theta_2)$, so repayment is weakly preferred. When the two payoffs coincide, the equilibrium convention selects repayment. Thus a repayment-feasible state in which default would liquidate is repaid.

If default would instead trigger extension, equity receives

$$\mathbb{E}[(Z(\theta_2, \theta_3) - D_1)^+ \mid \theta_2] = V_C(\theta_2) - \mathbb{E}[\min\{D_1, Z(\theta_2, \theta_3)\} \mid \theta_2] \geq V_C(\theta_2) - D_1.$$

Strategic default is therefore weakly optimal in this case. It is strictly optimal when the inequality is strict; when the payoffs coincide, the selection convention above chooses default. Hence, whenever default would lead to extension, the entrepreneur defaults and creditors extend.

Consequently, no repayment-feasible state is liquidated. When $V_C(\theta_2) < D_1$, repayment is infeasible and default is forced, so liquidation occurs exactly when at least one participating class objects under Assumption 2. $\square$

A.2 Proof of Proposition 3

Proof. Fix any $(D_0^s, D_0^u, D_1^u) \in \mathcal{E}^{\text{nc}}$. Define total junior face due at $t = 2$ by

$$D^u \equiv D_0^u + D_1^u,$$

and write $\mathbf{1}_{\text{liq}}(\theta_2; D_0^s, D^u)$ for the induced liquidation indicator. Consider the total realized resources delivered to claimholders (debt plus equity), denoted $\mathcal{R}$. If the firm liquidates at $t = 2$, resources equal $V_L(\theta_2)$. If the firm continues to $t = 3$, resources equal $Z(\theta_2, \theta_3)$. Thus,

$$\begin{aligned} \mathcal{R} &= \mathbf{1}_{\text{liq}}(\theta_2; \cdot) V_L(\theta_2) + (1 - \mathbf{1}_{\text{liq}}(\theta_2; \cdot)) Z(\theta_2, \theta_3) \\ &= G(\theta_2) + \mathbf{1}_{\text{liq}}(\theta_2; \cdot) L(\theta_2, 2) + (1 - \mathbf{1}_{\text{liq}}(\theta_2; \cdot)) (G(\theta_3) + L(\theta_3, 3)). \end{aligned}$$

Taking conditional expectations given θ_2 gives

$$\mathbb{E}[\mathcal{R} \mid \theta_2] = G(\theta_2) + V_3(\theta_2) - \mathbf{1}_{\text{liq}}(\theta_2; \cdot) \Delta(\theta_2).$$

Therefore,

$$\mathbb{E}[\mathcal{R}] = \mathbb{E}[G(\theta_2) + V_3(\theta_2)] - \Lambda(D_0^s, D^u).$$

The first term is finite by equation (3), depends only on primitives, and is independent of the financing policy. The private real distress cost in Assumption 3 is borne by the entrepreneur outside the bankruptcy estate. Hence

$$\mathbb{E}[\mathcal{R}] - \Psi(D_0^s + D^u) - I_0 - I_1 = \mathbb{E}[G(\theta_2) + V_3(\theta_2)] - I_0 - I_1 - \mathcal{T}(D_0^s, D^u).$$

Equivalently, let U_0 denote the entrepreneur's date-0 expected payoff, equal to expected payouts plus the continuation market value of equity net of the privately borne real distress cost. Competitive break-even at dates 0 and 1 implies

$$\mathbb{E}[\mathcal{R}] - \Psi(D_0^s + D^u) = I_0 + I_1 + U_0.$$

Because I_0 and I_1 are fixed primitives, maximizing U_0 over $\mathcal{E}^{\text{nc}}$ is equivalent to minimizing total loss over the same equilibrium set. The minimized object is $\mathcal{T}(D_0^s, D_0^u + D_1^u)$. The same identity applies to any restricted policy class after imposing that class's continuation and feasibility constraints. $\square$

A.2A Sufficient conditions for monotone total loss

This subsection provides sufficient conditions for Assumption 4. Write

$$\mathbf{D} \equiv (D_0^s, D^u)$$

for an outstanding face vector, and use the componentwise order $\mathbf{D}' \geq \mathbf{D}$ to mean $D_0^{s'} \geq D_0^s$ and $D^{u'} \geq D^u$.

Define the set of outstanding face vectors generated by feasible no-commitment continuations

as

$$\mathcal{G}^{\text{nc}} \equiv \{(D_0^s, D_0^u + D_1^u) : (D_0^s, D_0^u, D_1^u) \in \mathcal{E}^{\text{nc}}\}.$$

Let $\mathcal{A}^{\text{com}}$ denote the feasible committed issuance plans used in the commitment benchmark and define

$$\mathcal{G}^{\text{com}} \equiv \{(D_0^s, D_0^u + D_1^u) : (D_0^s, D_0^u, D_1^u) \in \mathcal{A}^{\text{com}}\}.$$

A set is order-convex if, whenever it contains two vectors $\mathbf{D} \leq \mathbf{D}'$, it also contains every $\tilde{\mathbf{D}}$ satisfying $\mathbf{D} \leq \tilde{\mathbf{D}} \leq \mathbf{D}'$. The policy-relevant domain $\mathcal{D}$ is the closed order-convex hull of $\mathcal{G}^{\text{nc}} \cup \mathcal{G}^{\text{com}}$. Thus the domain contains every committed and no-commitment face vector compared in the main analysis, together with the coordinatewise segments needed for monotonicity arguments; only vectors in the two generating sets need arise as equilibrium or committed policies.

For each $\mathbf{D} \in \mathcal{D}$, define the equilibrium liquidation region

$$\mathcal{L}(\mathbf{D}) \equiv \{\theta_2 : \mathbf{1}_{\text{liq}}(\theta_2; D_0^s, D^u) = 1\}.$$

Expected liquidation deadweight loss can then be written as

$$\Lambda(\mathbf{D}) \equiv \Lambda(D_0^s, D^u) = \int_{\mathcal{L}(\mathbf{D})} \Delta(\theta_2) dF_2(\theta_2),$$

where F_2 is the distribution of θ_2 . Total expected real loss is

$$\mathcal{T}(\mathbf{D}) \equiv \mathcal{T}(D_0^s, D^u) = \Lambda(\mathbf{D}) + \Psi(D_0^s + D^u).$$

Higher face may induce continuation in some states and thereby reduce Λ . Assumption 4 holds if any such decline is dominated by the increase in the private distress cost. The following lemma formalizes this requirement in marginal terms.

Lemma 9 (Marginal-loss dominance). *Suppose that Λ is absolutely continuous along every coordinate segment contained in $\mathcal{D}$, and that Ψ is locally absolutely continuous on every interval of total*

face attained along such a segment. If

$$\frac{\partial \Lambda(D_0^s, D^u)}{\partial D_0^s} + \Psi'(D_0^s + D^u) \geq 0, \quad (72)$$

$$\frac{\partial \Lambda(D_0^s, D^u)}{\partial D^u} + \Psi'(D_0^s + D^u) \geq 0 \quad (73)$$

almost everywhere along every such segment, then Assumption 4 holds.

Proof. Consider any $\mathbf{D} = (D_0^s, D^u)$ and $\mathbf{D}' = (D_0^{s'}, D^{u'})$ in $\mathcal{D}$ satisfying $\mathbf{D}' \geq \mathbf{D}$. Because $\mathcal{D}$ is order-convex, the intermediate vector $(D_0^{s'}, D^u)$ and the two coordinate segments joining these vectors belong to $\mathcal{D}$. Absolute continuity gives

$$\begin{aligned} \mathcal{T}(\mathbf{D}') - \mathcal{T}(\mathbf{D}) &= \int_{D_0^s}^{D_0^{s'}} \left[\frac{\partial \Lambda(x, D^u)}{\partial x} + \Psi'(x + D^u) \right] dx \\ &\quad + \int_{D^u}^{D^{u'}} \left[\frac{\partial \Lambda(D_0^{s'}, y)}{\partial y} + \Psi'(D_0^{s'} + y) \right] dy. \end{aligned}$$

Both integrands are nonnegative almost everywhere by (72)–(73). Hence $\mathcal{T}(\mathbf{D}') \geq \mathcal{T}(\mathbf{D})$. $\square$

A simpler uniform sufficient condition for (72)–(73) is the existence of a constant $\kappa \geq 0$ such that

$$\frac{\partial \Lambda(D_0^s, D^u)}{\partial D_0^s} \geq -\kappa, \quad \frac{\partial \Lambda(D_0^s, D^u)}{\partial D^u} \geq -\kappa$$

almost everywhere along every coordinate segment contained in $\mathcal{D}$, and

$$\Psi'(D) \geq \kappa$$

for almost every total face D attained along such segments. Thus, expected liquidation loss itself need not be monotone: Assumption 4 requires only that any marginal reduction in liquidation loss caused by higher face be no larger than the associated increase in private distress cost. These sufficient conditions are imposed only on the policy-relevant order-convex domain $\mathcal{D}$; they are not global restrictions on liquidation sets, because at sufficiently large face liquidation loss can decline even while the private distress cost continues to discipline total loss.

A.3 Proof of Proposition 4

Proof. Fix any feasible committed policy and write

$$D \equiv D_0^s + D_0^u + D_1^u$$

for its total face. Because $\mathcal{F}_1 = \mathcal{F}_0$, the risk-free rate is zero, and both debt issues are competitively priced, the aggregate market value of all debt issued under this policy is

$$I_0 + I_1 + P_1.$$

We first compare priority structures at the same total face D . In every repayment-feasible state, commitment prescribes full repayment, so aggregate creditor payoff equals D under either structure. In forced default, a one-layer creditor chooses between liquidation and extension to maximize aggregate creditor payoff. A pivotal class in a two-layer structure chooses one of the same two actions but need not maximize their aggregate payoff. Therefore, state by state, aggregate creditor payoff under one-layer debt is weakly greater than under the original two-layer policy. Competitive pricing implies

$$\Pi^{\text{one}}(D) \geq I_0 + I_1 + P_1 \geq I_0 + I_1. \quad (74)$$

At the same total face, Proposition 2 and priority-invariance of the distress cost give

$$\mathcal{T}(0, D) \leq \mathcal{T}(D_0^s, D_0^u + D_1^u). \quad (75)$$

By the definition of $D^{\text{com}}(I_1)$ and (27),

$$D^{\text{com}}(I_1) \leq D, \quad \Pi^{\text{one}}(D^{\text{com}}(I_1)) = I_0 + I_1.$$

Assumption 4 and (75) then imply

$$\mathcal{T}(0, D^{\text{com}}(I_1)) \leq \mathcal{T}(0, D) \leq \mathcal{T}(D_0^s, D_0^u + D_1^u). \quad (76)$$

Part (i): no interim funding. Suppose $I_1 = 0$. Issue one one-layer date-0 claim with face $D^{\text{com}}(0)$.

It raises exactly I_0 , requires no date-1 issue, and has $P_1 = 0$. Inequality (76) shows that this replacement weakly lowers total expected loss. Hence, starting from an optimal committed policy, an optimal policy can be chosen with one priority class and no date-1 issuance. With one class, the labels “senior” and “junior” are payoff-irrelevant, so the claim may be represented as either all senior or all junior.

Part (ii): positive interim funding. Suppose $I_1 > 0$. Define the common per-unit value of one-layer debt at the minimal commitment face by

$$p^{\text{one}} \equiv \frac{\Pi^{\text{one}}(D^{\text{com}}(I_1))}{D^{\text{com}}(I_1)} = \frac{I_0 + I_1}{D^{\text{com}}(I_1)} > 0.$$

Because $\mathcal{F}_1 = \mathcal{F}_0$, claims issued at dates 0 and 1 with the same priority and maturity carry this common competitive per-unit price. Construct an all-junior committed policy by issuing

$$D_0^{u,\text{one}} \equiv \frac{I_0}{p^{\text{one}}}, \quad D_1^{u,\text{one}} \equiv \frac{I_1}{p^{\text{one}}}.$$

These faces sum to $D^{\text{com}}(I_1)$, both financing constraints hold with equality, and $P_1 = 0$. Inequality (76) shows that the constructed all-junior policy weakly lowers total expected loss. The restricted-class accounting identity established in Appendix A.2 therefore permits an optimal committed policy to be chosen with $D_0^s = 0$, date-1 proceeds exactly equal to I_1 , and zero payout. $\square$

A.4 Proof of Proposition 5, part (i)

Proof. Fix the date-1 information set $\mathcal{F}_1$ and a candidate junior issue $y \equiv D_1^u \geq 0$. Let $X^u(y)$ denote the terminal payoff to the new junior claim and let $X^E(y)$ denote the terminal payoff to equity. Because $I_1 = 0$, competitive pricing gives

$$P_1 = p_1^u(y)y = \mathbb{E}[X^u(y) \mid \mathcal{F}_1].$$

The entrepreneur’s date-1 expected payoff can therefore be written as

$$U_1(y) = \mathbb{E}[X^u(y) + X^E(y) \mid \mathcal{F}_1] - \Psi(D_0^s + y).$$

Let $\mathcal{L}(y)$ denote the set of date-2 states in which liquidation occurs under issue y and the endogenous repayment/default rule of Lemma 1. The legacy senior class's liquidation-versus-extension comparison depends only on (D_0^s, θ_2) . Accordingly,

$$\mathcal{L}(0) \subseteq \mathcal{L}(y).$$

Indeed, every state liquidated under $y = 0$ is a forced-default state in which the legacy senior class objects, and adding subordinated face does not alter that class's comparison.

Outside $\mathcal{L}(y) \setminus \mathcal{L}(0)$, the combined payoff to the new junior claim and equity is unchanged relative to $y = 0$. If both policies repay, the combined payoff below the legacy senior claim is $V_C(\theta_2) - D_0^s$. If both extend, it is $\mathbb{E}[(Z(\theta_2, \theta_3) - D_0^s)^+ \mid \theta_2]$. The only remaining possibility is that $y = 0$ repays while $y > 0$ defaults and extends. When $G(\theta_2) \geq D_0^s$, we have $Z \geq D_0^s$ almost surely, so extension again gives $V_C(\theta_2) - D_0^s$. When $G(\theta_2) < D_0^s$, the legacy senior comparison is identical under both policies. If default leads to extension at $y > 0$, it also leads to extension at $y = 0$, and Lemma 1 implies that the $y = 0$ policy defaults as well. Hence the case in which $y = 0$ repays while $y > 0$ extends does not arise. Thus the combined residual payoff is unchanged in every state outside the newly liquidated set. Now fix $\theta_2 \in \mathcal{L}(y) \setminus \mathcal{L}(0)$. Such a state must satisfy $V_C(\theta_2) \geq D_0^s$. Otherwise the senior class would be cash-uncovered and pivotal under both policies; because its objection condition is unchanged, its failure to liquidate under $y = 0$ would, by Lemma 2, preclude liquidation after the junior issue as well.

Under $y = 0$, the entrepreneur either repays or strategically defaults and obtains extension. Her conditional payoff is therefore at least

$$(V_C(\theta_2) - D_0^s)^+,$$

because Jensen's inequality gives

$$\mathbb{E}[(Z(\theta_2, \theta_3) - D_0^s)^+ \mid \theta_2] \geq (V_C(\theta_2) - D_0^s)^+.$$

Under issue y , liquidation instead gives the combined junior-and-equity payoff

$$(V_L(\theta_2) - D_0^s)^+.$$

Assumption 1 implies $V_L(\theta_2) < V_C(\theta_2)$, so

$$(V_L(\theta_2) - D_0^s)^+ \leq (V_C(\theta_2) - D_0^s)^+,$$

with strict inequality whenever $V_C(\theta_2) > D_0^s$.

It follows that

$$\mathbb{E}[X^u(y) + X^E(y) \mid \mathcal{F}_1] \leq \mathbb{E}[X^E(0) \mid \mathcal{F}_1].$$

Assumption 3 also gives

$$\Psi(D_0^s + y) \geq \Psi(D_0^s).$$

Hence $U_1(y) \leq U_1(0)$ for every $y \geq 0$, so $D_1^u = P_1 = 0$ is a weak best response. The inequality is strict if newly induced liquidation occurs with positive probability on states satisfying $V_C(\theta_2) > D_0^s$. $\square$

A.5 Proof of Proposition 5, part (ii)

Proof. Any no-commitment equilibrium strategy profile can be replicated under commitment by committing at date 0 to the issuance and repayment/default policies selected sequentially. Hence the commitment value weakly exceeds the value of every no-commitment allocation. This relaxation step does not use Condition 1.

By Proposition 4(i), when $I_1 = 0$ there is a commitment-optimal one-layer allocation with total face $D^{\text{com}}(0)$ and no date-1 issuance. Issue this face as one all-senior date-0 claim. Because Condition 1 holds at $D^{\text{com}}(0)$,

$$\Pi^{\text{strat}}(D^{\text{com}}(0)) = \Pi^{\text{one}}(D^{\text{com}}(0)) = I_0.$$

Thus the issue finances the date-0 outlay.

Part (i) implies that $D_1^u = P_1 = 0$ is a date-1 weak best response. With one outstanding

creditor class, the liquidation rule is the one-layer rule, and the private distress cost is identical across commitment and no commitment because it depends only on total face. The constructed no-commitment allocation therefore attains the commitment upper bound and is globally optimal. $\square$

A.6 Strategic-default transmission, aggregate pricing, regularity, and proof of Proposition 6

Condition 2 (Regularity on the safe-seniority design domain). *(i) A nonempty compact set $\mathcal{X}_0$ exists, with*

$$\mathcal{X}_0 \subseteq \{(D_0^s, D_0^u) \in \mathbb{R}_+^2 : D_0^s + D_0^u \leq D_{\mathcal{I}}^+\}$$

that contains every date-0-feasible inherited pair relevant to the optimization and every pair $(S, \hat{D}_0^u(D; S))$ generated by the safe-seniority construction for $S \in [0, \bar{S}]$ and $D \in \mathcal{I}$. On this domain the date-1 feasible correspondence is nonempty, compact-valued, and continuous, and the reoptimized date-1 objective is continuous on its feasible graph.

(ii) The date-1 best-response graph has a nonempty closed subset satisfying the date-0 financing constraint; equilibrium prices and total expected loss are continuous on this subset.

Remark 1 (Extension to two-layer structures). *Fix $D \in \mathcal{I}$ and a two-layer structure with senior face $S \in [0, D]$.*

Suppose first that inequality (23) holds. If the junior class is pivotal, then

$$S \leq G(\theta_2) < D.$$

Proposition 1 gives the unified-creditor liquidation rule. The junior liquidation payoff is strictly positive because

$$D - S > 0, \quad V_L(\theta_2) - S = G(\theta_2) - S + L(\theta_2, 2) > 0.$$

Hence the junior class objects whenever the unified class objects.

If instead the senior class is pivotal, define

$$\mathcal{C}(q, \theta_2) \equiv \mathbb{E}[\min\{q, Z(\theta_2, \theta_3)\} \mid \theta_2].$$

Condition (23) implies

$$\min\{D, V_L(\theta_2)\} \geq C(D, \theta_2).$$

Because

$$V_L(\theta_2) \geq C(D, \theta_2) \geq C(S, \theta_2), \quad S \geq C(S, \theta_2),$$

we obtain

$$\min\{S, V_L(\theta_2)\} \geq C(S, \theta_2).$$

Moreover,

$$S > G(\theta_2) \geq 0, \quad V_L(\theta_2) > G(\theta_2),$$

so the senior liquidation payoff is strictly positive. The senior class therefore objects.

Thus, under clause (i), default triggers liquidation and Lemma 1 selects refinancing and repayment.

Suppose instead that equality (24) holds. Since $\min\{D, Z\} \leq D$ pointwise,

$$\mathbb{E}[\min\{D, Z\} \mid \theta_2] = D$$

implies

$$Z(\theta_2, \theta_3) \geq D \quad \text{almost surely.}$$

Every priority class is therefore fully repaid under extension. If a default would trigger liquidation, Lemma 1 selects repayment. Otherwise default leads to extension, which preserves the firm's continuation resources, incurs no forced-distress cost, and is payoff-equivalent to repayment for every claimant.

Lemma 10 (Aggregate pricing under safe seniority). *Fix $D \in \mathcal{I}$ and $0 \leq S \leq \underline{G}$. Aggregate no-commitment debt value equals $\Pi^{\text{one}}(D)$. If $D > S$, the junior pool has aggregate value $\Pi^{\text{one}}(D) - S > 0$.*

Proof. Because $S \leq \underline{G}$, the senior claim is fully cash-covered. If $D > S$, Proposition 1 gives the unified-creditor resolution rule; if $D = S$, the structure is already one-layer. Condition 1 and Remark 1 imply that strategic default does not reduce aggregate debt value. Hence aggregate debt value equals $\Pi^{\text{one}}(D)$.

If $D > S$, the junior pool has positive face. Under full repayment it receives $D - S > 0$. Under liquidation its residual pool is

$$V_L(\theta_2) - S = G(\theta_2) - S + L(\theta_2, 2) > 0,$$

and under extension its residual pool is

$$Z(\theta_2, \theta_3) - S = G(\theta_2) - S + G(\theta_3) + L(\theta_3, 3) > 0$$

almost surely. Its aggregate market value is therefore strictly positive. $\square$

Lemma 11 (Existence of an optimal no-commitment equilibrium). *Under Condition 2 and Condition 1 on $\mathcal{I}$, the date-0 no-commitment problem attains an optimum.*

Proof. Condition 2(i) and Berge's maximum theorem imply that the date-1 best-response correspondence is nonempty, compact-valued, and upper hemicontinuous on the compact inherited-face domain $\mathcal{X}_0$. Its graph is closed and contained in the compact feasible graph and is therefore compact.

Condition 2(ii) gives a nonempty closed date-0-feasible subset of that graph. Total expected loss is continuous on this subset and therefore attains its minimum. Proposition 3 implies that the minimizer is an optimal no-commitment equilibrium. $\square$

Proof of Proposition 6. Suppose $I_1 > 0$. By Lemma 11, fix an optimal no-commitment equilibrium.

Under clause (i) of Condition 1, Lemma 1 selects repayment in every repayment-feasible state. Under clause (ii), either the firm repays or default leads to an extension that delivers the full promised face almost surely, preserves the firm's continuation resources, causes no liquidation, and incurs no forced-distress cost. Thus, on $\mathcal{I}$, strategic default neither reduces aggregate debt value nor changes real loss relative to repayment.

Let

$$\bar{S} \equiv \min\{I_0, \underline{G}\}.$$

If $D_0^s \geq \bar{S}$, there is nothing to prove. Otherwise, $D_0^s < \bar{S}$, the senior debt is safe, and Lemma 10

implies

$$\Pi^{\text{one}}(D_1) = I_0 + I_1 + P_1. \quad (77)$$

Limited liability gives $\Pi^{\text{one}}(D_1) \leq D_1$. Hence, for every $S \in [D_0^s, \bar{S}]$,

$$D_1 \geq I_0 + I_1 + P_1 > I_0 \geq S,$$

so $D_1 > S$. The same argument applies to every feasible equilibrium total face generated by the construction below.

Fix $S \in [D_0^s, \bar{S}]$. For $D > S$, define

$$v(D; S) \equiv \frac{\Pi^{\text{one}}(D) - S}{D - S}.$$

If total face $D > S$ is anticipated at date 0, competitive pricing requires legacy junior face

$$\hat{D}_0^u(D; S) \equiv \frac{I_0 - S}{v(D; S)}. \quad (78)$$

At $D = D_1$, the corresponding new junior face is $(I_1 + P_1)/v(D_1; S)$. Equation (77) implies that these two junior faces sum with S to D_1 , while their market values are $I_0 - S$ and $I_1 + P_1$. Thus, the structure finances both investments at the original total face. Proposition 1 shows that it preserves the one-layer liquidation rule, and (28) shows that it preserves total real loss. At $S = D_0^s$, the construction reproduces the original equilibrium faces.

For each candidate total face D , the entrepreneur reoptimizes $P_1 \geq 0$ subject to the date-1 financing constraint. The comparisons below are therefore between maximized date-1 payoffs at each D , not between policies holding payout fixed.

Suppose that, after legacy debt is priced at anticipated total face D_1 , the entrepreneur chooses total face D'_1 . Legacy debt then has market value

$$S + \hat{D}_0^u(D_1; S)v(D'_1; S),$$

so the incremental dilution benefit relative to D_1 is

$$B^{\text{dev}}(D'_1; S, D_1) = (I_0 - S) \left[1 - \frac{v(D'_1; S)}{v(D_1; S)} \right]. \quad (79)$$

The resulting payoff gain and market value of the new issue are, respectively,

$$B^{\text{dev}}(D'_1; S, D_1) - [\mathcal{T}(0, D'_1) - \mathcal{T}(0, D_1)], \quad (80)$$

and

$$\Pi^{\text{one}}(D'_1) - I_0 + B^{\text{dev}}(D'_1; S, D_1). \quad (81)$$

For $D'_1 \geq D_1$, let

$$\rho(S) \equiv \frac{v(D'_1; S)}{v(D_1; S)} = \frac{(\Pi^{\text{one}}(D'_1) - S)(D_1 - S)}{(D'_1 - S)(\Pi^{\text{one}}(D_1) - S)} \leq 1.$$

Using $S \leq I_0$, $I_0 \leq \Pi^{\text{one}}(D_1)$, and $\Pi^{\text{one}}(D'_1) \leq D'_1$, direct differentiation gives

$$\begin{aligned} \frac{\partial B^{\text{dev}}(D'_1; S, D_1)}{\partial S} &= -(1 - \rho(S)) - (I_0 - S)\rho'(S) \\ &\leq \begin{cases} -(1 - \rho(S)), & \rho'(S) \geq 0, \\ -\frac{(D'_1 - \Pi^{\text{one}}(D'_1))(D'_1 - D_1)}{(D'_1 - S)^2}, & \rho'(S) < 0, \end{cases} \\ &\leq 0. \end{aligned} \quad (82)$$

Thus, after safe seniority is increased and legacy junior debt is repriced at D_1 , every upward deviation yields a weakly smaller dilution benefit. Since total real loss at each total face is invariant to safe seniority, equations (80) and (81) imply that its payoff gain and new-debt value are also weakly smaller. Hence no upward deviation can become newly feasible and profitable. Because D_1 remains feasible, a date-1 best response can be selected weakly below D_1 .

It remains to impose date-0 pricing at the selected best response. Let $\text{BR}_S(\cdot)$ denote the date-1 best-response correspondence in total face, restricted to $D \in \mathcal{I}$, given safe senior face S and the

legacy junior face. Starting with $D^{(0)} = D_1$, recursively choose

$$\widehat{D}_0^{u,(n)} = \widehat{D}_0^u(D^{(n)}; S), \quad D^{(n+1)} \in \text{BR}_S(\widehat{D}_0^{u,(n)}), \quad D^{(n+1)} \leq D^{(n)}. \quad (83)$$

The first such selection follows from the preceding argument. Suppose $D^{(n)} \in \text{BR}_S(\widehat{D}_0^{u,(n-1)})$. Because $v(D; S)$ is weakly decreasing in D , $D^{(n)} \leq D^{(n-1)}$ implies $\widehat{D}_0^{u,(n)} \leq \widehat{D}_0^{u,(n-1)}$. For any $D > D^{(n)}$, the payoff gain relative to $D^{(n)}$ under a generic legacy junior face D_0^u is

$$D_0^u[v(D^{(n)}; S) - v(D; S)] - [\mathcal{T}(0, D) - \mathcal{T}(0, D^{(n)})],$$

which is weakly increasing in D_0^u ; hence reducing the legacy junior face weakly lowers the gain from every upward deviation. Moreover, the market value of the new issue,

$$\Pi^{\text{one}}(D) - S - D_0^u v(D; S),$$

is nondecreasing in D : the hypotheses of Proposition 6 imply that Π^{one} is weakly increasing on $\mathcal{J}$, while concavity of Π^{one} implies that $v(D; S)$ is weakly decreasing in D . Hence every $D > D^{(n)}$ was already feasible under $\widehat{D}_0^{u,(n-1)}$. Reducing legacy junior face therefore cannot make any such choice strictly preferable to $D^{(n)}$, establishing the next selection in (83).

The sequence $\{D^{(n)}\}$ is nonincreasing and bounded below, so it converges to some $\tilde{D}_1 \leq D_1$. By continuity, $\widehat{D}_0^{u,(n)} \rightarrow \widehat{D}_0^u(\tilde{D}_1; S)$. Condition 2(i) and Berge's maximum theorem imply that the date-1 best-response correspondence is nonempty, compact-valued, and upper hemicontinuous on $\mathcal{X}_0$. It therefore has a closed graph, and

$$\tilde{D}_1 \in \text{BR}_S(\widehat{D}_0^u(\tilde{D}_1; S)).$$

Thus competitive date-0 pricing and date-1 optimality hold simultaneously, yielding an equilibrium with total face $\tilde{D}_1 \leq D_1$ and safe senior face S .

Take $S = \bar{S}$. Equation (28) and Assumption 4 imply

$$\mathcal{T}(\bar{S}, \tilde{D}_1 - \bar{S}) = \mathcal{T}(0, \tilde{D}_1) \leq \mathcal{T}(0, D_1) = \mathcal{T}(D_0^s, D_1 - D_0^s).$$

By Proposition 3, the entrepreneur's date-0 expected payoff cannot fall. Because the original structure was optimal, the constructed structure is also optimal and has safe senior face $\bar{S}$. If $I_0 \leq \underline{G}$, then $\bar{S} = I_0$ and (78) gives

$$\hat{D}_0^u(\tilde{D}_1; \bar{S}) = 0.$$

□

A.7 Proof of Corollary 1

Proof. By the domination property defining $\mathcal{I}$, restrict attention to conditional equilibria with total face $D \in \mathcal{I}$.

Fix

$$0 \leq S \leq \min\{\underline{G}, I_0\}.$$

By Lemma 10,

$$\Pi^{\text{one}}(D) = I_0 + I_1 + P_1 \geq I_0 + I_1.$$

Hence

$$D \geq D^{\text{com}} \equiv D^{\text{com}}(I_1).$$

Equation (27) gives

$$\Pi^{\text{one}}(D^{\text{com}}) = I_0 + I_1.$$

Limited liability implies

$$D^{\text{com}} \geq I_0 + I_1 > S.$$

Because legacy and new junior claims are *pari passu* and $\mathcal{F}_1 = \mathcal{F}_0$, their common unit price at the commitment face is

$$v(D^{\text{com}}; S) = \frac{\Pi^{\text{one}}(D^{\text{com}}) - S}{D^{\text{com}} - S} > 0.$$

Set

$$D_0^u = \frac{I_0 - S}{v(D^{\text{com}}; S)}, \quad D_1^u = \frac{I_1}{v(D^{\text{com}}; S)}.$$

Then

$$D_0^u + D_1^u = D^{\text{com}} - S,$$

and

$$S + v(D^{\text{com}}; S)D_0^u = I_0, \quad v(D^{\text{com}}; S)D_1^u = I_1.$$

Thus both financing constraints bind and $P_1 = 0$. By hypothesis this is a date-1 best response.

Any other feasible conditional equilibrium under the same S has $D \geq D^{\text{com}}$, so

$$D - S \geq D^{\text{com}} - S.$$

Assumption 4 therefore gives

$$\mathcal{T}(S, D - S) \geq \mathcal{T}(S, D^{\text{com}} - S).$$

Proposition 3 implies conditional date-0 optimality. Since $S \leq \underline{G}$, the liquidation rule and real loss coincide with the one-layer commitment benchmark at the same total face. The constructed equilibrium therefore matches that benchmark in total face, payout, creditor value, total loss, and entrepreneur payoff. Under clause (ii) of Condition 1, repayment/default timing may differ while remaining payoff-equivalent. $\square$

A.8 Derivation of the one-layer liquidation cutoff

When $0 < \theta_L(D) < \bar{\theta}(D) < 1$,

$$\Lambda_D(0) = \int_{\theta_L(D)}^{\bar{\theta}(D)} \Delta(\theta_2) d\theta_2 = \delta_0 [\bar{\theta}(D) - \theta_L(D)] + \frac{\delta_1}{2} [\bar{\theta}(D)^2 - \theta_L(D)^2]. \quad (84)$$

Proof. Fix $\theta_2 \in [0, 1]$ in the forced-default region, i.e.,

$$V_C(\theta_2) < D. \quad (85)$$

We compare the liquidation and continuation payoffs for the single creditor class, with ties resolved in favor of liquidation.

Under liquidation at $t = 2$, total resources equal $V_L(\theta_2)$, and the creditor receives $\min\{D, V_L(\theta_2)\}$.

By Assumption 1, inefficient liquidation implies $L(\theta_2, 2) < V_3(\theta_2)$, and hence

$$V_L(\theta_2) = G(\theta_2) + L(\theta_2, 2) < G(\theta_2) + V_3(\theta_2) = V_C(\theta_2).$$

Combining with (85) yields $V_L(\theta_2) < D$, so the liquidation payoff equals $V_L(\theta_2)$.

Under continuation to $t = 3$, the creditor receives $\min\{D, Z(\theta_2, \theta_3)\}$. Conditional on θ_2 ,

$$Z(\theta_2, \theta_3) \in \{V_C(\theta_2) + cx, V_C(\theta_2) - cx\} \quad \text{with probabilities } \frac{1}{2}, \frac{1}{2}.$$

Since $D \leq V_C(0) + cx \leq V_C(\theta_2) + cx$, we have $\min\{D, V_C(\theta_2) + cx\} = D$. Moreover, by (85), $V_C(\theta_2) - cx < D$, so $\min\{D, V_C(\theta_2) - cx\} = V_C(\theta_2) - cx$. Therefore the conditional expected continuation payoff equals

$$\mathbb{E}[\min\{D, Z(\theta_2, \theta_3)\} \mid \theta_2] = \frac{1}{2}D + \frac{1}{2}(V_C(\theta_2) - cx).$$

The creditor liquidates if and only if

$$V_L(\theta_2) \geq \frac{1}{2}D + \frac{1}{2}(V_C(\theta_2) - cx).$$

Multiplying by 2 and substituting the affine forms $V_L(\theta_2) = a_L + b_L\theta_2$ and $V_C(\theta_2) = a_C + b_C\theta_2$ yields

$$(2b_L - b_C)\theta_2 \geq D + a_C - cx - 2a_L. \tag{86}$$

Using $b_L = g + \lambda_2$, $b_C = 2g + \lambda_3$, and defining $d \equiv \lambda_2 - \frac{1}{2}\lambda_3 > 0$, this inequality becomes

$$2d\theta_2 \geq D + \ell_3 - 2\ell_2 - cx.$$

Dividing by $2d$ gives the cutoff

$$\theta_2 \geq \theta_L(D) \equiv \frac{D + \ell_3 - 2\ell_2 - cx}{2d},$$

which proves the claim. Differentiating shows $\partial\theta_L(D)/\partial D = 1/(2d) > 0$. □

A.9 Derivation of the two-layer liquidation cutoffs

For each θ_2 , define the (statewise) liquidation-advantage functions corresponding exactly to the model's liquidation conditions (17) (junior control) and (18) (senior control):

$$\Phi^u(\theta_2; D_1) \equiv V_L(\theta_2) - \mathbb{E}[\min\{D_1, Z(\theta_2, \theta_3)\} \mid \theta_2], \quad (87)$$

$$\Phi^s(\theta_2; D_0^s) \equiv \min\{D_0^s, V_L(\theta_2)\} - \mathbb{E}[\min\{D_0^s, Z(\theta_2, \theta_3)\} \mid \theta_2]. \quad (88)$$

The next lemma gives fully explicit formulas for the liquidation-advantage functions in (87)–(88). We maintain the capped-upside regime (46), applied below to the relevant promised face values.

Lemma 12. *Maintain Assumption 5 and fix promised faces (D_0^s, D_1) with $0 \leq D_0^s \leq D_1$. Maintain the capped-upside condition (46) applied to the total face value D_1 , i.e.*

$$D_1 \leq a_C + cx \quad (\text{equivalently, } D_1 \leq V_C(0) + cx). \quad (89)$$

Then, for every forced-default state θ_2 with $V_C(\theta_2) < D_1$,

$$\Phi^u(\theta_2; D_1) = V_L(\theta_2) - \frac{1}{2}D_1 - \frac{1}{2}(V_C(\theta_2) - cx). \quad (90)$$

Moreover, for every $\theta_2 \in [0, 1]$,

$$\mathbb{E}[\min\{D_0^s, Z(\theta_2, \theta_3)\} \mid \theta_2] = \begin{cases} D_0^s, & D_0^s \leq V_C(\theta_2) - cx, \\ \frac{1}{2}D_0^s + \frac{1}{2}(V_C(\theta_2) - cx), & V_C(\theta_2) - cx < D_0^s \leq V_C(\theta_2) + cx, \\ V_C(\theta_2), & D_0^s > V_C(\theta_2) + cx, \end{cases} \quad (91)$$

and hence

$$\Phi^s(\theta_2; D_0^s) = \min\{D_0^s, V_L(\theta_2)\} - \begin{cases} D_0^s, & D_0^s \leq V_C(\theta_2) - cx, \\ \frac{1}{2}D_0^s + \frac{1}{2}(V_C(\theta_2) - cx), & V_C(\theta_2) - cx < D_0^s \leq V_C(\theta_2) + cx, \\ V_C(\theta_2), & D_0^s > V_C(\theta_2) + cx, \end{cases} \quad (92)$$

Under (89) and $D_0^s \leq D_1$, the third line in (91)–(92) is never reached.

Proof. From (44) and Assumption 5, conditional on θ_2 ,

$$Z(\theta_2, \theta_3) \in \{V_C(\theta_2) + cx, V_C(\theta_2) - cx\} \quad \text{with probabilities } \frac{1}{2}, \frac{1}{2}.$$

Therefore,

$$\mathbb{E}[\min\{D_1, Z(\theta_2, \theta_3)\} \mid \theta_2] = \frac{1}{2} \min\{D_1, V_C(\theta_2) + cx\} + \frac{1}{2} \min\{D_1, V_C(\theta_2) - cx\}.$$

Under (89) and monotonicity of $V_C(\theta_2)$, $V_C(\theta_2) + cx \geq V_C(0) + cx \geq D_1$ for all $\theta_2 \in [0, 1]$, so the first minimum equals D_1 . If θ_2 is forced default, then $V_C(\theta_2) < D_1$, hence $V_C(\theta_2) - cx < D_1$ and the second minimum equals $V_C(\theta_2) - cx$. This yields (90).

Similarly,

$$\mathbb{E}[\min\{D_0^s, Z(\theta_2, \theta_3)\} \mid \theta_2] = \frac{1}{2} \min\{D_0^s, V_C(\theta_2) + cx\} + \frac{1}{2} \min\{D_0^s, V_C(\theta_2) - cx\},$$

and the cases in (91) follow directly. Substituting into (88) gives (92).

Finally, under (89) and $D_0^s \leq D_1$ we have $D_0^s \leq V_C(0) + cx \leq V_C(\theta_2) + cx$ for all θ_2 , so Case 3 cannot occur. $\square$

Proof of equations (50)–(52). By Lemma 1, liquidation can occur only in forced-default states $\theta_2 < \bar{\theta}(D_1)$, so we restrict attention to those states.

(i) *Senior control.* When $\theta_2 < \theta_S(D_0^s)$, the liquidation decision is governed by the senior-control condition (18), i.e. $\Phi^s(\theta_2; D_0^s) \geq 0$ with Φ^s defined in (88). Under the global-cap restriction and $D_0^s \leq D_1$, Lemma 12 implies the continuation payoff is given by the first two cases of (91).

If $V_L(\theta_2) \geq D_0^s$ (equivalently $\theta_2 \geq \theta_F^s(D_0^s)$), then $\min\{D_0^s, V_L(\theta_2)\} = D_0^s$ and the continuation payoff is at most D_0^s , so liquidation is (weakly) optimal (ties broken toward liquidation).

If instead $V_L(\theta_2) < D_0^s$, then liquidation is possible only under Case 2 of (91), in which the continuation payoff equals $\frac{1}{2}D_0^s + \frac{1}{2}(V_C(\theta_2) - cx)$. Then $\Phi^s(\theta_2; D_0^s) \geq 0$ is equivalent to

$$V_L(\theta_2) \geq \frac{1}{2}D_0^s + \frac{1}{2}(V_C(\theta_2) - cx),$$

which yields the cutoff $\theta_2 \geq \theta_L^s(D_0^s)$ exactly as in equation (49) (with D replaced by D_0^s).

Combining the two subcases gives liquidation iff $\theta_2 \geq \min\{\theta_F^s(D_0^s), \theta_L^s(D_0^s)\} = \underline{\theta}^s(D_0^s)$.

(ii) *Junior control.* When $\theta_2 \geq \theta_S(D_0^s)$, liquidation is governed by the junior-control condition (17), i.e. $\Phi^u(\theta_2; D_1) \geq 0$ with Φ^u in (87). Lemma 12 yields (90), and solving $\Phi^u(\theta_2; D_1) \geq 0$ produces the cutoff $\theta_2 \geq \theta_L^u(D_1)$, which coincides with the one-layer cutoff in equation (49) (with $D = D_1$). $\square$

A.10 One-layer continuation-risk derivative

For compactness in Sections A.10–A.11, write

$$\Lambda^{\text{one}}(D) \equiv \Lambda_D(0), \quad \Lambda^{\text{two}}(D_0^s, D^u) \equiv \Lambda(D_0^s, D^u).$$

Proof of Lemma 4. Under the maintained interior condition $0 < \theta_L(D) < \bar{\theta}(D) < 1$, no support clipping occurs and

$$\Lambda^{\text{one}}(D) = \int_{\theta_L(D)}^{\bar{\theta}(D)} \Delta(\theta_2) d\theta_2.$$

Under Assumption 5, $\bar{\theta}(D)$ and $\Delta(\cdot)$ do not depend on x , while $\theta_L(D) = (D + \ell_3 - 2\ell_2 - cx)/(2d)$ satisfies $\partial\theta_L(D)/\partial x = -c/(2d)$. By the Leibniz rule,

$$\frac{\partial\Lambda^{\text{one}}(D)}{\partial x} = -\Delta(\theta_L(D)) \frac{\partial\theta_L(D)}{\partial x} = \Delta(\theta_L(D)) \cdot \frac{c}{2d}.$$

Assumption 1 implies $\Delta(\theta_2) > 0$, and $c > 0$, $d > 0$, hence the derivative is strictly positive. $\square$

A.11 Two-layer continuation-risk derivatives and numerical implementation

A.11A Closed-form pricing and all-junior issuance under Assumption 5

This subsection records the closed-form objects used in the numerical solution of Sections 3.3.2–3.3.3. The formulas below are derived on the interior capped branch

$$D \leq D_{\text{cap}} \equiv a_C + cx, \quad 0 < \theta_L(D) < \bar{\theta}(D) < 1.$$

For constrained optimization, we also evaluate boundary candidates using the continuous extension of these branch formulas. Define

$$q_D \equiv \frac{1}{2d}, \quad \beta \equiv \frac{a_C - 2a_L - cx}{2d}, \quad \theta_L(D) = q_D D + \beta.$$

One-layer debt value. For one-layer debt of face value D , the expected creditor payoff is

$$\begin{aligned} \Pi^{\text{one}}(D) = & D(1 - \bar{\theta}(D)) + \int_0^{\theta_L(D)} \frac{1}{2}(D + a_C - cx + b_C\theta) d\theta \\ & + \int_{\theta_L(D)}^{\bar{\theta}(D)} (a_L + b_L\theta) d\theta. \end{aligned} \quad (93)$$

The first term is repayment in nondefault states, the second term is extension in forced-default states in which creditors continue, and the third term is liquidation in forced-default states in which creditors liquidate. Evaluating the integrals gives the quadratic

$$\Pi^{\text{one}}(D) = A_2 D^2 + A_1 D + A_0, \quad (94)$$

where

$$A_2 = \frac{q_D^2 b_C}{4} - \frac{q_D^2 b_L}{2} + \frac{q_D}{2} - \frac{1}{b_C} + \frac{b_L}{2b_C^2}, \quad (95)$$

$$A_1 = \frac{a_C q_D}{2} + \frac{a_C}{b_C} - \frac{a_C b_L}{b_C^2} - a_L q_D + \frac{a_L}{b_C} + \frac{q_D b_C \beta}{2} - q_D b_L \beta - \frac{q_D c x}{2} + \frac{\beta}{2} + 1, \quad (96)$$

$$A_0 = \frac{a_C^2 b_L}{2b_C^2} - \frac{a_C a_L}{b_C} + \frac{a_C \beta}{2} - a_L \beta + \frac{b_C \beta^2}{4} - \frac{b_L \beta^2}{2} - \frac{\beta c x}{2}. \quad (97)$$

Similarly, equation (84) can be written as

$$\Lambda^{\text{one}}(D) = B_2 D^2 + B_1 D + B_0, \quad (98)$$

where

$$B_2 = \frac{\delta_1}{2b_C^2} - \frac{q_D^2 \delta_1}{2}, \quad (99)$$

$$B_1 = -\frac{a_C \delta_1}{b_C^2} - q_D \beta \delta_1 - q_D \delta_0 + \frac{\delta_0}{b_C}, \quad (100)$$

$$B_0 = \frac{a_C^2 \delta_1}{2b_C^2} - \frac{a_C \delta_0}{b_C} - \frac{\beta^2 \delta_1}{2} - \beta \delta_0. \quad (101)$$

Commitment benchmark. Under commitment, the firm uses one-layer debt and sets $P_1 = 0$.

Therefore total promised face D^{com} is the smallest admissible solution to

$$\Pi^{\text{one}}(D) = I_0 + I_1. \quad (102)$$

On the quadratic branch, the candidate roots are the real elements of

$$D^{\text{com}} \in \left\{ \frac{-A_1 \pm \sqrt{A_1^2 - 4A_2(A_0 - I_0 - I_1)}}{2A_2} \right\} \cap \{D \in [0, D_{\text{cap}}] : 0 \leq \theta_L(D) \leq \bar{\theta}(D) \leq 1\}, \quad (103)$$

with the economically relevant root being the smallest admissible one. If $A_2 = 0$, the same condition is solved using the corresponding linear equation. The committed date-0 and date-1 junior faces are

$$D_0^{u,\text{com}} = \frac{I_0}{I_0 + I_1} D^{\text{com}}, \quad D_1^{u,\text{com}} = \frac{I_1}{I_0 + I_1} D^{\text{com}}, \quad P_1^{\text{com}} = 0. \quad (104)$$

All-junior no-commitment benchmark under the distress-cost completion. If the firm issues only junior debt at $t = 0$, total face due at $t = 2$ is

$$D = D_0^u + D_1^u.$$

Because legacy and new junior debt are pari passu, legacy junior receives the fraction D_0^u/D of the one-layer creditor payoff. Recall from Section 3.2 that $\Pi^{\text{strat}}(D)$ denotes one-layer creditor value

under the endogenous repayment/default rule. The date-1 problem is

$$\min_{D \geq D_0^u} \left\{ \frac{D_0^u}{D} \Pi^{\text{strat}}(D) + \Lambda^{\text{one}}(D) + \Psi(D) \right\} \quad \text{s.t.} \quad \left(1 - \frac{D_0^u}{D} \right) \Pi^{\text{strat}}(D) \geq I_1. \quad (105)$$

Coercivity in Assumption 3 makes the objective in (105) diverge as $D \rightarrow \infty$, while the remaining terms are bounded below. Hence the coercive private distress cost eliminates the large-face escape and a finite minimizer exists whenever the feasible set is nonempty.

At an interior regular optimum with a slack financing constraint,

$$\frac{D_0^u}{D^2} \left[D(\Pi^{\text{strat}})'(D) - \Pi^{\text{strat}}(D) \right] + (\Lambda^{\text{one}})'(D) + \Psi'(D) = 0. \quad (106)$$

Date-0 competitive pricing gives

$$I_0 = \frac{D_0^u}{D} \Pi^{\text{strat}}(D), \quad D_0^u = \frac{I_0 D}{\Pi^{\text{strat}}(D)}. \quad (107)$$

Substituting (107) into (106) yields the one-dimensional equilibrium condition

$$\frac{I_0}{D} \left(\frac{D(\Pi^{\text{strat}})'(D)}{\Pi^{\text{strat}}(D)} - 1 \right) + (\Lambda^{\text{one}})'(D) + \Psi'(D) = 0. \quad (108)$$

Under Condition 1, $\Pi^{\text{strat}}(D) = \Pi^{\text{one}}(D)$ on the relevant branch. Under the quadratic specification in equation (62) and $D \in [a_C, a_C + b_C]$,

$$\Psi'(D) = \frac{\chi(D - a_C)^2}{2b_C}.$$

Write

$$\omega_\chi \equiv \frac{\chi}{2b_C}, \quad M_2 \equiv \omega_\chi, \quad M_1 \equiv 2B_2 - 2\omega_\chi a_C, \quad M_0 \equiv B_1 + \omega_\chi a_C^2,$$

so that

$$(\Lambda^{\text{one}})'(D) + \Psi'(D) = M_2 D^2 + M_1 D + M_0.$$

Using $\Pi^{\text{one}}(D) = A_2 D^2 + A_1 D + A_0$, equation (108) is equivalent to the quintic

$$Q_5 D^5 + Q_4 D^4 + Q_3 D^3 + Q_2 D^2 + Q_1 D + Q_0 = 0, \quad (109)$$

where

$$\begin{aligned}
Q_5 &= A_2 M_2, \\
Q_4 &= A_2 M_1 + A_1 M_2, \\
Q_3 &= A_2 M_0 + A_1 M_1 + A_0 M_2, \\
Q_2 &= A_1 M_0 + A_0 M_1 + I_0 A_2, \\
Q_1 &= A_0 M_0, \\
Q_0 &= -I_0 A_0.
\end{aligned}$$

The polynomial characterizes only interior regular candidates on the maintained capped-upside branch. The constrained solution also evaluates the zero-payout financing boundary, all structural and branch boundaries, and candidates outside the closed-form branch using the primitive payoff evaluator. The reported equilibrium is the lowest-objective feasible policy located by the search described in Appendix A; the closed-form branch bound is not imposed as an economic constraint.

Given the selected total face D^{AJ} , the equilibrium faces and payout are

$$D_0^{u,AJ} = \frac{I_0 D^{AJ}}{\Pi^{\text{strat}}(D^{AJ})}, \quad D_1^{u,AJ} = D^{AJ} - D_0^{u,AJ}, \quad P_1^{AJ} = \Pi^{\text{strat}}(D^{AJ}) - I_0 - I_1.$$

A.11B Branchwise monotonicity conditions under Assumption 5

This subsection gives branchwise derivative conditions for the capped-upside tractable example. The conditions are necessary and sufficient for *local* monotonicity on open, order-convex subregions of the fixed-ordering, support-interior branches stated below. They should be read together with the support and ordering restrictions defining each branch. At branch boundaries and kinks, the corresponding one-sided derivatives must be checked separately.

One-layer branch. On the interior one-layer branch $0 < \theta_L(D) < \bar{\theta}(D) < 1$,

$$\frac{\partial \Lambda^{\text{one}}(D)}{\partial D} = \frac{\Delta(\bar{\theta}(D))}{b_C} - \frac{\Delta(\theta_L(D))}{2d}.$$

Therefore, Λ^{one} is weakly increasing on an interval $\mathcal{I}$ contained in this branch if and only if

$$\frac{\Delta(\bar{\theta}(D))}{b_C} \geq \frac{\Delta(\theta_L(D))}{2d} \quad \text{for all } D \in \mathcal{I}. \quad (110)$$

Two-layer early-switch branch. Let $D_1 \equiv D_0^s + D^u$. Define the senior-control lower liquidation boundary by

$$\underline{\theta}^s(D_0^s) \equiv \min\{\theta_F^s(D_0^s), \theta_L^s(D_0^s)\}.$$

Let

$$\eta_s(D_0^s) \equiv \frac{\partial \underline{\theta}^s(D_0^s)}{\partial D_0^s} = \begin{cases} 1/b_L, & \text{if } \theta_F^s(D_0^s) < \theta_L^s(D_0^s), \\ 1/(2d), & \text{if } \theta_L^s(D_0^s) < \theta_F^s(D_0^s). \end{cases}$$

At the kink $\theta_F^s(D_0^s) = \theta_L^s(D_0^s)$, the relevant one-sided derivatives must be checked separately.

Consider an active early-switch branch satisfying

$$0 < \underline{\theta}^s(D_0^s) < \theta_S(D_0^s) < \theta_L^u(D_1) < \bar{\theta}(D_1) < 1.$$

On this branch,

$$\Lambda^{\text{two}}(D_0^s, D^u) = \int_{\underline{\theta}^s(D_0^s)}^{\theta_S(D_0^s)} \Delta(\theta) d\theta + \int_{\theta_L^u(D_1)}^{\bar{\theta}(D_1)} \Delta(\theta) d\theta.$$

Hence

$$\frac{\partial \Lambda^{\text{two}}}{\partial D^u} = \frac{\Delta(\bar{\theta}(D_1))}{b_C} - \frac{\Delta(\theta_L^u(D_1))}{2d}, \quad (111)$$

$$\frac{\partial \Lambda^{\text{two}}}{\partial D_0^s} = \frac{\Delta(\theta_S(D_0^s))}{g} - \eta_s(D_0^s) \Delta(\underline{\theta}^s(D_0^s)) + \frac{\Delta(\bar{\theta}(D_1))}{b_C} - \frac{\Delta(\theta_L^u(D_1))}{2d}. \quad (112)$$

Therefore, on any open order-convex subregion of this early-switch branch that does not cross the kink $\theta_F^s(D_0^s) = \theta_L^s(D_0^s)$, Λ^{two} is weakly increasing in both arguments if and only if the right-hand sides of (111) and (112) are both weakly nonnegative throughout that subregion.

Two-layer late-switch branch. Now consider an active late-switch branch satisfying

$$0 < \underline{\theta}^s(D_0^s) < \theta_L^u(D_1) < \theta_S(D_0^s) < \bar{\theta}(D_1) < 1.$$

On this branch, the senior-control liquidation interval and the junior-control liquidation interval connect, so the total liquidation set is

$$[\underline{\theta}^s(D_0^s), \theta_S(D_0^s)) \cup [\theta_S(D_0^s), \bar{\theta}(D_1)) = [\underline{\theta}^s(D_0^s), \bar{\theta}(D_1))$$

up to the endpoint $\theta_S(D_0^s)$, which is irrelevant for the integral. Therefore,

$$\Lambda^{\text{two}}(D_0^s, D^u) = \int_{\underline{\theta}^s(D_0^s)}^{\bar{\theta}(D_1)} \Delta(\theta) d\theta.$$

Hence

$$\frac{\partial \Lambda^{\text{two}}}{\partial D^u} = \frac{\Delta(\bar{\theta}(D_1))}{b_C}, \quad (113)$$

$$\frac{\partial \Lambda^{\text{two}}}{\partial D_0^s} = \frac{\Delta(\bar{\theta}(D_1))}{b_C} - \eta_s(D_0^s) \Delta(\underline{\theta}^s(D_0^s)). \quad (114)$$

Because $\Delta(\cdot) > 0$, the derivative with respect to D^u is strictly positive on this branch. Therefore, on any open order-convex subregion of this late-switch branch that does not cross the kink $\theta_F^s(D_0^s) = \theta_L^s(D_0^s)$, Λ^{two} is weakly increasing in both arguments if and only if the right-hand side of (114) is weakly nonnegative throughout that subregion.

The conditions in this subsection are branchwise conditions. Globally, one must check every active ordering branch reached by the equilibrium policy and verify weakly nonnegative one-sided derivatives at kinks where the ordering of $\underline{\theta}^s(D_0^s)$, $\theta_S(D_0^s)$, $\theta_L^u(D_1)$, and $\bar{\theta}(D_1)$ changes, and also at the kink where $\theta_F^s(D_0^s) = \theta_L^s(D_0^s)$. Outside the capped-upside branch, the continuation payoff changes form and these formulas must be replaced by the general sufficient conditions in Appendix A.2A.

Early-switch fragmentation derivative

Proof. Fix (D_0^s, D_1) such that $0 < \theta_L^s(D_0^s) < \theta_S(D_0^s) < \bar{\theta}(D_1)$, $\theta_S(D_0^s) \leq \theta_L^u(D_1)$, and $\theta_L^s(D_0^s) < \theta_F^s(D_0^s)$. By equations (50)–(52), the fragmentation wedge can be written as

$$\text{Frag}(D_0^s, D_1) = \Lambda^{\text{two}}(D_0^s, D_1) - \Lambda^{\text{one}}(D_1) = \int_{\theta_L^s(D_0^s)}^{\theta_S(D_0^s)} \Delta(\theta_2) d\theta_2,$$

because the interval $[\theta_L^u, \bar{\theta}(D_1))$ cancels when subtracting the one-layer deadweight loss at the same total face D_1 . Define $H_\Delta(t) \equiv \int_0^t \Delta(s) ds$, so that $H'_\Delta(t) = \Delta(t)$. This gives $\text{Frag} = H_\Delta(\theta_S) - H_\Delta(\theta_L^s)$.

Differentiate with respect to x . Since θ_S does not depend on x ,

$$\frac{\partial \text{Frag}}{\partial x} = -\frac{\partial}{\partial x} H_\Delta(\theta_L^s) = -H'_\Delta(\theta_L^s) \cdot \frac{\partial \theta_L^s}{\partial x} = -\Delta(\theta_L^s) \cdot \frac{\partial \theta_L^s}{\partial x},$$

using $H'_\Delta(t) = \Delta(t)$.

Recall $\theta_L^s(D_0^s) = \frac{D_0^s + \ell_3 - 2\ell_2 - cx}{2d}$ with $c \equiv g + \lambda_3$. Hence

$$\frac{\partial \theta_L^s(D_0^s)}{\partial x} = -\frac{c}{2d}.$$

Therefore,

$$\frac{\partial \text{Frag}(D_0^s, D_1)}{\partial x} = -\Delta(\theta_L^s(D_0^s)) \cdot \frac{\partial \theta_L^s(D_0^s)}{\partial x} = \Delta(\theta_L^s(D_0^s)) \cdot \frac{c}{2d} > 0.$$

□

Late-switch fragmentation derivative

Proof. Under $\theta_L^s(D_0^s) < \theta_F^s(D_0^s)$ we have $\underline{\theta}^s(D_0^s) = \theta_L^s(D_0^s)$. Moreover, the maintained inequalities

$$\theta_L^s(D_0^s) < \theta_L^u(D_1) < \min\{\theta_S(D_0^s), \bar{\theta}(D_1)\}$$

imply $\min\{\theta_S(D_0^s), \theta_L^u(D_1), \bar{\theta}(D_1)\} = \theta_L^u(D_1)$. Therefore, by equation (53),

$$\text{Frag}(D_0^s, D_1) = \int_{\theta_L^s(D_0^s)}^{\theta_L^u(D_1)} \Delta(\theta_2) d\theta_2.$$

Differentiating with respect to x (Leibniz rule) and using $\partial_x \theta_L^s(D_0^s) = \partial_x \theta_L^u(D_1) = -c/(2d)$ (equation (49)) yields

$$\frac{\partial}{\partial x} \text{Frag}(D_0^s, D_1) = \Delta(\theta_L^u(D_1)) \partial_x \theta_L^u(D_1) - \Delta(\theta_L^s(D_0^s)) \partial_x \theta_L^s(D_0^s) = \frac{c}{2d} \left(\Delta(\theta_L^s(D_0^s)) - \Delta(\theta_L^u(D_1)) \right).$$

If $\Delta(\theta_2) = \delta_0 + \delta_1 \theta_2$ is affine, then $\theta_L^u(D_1) > \theta_L^s(D_0^s)$ implies $\Delta(\theta_L^s) - \Delta(\theta_L^u) = -\delta_1(\theta_L^u - \theta_L^s)$, so

$$\text{sign}(\partial_x \text{Frag}) = -\text{sign}(\delta_1).$$

□

A.11C Proof of Lemma 5

Proof. Fix θ_2 and let j satisfy

$$C_{j-1} \leq G(\theta_2) < C_j.$$

For any resource pool $X \geq 0$, the payoff to class i , whose priority interval is $(C_{i-1}, C_i]$, is

$$r_i(X) = (\min\{C_i, X\} - C_{i-1})^+.$$

Equivalently,

$$r_i(X) = \min\{C_i, X\} - \min\{C_{i-1}, X\}.$$

Let

$$V_L(\theta_2) = G(\theta_2) + L(\theta_2, 2)$$

denote the liquidation pool, and let

$$Z(\theta_2, \theta_3) = G(\theta_2) + G(\theta_3) + L(\theta_3, 3)$$

denote the continuation pool. Since $L(\theta_2, 2) > 0$, $G(\theta_3) \geq 0$, and $L(\theta_3, 3) > 0$, we have

$$V_L(\theta_2) > G(\theta_2)$$

and

$$Z(\theta_2, \theta_3) \geq G(\theta_2) \quad \text{a.s. conditional on } \theta_2.$$

Because

$$C_{j-1} \leq G(\theta_2) < C_j,$$

the lower block of claims up to C_{j-1} is paid under both liquidation and continuation. Therefore class j 's liquidation payoff is

$$\pi_j^L(\theta_2) = r_j(V_L(\theta_2)) = \min\{C_j, V_L(\theta_2)\} - C_{j-1},$$

and its expected continuation payoff is

$$\pi_j^E(\theta_2) = \mathbb{E}[r_j(Z(\theta_2, \theta_3)) \mid \theta_2] = \mathbb{E}[\min\{C_j, Z(\theta_2, \theta_3)\} \mid \theta_2] - C_{j-1}.$$

Moreover,

$$\pi_j^L(\theta_2) > 0,$$

because $V_L(\theta_2) > G(\theta_2) \geq C_{j-1}$ and $C_j > G(\theta_2)$.

By Assumption 2, class j objects to continuation if and only if

$$\pi_j^L(\theta_2) \geq \pi_j^E(\theta_2).$$

Substituting the expressions above, this is equivalent to

$$\min\{C_j, V_L(\theta_2)\} \geq \mathbb{E}[\min\{C_j, Z(\theta_2, \theta_3)\} \mid \theta_2].$$

If this inequality holds, class j objects. Since class j is a participating creditor class, Assumption 2 implies liquidation.

Now suppose class j does not object. Since $\pi_j^L(\theta_2) > 0$, Assumption 2 implies

$$\pi_j^L(\theta_2) < \pi_j^E(\theta_2),$$

or equivalently,

$$\min\{C_j, V_L(\theta_2)\} < \mathbb{E}[\min\{C_j, Z(\theta_2, \theta_3)\} \mid \theta_2].$$

This strict inequality implies

$$V_L(\theta_2) < C_j.$$

Indeed, if $V_L(\theta_2) \geq C_j$, then

$$\min\{C_j, V_L(\theta_2)\} = C_j,$$

whereas

$$\min\{C_j, Z(\theta_2, \theta_3)\} \leq C_j \quad \text{a.s.}$$

so

$$\mathbb{E}[\min\{C_j, Z(\theta_2, \theta_3)\} \mid \theta_2] \leq C_j.$$

This contradicts the strict inequality above. Hence $V_L(\theta_2) < C_j$.

Now consider any junior class $k > j$. Since the priority cutoffs are increasing,

$$C_{k-1} \geq C_j.$$

Therefore,

$$V_L(\theta_2) < C_j \leq C_{k-1}.$$

Hence class k receives zero under liquidation:

$$r_k(V_L(\theta_2)) = (\min\{C_k, V_L(\theta_2)\} - C_{k-1})^+ = 0.$$

By Assumption 2, a participating class with zero liquidation payoff does not block continuation.

Thus no junior class $k > j$ objects.

Finally, any class $i < j$ is fully cash-covered because

$$C_i \leq C_{j-1} \leq G(\theta_2).$$

Such classes are taken out by contemporaneous cash and do not participate. Therefore, if class j does not object, no participating class objects, and Assumption 2 implies continuation.

Thus liquidation occurs if and only if the first cash-uncovered class objects. This proves that the first cash-uncovered class is the endogenous fulcrum creditor. $\square$

A.11D Closed forms for the quadratic refinancing-shortfall cost

Under Assumption 5,

$$V_C(\theta_2) = a_C + b_C \theta_2, \quad a_C = 2g_0 + \ell_3, \quad b_C = 2g + \lambda_3.$$

For $D \in [a_C, a_C + b_C]$, the quadratic refinancing-shortfall cost in equation (62) is

$$\Psi(D) = \frac{\chi}{2} \int_0^{(D-a_C)/b_C} (D - a_C - b_C \theta)^2 d\theta = \frac{\chi(D - a_C)^3}{6b_C},$$

and

$$\Psi'(D) = \frac{\chi(D - a_C)^2}{2b_C}.$$

For $D < a_C$, $\Psi(D) = 0$. For $D > a_C + b_C$, the expression is the corresponding full-support integral

$$\Psi(D) = \frac{\chi}{2} \int_0^1 (D - a_C - b_C \theta)^2 d\theta.$$

A.11E Linear refinancing-shortfall alternative

As an alternative specification, consider

$$\phi_{\text{lin}}(z) = \chi z, \quad z \geq 0, \quad \chi > 0, \quad \text{so that} \quad \Psi_{\text{lin}}(D) = \mathbb{E}[\phi_{\text{lin}}((D - V_C(\theta_2))_+)].$$

Under Assumption 5, for $D \in [a_C, a_C + b_C]$,

$$\Psi_{\text{lin}}(D) = \chi \int_0^{(D-a_C)/b_C} (D - a_C - b_C \theta) d\theta = \frac{\chi(D - a_C)^2}{2b_C},$$

and

$$\Psi'_{\text{lin}}(D) = \frac{\chi(D - a_C)}{b_C}.$$

For $D < a_C$, $\Psi_{\text{lin}}(D) = 0$. For $D > a_C + b_C$,

$$\Psi_{\text{lin}}(D) = \chi \int_0^1 (D - a_C - b_C \theta) d\theta = \chi \left(D - a_C - \frac{b_C}{2} \right).$$

The linear form is continuous, priority-invariant, resolution-invariant, and coercive, but its marginal cost has a kink at the forced-default boundary. It therefore provides a natural proportional-loss alternative, while the quadratic form is more convenient for smooth closed-form first-order conditions.

A.11F Numerical implementation and verification

All policies are evaluated with a common terminal-payoff evaluator built from the primitive pools $V_L(\theta_2)$, $Z^+(\theta_2)$, and $Z^-(\theta_2)$. Unit prices are expected per-unit creditor payoffs under the equilibrium resolution rule and contain no distress cost; the private cost is subtracted only from the entrepreneur's objective. The date-1 search enumerates every grid-detected feasible component, financing-boundary root (including tangential roots), structural boundary, and locally refined minimum. The date-0 search likewise compares sign-changing, tangential, and structural budget roots and selects the equilibrium root with the lowest total loss.

The baseline verification has five parts. First, the affine admissibility margins are $g_0 - gx = 0.100$, $\ell_2 - \lambda_2 x = 0.062$, and $\ell_3 - \lambda_3 x = 0.180$, and $\Delta(\theta_2) = 0.20 - 0.02\theta_2 > 0$ on $[0, 1]$. Second, Condition 1 is verified analytically over the full baseline face domain, not only at the reported equilibria. If $D > a_C + b_C = 2.85$, there are no repayment-feasible states and the condition is vacuous. For $D \leq 2.85$, whenever downside continuation already covers face, clause (ii) holds. Otherwise the difference between liquidation value and expected capped continuation value is piecewise affine. Before liquidation recovery reaches face, its minimum at the repayment boundary is

$$\min \left\{ \frac{cx}{2}, \frac{20D + 177}{1300} \right\} > 0;$$

after liquidation recovery reaches face it equals $\frac{1}{2}(D - V_C(\theta_2) + cx) \geq 0$ until clause (ii) begins. Hence clause (i) or clause (ii) holds in every repayment-feasible state for every baseline face. At the commitment face, the reported cutoff values remain $\theta_L(D^{\text{com}}) \simeq 0.156$ and $\bar{\theta}(D^{\text{com}}) \simeq 0.407$. Remark 1 then transmits the result to the two-layer structures used in the design analysis.

Third, the closed-form branch slack $D_{\text{cap}} - D_1$ is positive for every reported policy. At the fixed all-senior total face reported in Table 1, $D_1 = 2.098075$, the relevant cutoffs satisfy

$$\theta_L^u(D_1) = 0.1702 < \bar{\theta}(D_1) = 0.4216 < 1.$$

Moreover, both $\theta_L^s(S)$ and $\theta_F^s(S)$ are negative on the relevant interval. Hence the boundary ordering in Section 3.3.2,

$$\underline{\theta}^s(S) \leq 0 < \theta_S(S) < \min\{\theta_L^u(D_1), \bar{\theta}(D_1), 1\},$$

holds for

$$S \in (g_0, g_0 + g\theta_L^u(D_1)) \simeq (g_0, 0.6351).$$

The reported solutions are also stable when the numerical search ceiling is increased. Fourth, Assumption 4 is evaluated numerically by finite differences on the closed order-convex rectangle containing the reported design locus; the replication output reports the smallest coordinatewise increment on the evaluation grid. Fifth, the continuation-favoring locus in Figure 4 uses the same primitive payoff evaluator, root enumeration, and tolerances as the baseline locus, while applying the alternative strict-objection convention to every cash-uncovered class. Solver tolerances, budget residuals, candidate counts, and objective gaps are reported in the replication package.

All numerical series underlying the manuscript-facing tables, Figure 5, its caption, and the numerical text originate from the same reloaded canonical CSV files in `outputs/output.data`. For the submission package, the reviewed LaTeX fragments are frozen and inlined in the master source. The final Figure 5 PDF applies only a rendering repair to the certified series—it displays the payout discontinuity at the certified switch explicitly and relocates the peak-wedge annotation—without changing any economic value or recomputing any model result from a separate data source. The retained production nodes around the gap are

$$I_1 \in \{0.76, 0.77, 0.78, 0.79, 0.80, 0.81, 0.82, 0.83, 0.84, 0.85, 0.86\};$$

the three certified transition estimates are stored separately and enter the figure only as vertical markers.

The reported implementation interval uses selected-equilibrium implementation: the selected conditional equilibrium must match commitment in total face within 2×10^{-6} , payout within 2×10^{-6} , and total loss within 2×10^{-7} . The variables `CommitmentTargetWeakBR` and `Date1DeviationGain` separately test weak-best-response membership. Full-domain implementation is scanned on $[0, I_0]$; the safe-capacity restriction is reported separately. Every endpoint is certified by a symmetric bracket of width at most 10^{-6} , and contiguity is enforced by assertion.

The dense implementation-set diagnostic used to distinguish the full-domain scan from the safe-capacity restriction is retained in the replication package as `tables/Table_Fig7_gap_dense.tex` rather than included as an additional manuscript figure. The manuscript therefore contains no

separate Figure 8 for this diagnostic.

The three transition brackets are

$$I_1^{\text{enter}} \in [0.781990814209, 0.781991577148],$$

$$I_1^{\text{switch}} \in [0.813406982422, 0.813407592773],$$

$$I_1^{\text{exit}} \in [0.859513854980, 0.859514617920].$$

Entry reproduces the closed form $\Pi^{\text{one}}(2\ell_2 - \ell_3 + cx) - I_0 = 0.781990533$; the numerical-minus-analytic difference is 6.63×10^{-7} . The switch equates the risky-partial and safe-partial minimized losses. Full-resolution endpoint re-solves preserve the branch ordering.

At $I_1 = 0.80$ and $S = I_0$, commitment-face financing slack is -1.69×10^{-4} . At $I_1 = 0.85$ and $S = g_0 = 0.55$, the date-1 deviation gain is 7.16×10^{-5} ; the best-response face is 2.003146, compared with $D^{\text{com}} = 1.986994$, and payout is 0.010905. These diagnostics identify the two distinct failures of the conditional implementation hypothesis.

On $I_1 \in \{0.82, 0.83, 0.84, 0.85\}$, the selected safe-partial policy is frozen at the safe-capacity point $S^* = g_0 = 0.55$ (numerically 0.549999) and $D_1^* = 2.004126$, with constant issue proceeds 0.859518 and total loss 0.066737 up to numerical tolerance. The fitted payout slope is -1.000000000 ; its zero crossing differs from the certified gap exit by 3.48×10^{-6} .

The independent primitive audit evaluates all 60 combinations formed by two tie conventions, five senior faces, and six total faces using a separate 200,001-node midpoint integration. This audit independently checks state integration and creditor voting logic. It deliberately shares the economic payoff definitions with the production code and therefore does not constitute an independent derivation of the model. The separate analytical checks cover the closed-form entry, the fragmentation-wedge integral, and the safe-partial accounting identity.

A.12 Proof of Lemma 6

Proof. Define

$$f(C) := \min\{C, V_L(\theta_2)\} - \mathbb{E}\left[\min\{C, Z(\theta_2, \theta_3)\} \middle| \theta_2\right] \quad (115)$$

as the liquidation advantage of tranche C . Fix $C' \in [0, C]$ and suppose $f(C) \geq 0$. If $C' \leq V_L(\theta_2)$, then $f(C') \geq 0$. Otherwise $C' > V_L(\theta_2)$, and since $C' \leq C$ this also implies $C > V_L(\theta_2)$. In this region $\min\{C', V_L(\theta_2)\} = \min\{C, V_L(\theta_2)\} = V_L(\theta_2)$, so

$$f(C') = V_L(\theta_2) - \mathbb{E}[\min\{C', Z(\theta_2, \theta_3)\} \mid \theta_2], \quad f(C) = V_L(\theta_2) - \mathbb{E}[\min\{C, Z(\theta_2, \theta_3)\} \mid \theta_2].$$

Because $C' \leq C$, truncation is monotone: $\min\{C', Z(\theta_2, \theta_3)\} \leq \min\{C, Z(\theta_2, \theta_3)\}$ for every z , hence $\min\{C', Z\} \leq \min\{C, Z(\theta_2, \theta_3)\}$ almost surely, and

$$\mathbb{E}[\min\{C', Z(\theta_2, \theta_3)\} \mid \theta_2] \leq \mathbb{E}[\min\{C, Z(\theta_2, \theta_3)\} \mid \theta_2].$$

Subtracting from $V_L(\theta_2)$ reverses the inequality, yielding $f(C') \geq f(C) \geq 0$. $\square$

A.13 Priority-refinement monotonicity, safe refinement, and conditional coarsening

Proof. Under weak pivot-control, liquidation is equivalent to the condition $f(C_{\text{pivot}}(\cdot)) \geq 0$ evaluated at the pivot cap. Now fix $\mathcal{P}$ and a refinement $\mathcal{P}'$. Refinement can only add caps, so the first cap strictly above $G(\theta_2)$ cannot move upward; in particular,

$$C_{\text{pivot}}(\mathcal{P}') \leq C_{\text{pivot}}(\mathcal{P}).$$

Moreover, the set $\{C : f(C) \geq 0\}$ is downward closed: if $f(C) \geq 0$ at some cap C , then $f(C') \geq 0$ for every lower cap $C' \leq C$ (Lemma 6). Therefore, if liquidation occurs under $\mathcal{P}$, so that $f(C_{\text{pivot}}(\mathcal{P})) \geq 0$, then the weakly lower pivot cap under $\mathcal{P}'$ also satisfies $f(C_{\text{pivot}}(\mathcal{P}')) \geq 0$. Applying the weak pivot-control rule again implies liquidation under $\mathcal{P}'$. $\square$

Proof of Corollary 3. Fix a date-2 state θ_2 .

Suppose first that $G(\theta_2) \geq D$. Then no class is cash-uncovered under either partition, so the default-stage participation sets coincide.

Now suppose that $G(\theta_2) < D$. Every cutoff added in passing from $\mathcal{P}$ to $\mathcal{P}'$ satisfies

$$C \leq \underline{G} \leq G(\theta_2).$$

Hence no newly inserted cutoff belongs to the set of cutoffs strictly above contemporaneous cash. Therefore,

$$\{C \in \mathcal{P}' : C > G(\theta_2)\} = \{C \in \mathcal{P} : C > G(\theta_2)\},$$

and consequently

$$C_{\text{pivot}}(\mathcal{P}') = C_{\text{pivot}}(\mathcal{P}).$$

Lemma 5 implies that the two partitions induce the same liquidation–continuation decision in every default state.

Total face and aggregate resources are also unchanged. The entrepreneur’s payoff from full repayment is therefore the same under the two partitions. Because the default-stage outcome and aggregate resources under default are the same, her payoff from default is also the same. The repayment/default selection in Definition 1 is consequently unchanged. Thus the two partitions induce the same repayment/default policy, liquidation set, and expected liquidation deadweight loss.

State by state, aggregate creditor payoff equals the same total payment from the same resource pool under both partitions. Aggregate debt value is therefore identical. Finally, any tranche with upper cumulative cutoff $C \leq \underline{G}$ satisfies

$$C \leq \underline{G} \leq G(\theta_2)$$

in every state. Such a tranche is repaid in full from contemporaneous cash, including in default states. Since the risk-free rate is normalized to zero, its competitive unit price is one. $\square$

We next connect the fixed-face refinement result to ex ante design. Suppose the date-0 structure contains total protected face $S \geq 0$ and a bottom junior class with face $U \geq 0$, and the firm subsequently issues additional junior face $y \geq 0$. Let

$$D \equiv S + U + y.$$

Let $\hat{\mathcal{P}}$ denote the resulting terminal partition when the protected block $[0, S]$ is divided into multiple

priority classes, while $(S, D]$ remains a single junior class. Let

$$\tilde{\mathcal{P}} \equiv \{0, S, D\},$$

with duplicate cutoffs omitted, denote the corresponding coarsened partition. Thus, the two structures have the same total protected face and the same junior pool, but $\hat{\mathcal{P}}$ subdivides the protected block.

For any terminal partition $\mathcal{Q}$ of total face D , let $\mathcal{L}(\mathcal{Q}; D)$ denote its liquidation set and $\Lambda(\mathcal{Q}; D)$ its expected liquidation deadweight loss.

Corollary 4 (Conditional dominance of coarsening). *Holding S , U , and y fixed,*

$$\mathcal{L}(\hat{\mathcal{P}}; D) \supseteq \mathcal{L}(\tilde{\mathcal{P}}; D)$$

and

$$\Lambda(\tilde{\mathcal{P}}; D) \leq \Lambda(\hat{\mathcal{P}}; D).$$

The second inequality is strict if

$$\Pr\left(\mathcal{L}(\hat{\mathcal{P}}; D) \setminus \mathcal{L}(\tilde{\mathcal{P}}; D)\right) > 0.$$

Proof. Fix (S, U, y) and let $D = S + U + y$. By construction, both $\hat{\mathcal{P}}$ and $\tilde{\mathcal{P}}$ are partitions of the same total promised face D .

If $\hat{\mathcal{P}} = \tilde{\mathcal{P}}$, then all claims hold trivially. Otherwise, $\hat{\mathcal{P}}$ is a strict refinement of $\tilde{\mathcal{P}}$ obtained by subdividing the protected block $[0, S]$ into multiple layers. Proposition 7 therefore applies: at a fixed total promised face, refinement weakly expands the liquidation set under weak pivot-control. This gives

$$\mathcal{L}(\hat{\mathcal{P}}; D) \supseteq \mathcal{L}(\tilde{\mathcal{P}}; D),$$

which proves the liquidation-set inclusion.

By Assumption 1 (inefficient liquidation), liquidation destroys value state-by-state, meaning the deadweight-loss kernel is strictly positive in every forced-default state. Expected deadweight loss is therefore obtained by integrating this strictly positive kernel over the liquidation set. Since the

liquidation set under $\widehat{\mathcal{P}}$ weakly contains that under $\widetilde{\mathcal{P}}$, it follows that

$$\Lambda(\widetilde{\mathcal{P}}; D) \leq \Lambda(\widehat{\mathcal{P}}; D),$$

which proves the weak inequality.

Finally, if the two liquidation sets differ on a set of positive probability, the difference is integrated against a strictly positive deadweight-loss kernel. Therefore, the expected deadweight loss under $\widehat{\mathcal{P}}$ must be strictly greater, making the inequality strict. This proves the strictness claim. $\square$

A.14 Proof of Proposition 7(ii)

Proof. Fix a forced-default state θ_2 and write $C_{\text{pivot}} := C_{\text{pivot}}(\mathcal{P})$. By definition of $\text{mesh}(\mathcal{P})$ and C_{pivot} ,

$$0 < C_{\text{pivot}} - G(\theta_2) \leq \text{mesh}(\mathcal{P}).$$

If $\text{mesh}(\mathcal{P}) \leq V_L(\theta_2) - G(\theta_2)$, then $C_{\text{pivot}} \leq V_L(\theta_2)$. Therefore the pivotal block is fully repaid under liquidation:

$$\min\{C_{\text{pivot}}, V_L(\theta_2)\} = C_{\text{pivot}}.$$

Since $\min\{C_{\text{pivot}}, Z(\theta_2, \theta_3)\} \leq C_{\text{pivot}}$ almost surely, it follows that

$$\mathbb{E}\left[\min\{C_{\text{pivot}}, Z(\theta_2, \theta_3)\} \mid \theta_2\right] \leq C_{\text{pivot}},$$

so the weak pivot-control liquidation condition holds.

For the sequence $(\mathcal{P}_n)$, note that $V_L(\theta_2) - G(\theta_2) = L(\theta_2, 2) > 0$. Since $\text{mesh}(\mathcal{P}_n) \rightarrow 0$, there exists $N(\theta_2)$ such that $\text{mesh}(\mathcal{P}_n) \leq L(\theta_2, 2)$ for all $n \geq N(\theta_2)$, and the first part implies liquidation for all such n . $\square$

A.15 Proof of Proposition 8

Proof. Let $s = D^{\text{sec}}$, $u = D^{\text{unsec}}$, and $D = s + u$. For collateral proceeds $c \geq 0$ and an unencumbered residual pool $y \geq 0$, define the unified and secured creditor recoveries by

$$\mathcal{T}(c, y) \equiv \min\{D, c + y\}$$

and

$$\mathcal{S}(c, y) \equiv \min\{s, c\} + \mathcal{R}_{\text{def}}((s - c)^+, y; u).$$

For fixed residual pool y , these recoveries have the piecewise representations

$$\mathcal{T}(c, y) = \min\{D, c + y\}$$

and

$$\mathcal{S}(c, y) = \begin{cases} c + \frac{s - c}{D - c} y, & c < s, \ c + y < D, \\ s, & c < s, \ c + y \geq D, \\ s, & c \geq s. \end{cases}$$

Both functions are continuous and piecewise continuously differentiable in c .

For a fixed total pool $x = c + y$, $\mathcal{S}(c, x - c)$ is weakly increasing in c . If $x \geq D$, the secured claim is fully repaid. If $x < D$ and $c < s$, then

$$\mathcal{S}(c, x - c) = s - \frac{s - c}{D - c}(D - x),$$

and $(s - c)/(D - c)$ is weakly decreasing in c . Giving a larger portion of a fixed pool collateral priority therefore weakly raises the secured creditor's recovery. For $c \geq s$, the secured claim is fully repaid and $\mathcal{S}(c, x - c) = s$. Hence the monotonicity of $c \mapsto \mathcal{S}(c, x - c)$ extends to the entire range of c , including the case in which the comparison point below satisfies $\alpha W \geq s$.

Let $W = G_3 + L_3$, $l = L_2^A = \alpha L_2$, and

$$r(W) \equiv G_2 + (1 - \alpha)W.$$

Under partial liquidation, collateral proceeds are l and the residual pool is $r(W)$. Under continuation, collateral proceeds are αL_3 and the residual pool is $r(W) + \alpha G_3$. Because

$$\alpha L_3 + (r(W) + \alpha G_3) = \alpha W + r(W),$$

the fixed-pool monotonicity comparison implies

$$\mathcal{S}(\alpha L_3, r(W) + \alpha G_3) \leq \mathcal{S}(\alpha W, r(W)),$$

while

$$\mathcal{T}(\alpha L_3, r(W) + \alpha G_3) = \mathcal{T}(\alpha W, r(W)).$$

Thus it is sufficient to compare partial liquidation with a hypothetical continuation outcome in which all of component A 's date-3 payoff receives collateral priority.

Define

$$\kappa \equiv \begin{cases} 0, & l \geq s, \\ \frac{s-l}{D-l} + \frac{u(D-G_2-L_2)^+}{(D-l)^2}, & l < s. \end{cases}$$

Because $(D - G_2 - L_2)^+ \leq D - l$, we have $\kappa \in [0, 1]$. The coefficient κ is a state-independent comparison slope that bounds the secured creditor's response to changes in collateral proceeds relative to the unified creditor's response. We next show that, for every realization $W = w$,

$$\mathcal{S}(l, r(w)) - \mathcal{S}(\alpha w, r(w)) \geq \kappa [\mathcal{T}(l, r(w)) - \mathcal{T}(\alpha w, r(w))]. \quad (116)$$

If $l \geq s$, the secured creditor is fully repaid under partial liquidation, so the left-hand side of equation (116) is nonnegative. The result follows from $\kappa = 0$. Suppose instead that $l < s$. Away from the kink points, the piecewise derivatives are

$$\frac{\partial \mathcal{T}(c, y)}{\partial c} = \mathbf{1}\{c + y < D\}$$

and

$$\frac{\partial \mathcal{S}(c, y)}{\partial c} = \begin{cases} m(c, y) \equiv 1 - \frac{uy}{(D-c)^2}, & c < s, \ c + y < D, \\ 0, & c \geq s \text{ or } c + y \geq D. \end{cases}$$

The kink points have measure zero and do not affect the integral comparisons. On its active branch, $m(c, y)$ is nonnegative and weakly decreasing in both c and y .

First suppose $w < L_2$. Then $\alpha w < l < s$, so the path $c \in [\alpha w, l]$ never crosses $c = s$. If $G_2 + L_2 < D$, then

$$\kappa = 1 - \frac{ur(L_2)}{(D-l)^2} = m(l, r(L_2)).$$

On every portion of the path satisfying $c + r(w) < D$, monotonicity of m in both arguments gives

$$m(c, r(w)) \geq m(l, r(L_2)) = \kappa.$$

If instead $G_2 + L_2 \geq D$, then $\kappa = (s-l)/(D-l)$. On every portion satisfying $c + r(w) < D$,

$$m(c, r(w)) \geq 1 - \frac{u}{D-c} = \frac{s-c}{D-c} \geq \frac{s-l}{D-l} = \kappa.$$

Once $c + r(w) \geq D$, both $\mathcal{S}$ and $\mathcal{T}$ are flat. Integrating the derivative inequality from αw to l proves equation (116) for $w < L_2$.

Next suppose $w > L_2$. If $G_2 + L_2 \geq D$, then $l + r(w) > D$ and both structures repay the unified and secured claims in full, so both sides of equation (116) are zero. Suppose instead that $G_2 + L_2 < D$. On the region satisfying $c < s$ and $c + r(w) < D$,

$$m(c, r(w)) \leq m(l, r(L_2)) = \kappa,$$

because $c \geq l$ and $r(w) > r(L_2)$. If $c \geq s$ while $c + r(w) < D$, then

$$\frac{\partial \mathcal{S}(c, r(w))}{\partial c} = 0 \leq \kappa \frac{\partial \mathcal{T}(c, r(w))}{\partial c}.$$

Once $c + r(w) \geq D$, both recoveries are flat. Integrating from l to αw and reversing signs proves equation (116) for $w > L_2$. The equality is immediate when $w = L_2$.

Taking expectations in equation (116) and using the fixed-total-pool comparison gives

$$U_{\text{sec}}^{PL} - U_{\text{sec}}^C \geq \kappa [U_{\text{one}}^{PL}(D) - U_{\text{one}}^C(D)].$$

If the one-layer class objects to continuation, its partial-liquidation payoff is strictly positive and the expression in brackets is nonnegative. The displayed inequality therefore gives $U_{\text{sec}}^{PL} \geq U_{\text{sec}}^C$. Moreover, $L_2^A = \alpha L_2 > 0$, so the secured creditor's partial-liquidation payoff is strictly positive. It consequently objects under the collateral enforcement rule and, by its exclusive enforcement right, liquidates A . $\square$

A.16 Proof of Proposition 9

Proof. For $h \in [0, D^{\text{sec}}]$ and $D^{\text{unsec}} > 0$,

$$0 \leq \mathcal{R}_{\text{def}}(h, Y; D^{\text{unsec}}) \leq \frac{D^{\text{sec}}}{D^{\text{unsec}}} Y.$$

Because Y^{PL} and Y^C have finite expectations, the expected deficiency recoveries in equations (70) and (71) converge to zero as $D^{\text{unsec}} \rightarrow \infty$. Hence

$$U_{\text{sec}}^{PL} \longrightarrow \min\{D^{\text{sec}}, L_2^A\}, \quad U_{\text{sec}}^C \longrightarrow \mathbb{E}[\min\{D^{\text{sec}}, L_3^A\}],$$

which proves the first limit.

For the one-layer structure, $D^{\text{sec}} + D^{\text{unsec}} \rightarrow \infty$. Monotone convergence therefore gives, for $q \in \{PL, C\}$,

$$U_{\text{one}}^q(D^{\text{sec}} + D^{\text{unsec}}) = \mathbb{E}[\min\{D^{\text{sec}} + D^{\text{unsec}}, X^q\}] \longrightarrow \mathbb{E}[X^q].$$

By proportionality,

$$\mathbb{E}[X^C - X^{PL}] = \alpha(\mathbb{E}[G_3 + L_3] - L_2) > 0,$$

which proves the second limit. If $\Gamma_{\text{sec}} > 0$, the two limiting margins have opposite signs. Thus both stated inequalities hold for all sufficiently large D^{unsec} . The secured creditor's partial-liquidation payoff is then strictly positive, so it objects under Assumption 2, while the one-layer class strictly prefers continuation.

An explicit sufficient bound. The limit argument establishes the existence of a finite threshold. A convenient sufficient bound for the secured-creditor side is obtained as follows. For any $h \in [0, D^{\text{sec}}]$, any $Y \geq 0$, and any $D^{\text{unsec}} > 0$,

$$\mathcal{R}_{\text{def}}(h, Y; D^{\text{unsec}}) = \frac{h}{h + D^{\text{unsec}}} \min\{Y, h + D^{\text{unsec}}\} \leq \frac{D^{\text{sec}}}{D^{\text{unsec}}} Y.$$

Hence

$$\mathbb{E}[\mathcal{R}_{\text{def}}((D^{\text{sec}} - L_3^A)^+, Y^C; D^{\text{unsec}})] \leq \frac{D^{\text{sec}}}{D^{\text{unsec}}} \mathbb{E}[Y^C],$$

which is finite because

$$Y^C \leq G_2 + G_3 + L_3$$

and G_3, L_3 have finite conditional expectations. Since recoveries are nonnegative,

$$\mathbb{E}[\mathcal{R}_{\text{def}}((D^{\text{sec}} - L_2^A)^+, Y^{PL}; D^{\text{unsec}})] \geq 0.$$

Therefore,

$$U_{\text{sec}}^{PL} - U_{\text{sec}}^C \geq \Gamma_{\text{sec}} - \frac{D^{\text{sec}}}{D^{\text{unsec}}} \mathbb{E}[Y^C].$$

Set

$$\overline{D}^{\text{unsec}} = \frac{2D^{\text{sec}} \mathbb{E}[Y^C]}{\Gamma_{\text{sec}}}.$$

For every $D^{\text{unsec}} > \overline{D}^{\text{unsec}}$,

$$U_{\text{sec}}^{PL} - U_{\text{sec}}^C > \Gamma_{\text{sec}} - \frac{\Gamma_{\text{sec}}}{2} = \frac{\Gamma_{\text{sec}}}{2} > 0.$$

Thus the secured creditor strictly prefers partial liquidation. Since the secured creditor is pivotal, it enforces.

For social inefficiency,

$$\begin{aligned} \mathbb{E}[X^C - X^{PL}] &= \mathbb{E}[G_2 + G_3 + L_3 - (L_2^A + G_2 + G_3^B + L_3^B)] \\ &= \mathbb{E}[G_3^A + L_3^A] - L_2^A \\ &= \alpha (\mathbb{E}[G_3 + L_3] - L_2) > 0. \end{aligned}$$

A planner maximizing expected total surplus would continue. The secured creditor nevertheless enforces when $D^{\text{unsec}} > \overline{D}^{\text{unsec}}$, so the enforcement decision is privately optimal for the secured creditor and socially inefficient. $\square$

A.17 Proof of Lemma 7

Proof. We first record two facts.

First, the recovery bound

$$\mathcal{R}_{\text{def}}(h, Y; D^{\text{unsec}}) \leq \min\{h, Y\}$$

holds for all $h, Y, D^{\text{unsec}} \geq 0$. If $h + D^{\text{unsec}} = 0$, then $\mathcal{R}_{\text{def}}(h, Y; D^{\text{unsec}}) = 0$ and the bound is immediate. If $h + D^{\text{unsec}} > 0$ and $Y \leq h + D^{\text{unsec}}$, then

$$\mathcal{R}_{\text{def}}(h, Y; D^{\text{unsec}}) = \frac{hY}{h + D^{\text{unsec}}} \leq \min\{h, Y\}.$$

If $Y > h + D^{\text{unsec}}$, then

$$\mathcal{R}_{\text{def}}(h, Y; D^{\text{unsec}}) = h \leq \min\{h, Y\}.$$

Second, the identity

$$\min\{D, a\} + \min\{(D - a)^+, b\} = \min\{D, a + b\}$$

holds for all $D, a, b \geq 0$.

For partial liquidation, apply the recovery bound and the identity pointwise with $D = D^{\text{sec}}$, $a = L_2^A$, and $b = Y^{PL}$:

$$\min\{D^{\text{sec}}, L_2^A\} + \mathcal{R}_{\text{def}}((D^{\text{sec}} - L_2^A)^+, Y^{PL}; D^{\text{unsec}}) \leq \min\{D^{\text{sec}}, L_2^A + Y^{PL}\} = \min\{D^{\text{sec}}, X^{PL}\}.$$

Taking expectations gives

$$U_{\text{sec}}^{PL} \leq U_{\text{sen}}^{PL}.$$

For continuation, apply the same argument state by state with $a = L_3^A$ and $b = Y^C$:

$$\min\{D^{\text{sec}}, L_3^A\} + \mathcal{R}_{\text{def}}((D^{\text{sec}} - L_3^A)^+, Y^C; D^{\text{unsec}}) \leq \min\{D^{\text{sec}}, L_3^A + Y^C\}.$$

Since

$$L_3^A + Y^C = L_3^A + G_2 + G_3 + L_3^B = G_2 + G_3 + L_3 = X^C,$$

we have

$$\min\{D^{\text{sec}}, L_3^A + Y^C\} = \min\{D^{\text{sec}}, X^C\}.$$

Taking expectations gives

$$U_{\text{sec}}^C \leq U_{\text{sen}}^C.$$

If $L_2^A \geq D^{\text{sec}}$, then $\min\{D^{\text{sec}}, L_2^A\} = D^{\text{sec}}$ and the secured deficiency vanishes, so

$$U_{\text{sec}}^{PL} = D^{\text{sec}}.$$

Also, $X^{PL} = L_2^A + Y^{PL} \geq L_2^A \geq D^{\text{sec}}$ state by state, so

$$U_{\text{sen}}^{PL} = D^{\text{sec}}.$$

Combining with the continuation comparison,

$$U_{\text{sec}}^{PL} - U_{\text{sec}}^C = D^{\text{sec}} - U_{\text{sec}}^C \geq D^{\text{sec}} - U_{\text{sen}}^C = U_{\text{sen}}^{PL} - U_{\text{sen}}^C.$$

Since $U_{\text{sen}}^C \leq D^{\text{sec}}$, the senior margin is nonnegative.

For strictness, on the event

$$\{L_3^A < D^{\text{sec}}, 0 < Y^C < D^{\text{sec}} - L_3^A + D^{\text{unsec}}\}$$

with $D^{\text{unsec}} > 0$, let

$$h = D^{\text{sec}} - L_3^A > 0.$$

The senior creditor's pointwise payoff is

$$\min\{D^{\text{sec}}, L_3^A + Y^C\} = L_3^A + \min\{h, Y^C\}.$$

The secured creditor's pointwise payoff is

$$L_3^A + \mathcal{R}_{\text{def}}(h, Y^C; D^{\text{unsec}}).$$

If $Y^C \leq h$, then

$$\mathcal{R}_{\text{def}}(h, Y^C; D^{\text{unsec}}) = \frac{hY^C}{h + D^{\text{unsec}}} < Y^C = \min\{h, Y^C\},$$

because $D^{\text{unsec}} > 0$ and $Y^C > 0$. If $h < Y^C < h + D^{\text{unsec}}$, then

$$\mathcal{R}_{\text{def}}(h, Y^C; D^{\text{unsec}}) = \frac{hY^C}{h + D^{\text{unsec}}} < h = \min\{h, Y^C\}.$$

Thus the senior pointwise payoff strictly exceeds the secured pointwise payoff on the stated positive-probability event. Taking expectations gives

$$U_{\text{sec}}^C < U_{\text{sen}}^C,$$

which makes the enforcement-preference margin inequality strict. □

B Bargaining interpretation and limited-renegotiation robustness

This appendix provides a bargaining interpretation of Assumption 2 and a state-by-state robustness check allowing bounded consent transfers. The organizing comparison is between the available bargaining slack and the pivot class's liquidation advantage. If the available slack is no larger than that liquidation advantage, then even allocating all feasible consent transfers to the pivot class cannot induce it to support continuation. Thus the pivot-driven liquidation outcome is robust to limited renegotiation in that state.

Fix a forced-default state θ_2 with $V_C(\theta_2) < D_1$, where repayment feasibility is characterized by (8). Since $V_C(\theta_2) \geq G(\theta_2)$, this implies $G(\theta_2) < D_1$, so at least one priority class is cash-uncovered. Liquidation at $t = 2$ yields the distributable pool $V_L(\theta_2)$ in (10). Continuation is implemented through the same maturity extension used in the main text, preserving relative priority among claims not taken out by contemporaneous cash, and yields random resources $Z(\theta_2, \theta_3)$ in (11).

By (11) and the definition of $V_3(\theta_2)$ in (1),

$$\mathbb{E}[Z(\theta_2, \theta_3) \mid \theta_2] = G(\theta_2) + V_3(\theta_2).$$

Therefore the expected incremental total surplus from continuation relative to liquidation is

$$\mathbb{E}[Z(\theta_2, \theta_3) \mid \theta_2] - V_L(\theta_2) = V_3(\theta_2) - L(\theta_2, 2) = \Delta(\theta_2),$$

where the final equality uses (19). By Assumption 1 and (4), $\Delta(\theta_2) > 0$.

Let

$$\mathcal{P} = \{0 = C_0 < C_1 < \dots < C_K = D_1\}$$

denote the relevant priority partition. The set of cash-uncovered classes in state θ_2 is

$$\mathcal{U}(\mathcal{P}, \theta_2) \equiv \{k \in \{1, \dots, K\} : C_k > G(\theta_2)\}, \quad \mathcal{K}(\mathcal{P}, \theta_2) \equiv \#\mathcal{U}(\mathcal{P}, \theta_2). \quad (117)$$

Thus, $\mathcal{U}(\mathcal{P}, \theta_2)$ is the set of cash-uncovered classes, and $\mathcal{K}(\mathcal{P}, \theta_2)$ denotes its cardinality, i.e., the number of such classes. Classes with $C_k \leq G(\theta_2)$ are fully covered by contemporaneous cash, are taken out by that cash, and do not participate in the continuation-consent decision. Thus only

classes in $\mathcal{U}(\mathcal{P}, \theta_2)$ must be brought into a continuation agreement.

We model two limits on renegotiation: limited contractibility/pledgeability, meaning only a fraction of continuation surplus can be pledged or verified for enforceable consent payments, and a coordination/holdout burden increasing in the number of cash-uncovered priority classes that must consent. Let $\kappa_0 \in [0, 1]$ denote the contractible fraction of the continuation surplus and let $\zeta \geq 0$ denote a per-additional-voting-class coordination/holdout burden. Define the net contractible slack available to fund consent payments as

$$\mathcal{B}(\theta_2; \mathcal{K}(\mathcal{P}, \theta_2)) \equiv \max \left\{ 0, \kappa_0 \Delta(\theta_2) - \zeta (\mathcal{K}(\mathcal{P}, \theta_2) - 1) \right\}. \quad (118)$$

By construction, $\mathcal{B}(\theta_2; \mathcal{K}(\mathcal{P}, \theta_2))$ is weakly decreasing in the number of cash-uncovered classes that must consent, holding θ_2 fixed.

To avoid requiring the entrepreneur to raise cash at $t = 2$, consent payments are modeled as date-3 contingent transfers promised at $t = 2$ and paid at $t = 3$ only if continuation occurs. Formally, a continuation proposal consists of the maturity extension used in the baseline model and a nonnegative schedule $\{t_k(\theta_2, \theta_3)\}_{k \in \mathcal{U}(\mathcal{P}, \theta_2)}$ of date-3 contingent consent transfers. Since agents are risk-neutral in the baseline model, each voting class evaluates such transfers by their conditional expectation given θ_2 . The transfers are feasible only up to the available net contractible slack:

$$\begin{aligned} t_k(\theta_2, \theta_3) &\geq 0 && \forall k \in \mathcal{U}(\mathcal{P}, \theta_2) \text{ and all realizations of } \theta_3, \\ \mathbb{E} \left[\sum_{k \in \mathcal{U}(\mathcal{P}, \theta_2)} t_k(\theta_2, \theta_3) \mid \theta_2 \right] &\leq \mathcal{B}(\theta_2; \mathcal{K}(\mathcal{P}, \theta_2)). \end{aligned} \quad (119)$$

Each cash-uncovered priority class accepts or rejects. Continuation occurs if and only if all cash-uncovered classes accept; otherwise liquidation occurs. If continuation occurs, baseline extension payoffs apply plus the realized contingent transfers.

For $k \in \mathcal{U}(\mathcal{P}, \theta_2)$, define class k 's liquidation payoff and expected extension payoff under absolute priority by

$$\Pi_k^L(\theta_2) \equiv (\min\{C_k, V_L(\theta_2)\} - C_{k-1})^+, \quad (120)$$

$$\Pi_k^E(\theta_2) \equiv \mathbb{E} \left[(\min\{C_k, Z(\theta_2, \theta_3)\} - C_{k-1})^+ \mid \theta_2 \right]. \quad (121)$$

For each cash-uncovered class k ,

$$\text{class } k \text{ accepts} \iff \left(\Pi_k^L(\theta_2) = 0 \right) \quad \text{or} \quad \left(\Pi_k^E(\theta_2) + \mathbb{E}[t_k(\theta_2, \theta_3) \mid \theta_2] > \Pi_k^L(\theta_2) \right). \quad (122)$$

Thus ties with strictly positive liquidation payoff are broken toward liquidation.

For a cutoff C , define the cumulative liquidation advantage

$$\Gamma(\theta_2, C) \equiv \min\{C, V_L(\theta_2)\} - \mathbb{E}[\min\{C, Z(\theta_2, \theta_3)\} \mid \theta_2]. \quad (123)$$

For any partition $\mathcal{P}$, let j be its pivot index, so that $C_{j-1} \leq G(\theta_2) < C_j = C_{\text{pivot}}(\mathcal{P})$. Because $V_L(\theta_2) > G(\theta_2) \geq C_{j-1}$ and $Z(\theta_2, \theta_3) \geq G(\theta_2) \geq C_{j-1}$ almost surely, the block of claims below C_{j-1} is fully repaid under both liquidation and continuation. Hence $\Gamma(\theta_2, C_j)$ is also the pivot tranche's liquidation advantage: the common lower-block payoff C_{j-1} cancels from the pivot tranche's liquidation and continuation payoffs.

Proposition 10. *Fix a forced-default state θ_2 with $G(\theta_2) < D_1$ and a priority partition $\mathcal{P}$. Let $C_{\text{pivot}}(\mathcal{P})$ be the pivot cutoff defined in (64). Maintain the bargaining protocol above, so that rejection by any cash-uncovered class, and therefore rejection by the pivot class, is sufficient to block continuation.*

If

$$\mathcal{B}(\theta_2; \mathcal{K}(\mathcal{P}, \theta_2)) \leq \Gamma(\theta_2, C_{\text{pivot}}(\mathcal{P})), \quad (124)$$

then no feasible continuation proposal can induce the pivot class to accept, and the firm is liquidated at $t = 2$ in state θ_2 .

Consequently, let $\mathcal{P}'$ be a refinement of $\mathcal{P}$, and suppose that θ_2 lies in a state where $\mathcal{P}$ continues but $\mathcal{P}'$ liquidates under the baseline pivot-control rule. If

$$\mathcal{B}(\theta_2; \mathcal{K}(\mathcal{P}', \theta_2)) \leq \Gamma(\theta_2, C_{\text{pivot}}(\mathcal{P}')), \quad (125)$$

then θ_2 remains in the fragmentation region under limited renegotiation: the coarse partition $\mathcal{P}$ continues, while the refinement $\mathcal{P}'$ still liquidates.

Proof. Let j denote the pivot index under $\mathcal{P}$, so that $C_{j-1} \leq G(\theta_2) < C_j = C_{\text{pivot}}(\mathcal{P})$. Then

$j \in \mathcal{U}(\mathcal{P}, \theta_2)$. Because

$$V_L(\theta_2) = G(\theta_2) + L(\theta_2, 2) > G(\theta_2) \geq C_{j-1},$$

and

$$Z(\theta_2, \theta_3) = G(\theta_2) + G(\theta_3) + L(\theta_3, 3) \geq G(\theta_2) \geq C_{j-1} \quad \text{a.s.},$$

the block of claims below C_{j-1} is fully repaid under both liquidation and continuation. Hence the pivot tranche's liquidation payoff and expected continuation payoff are

$$\Pi_{\text{pivot}}^L(\theta_2) = \min\{C_{\text{pivot}}(\mathcal{P}), V_L(\theta_2)\} - C_{j-1},$$

and

$$\Pi_{\text{pivot}}^E(\theta_2) = \mathbb{E}[\min\{C_{\text{pivot}}(\mathcal{P}), Z(\theta_2, \theta_3)\} \mid \theta_2] - C_{j-1}.$$

Moreover, since both $C_{\text{pivot}}(\mathcal{P})$ and $V_L(\theta_2)$ are strictly larger than $G(\theta_2)$, while $G(\theta_2) \geq C_{j-1}$, we have $\Pi_{\text{pivot}}^L(\theta_2) > 0$. Subtracting gives

$$\Pi_{\text{pivot}}^L(\theta_2) - \Pi_{\text{pivot}}^E(\theta_2) = \Gamma(\theta_2, C_{\text{pivot}}(\mathcal{P})).$$

Let

$$T_{\text{pivot}}(\theta_2) \equiv \mathbb{E}[t_j(\theta_2, \theta_3) \mid \theta_2]$$

denote the pivot class's expected consent transfer. Because $\Pi_{\text{pivot}}^L(\theta_2) > 0$, the zero-liquidation-payoff clause in (122) is irrelevant for the pivot class. Thus the pivot class accepts continuation only if

$$\Pi_{\text{pivot}}^E(\theta_2) + T_{\text{pivot}}(\theta_2) > \Pi_{\text{pivot}}^L(\theta_2),$$

or equivalently only if

$$T_{\text{pivot}}(\theta_2) > \Gamma(\theta_2, C_{\text{pivot}}(\mathcal{P})). \tag{126}$$

Feasibility of consent transfers and nonnegativity of each transfer imply

$$T_{\text{pivot}}(\theta_2) \leq \mathbb{E} \left[\sum_{k \in \mathcal{U}(\mathcal{P}, \theta_2)} t_k(\theta_2, \theta_3) \mid \theta_2 \right] \leq \mathcal{B}(\theta_2; \mathcal{K}(\mathcal{P}, \theta_2)).$$

Therefore, if (124) holds, even allocating the entire available bargaining slack to the pivot class cannot satisfy (126). The pivot class rejects, and the unanimity requirement over cash-uncovered classes implies liquidation.

For the consequence, suppose θ_2 is a state where the coarse partition $\mathcal{P}$ continues but the refinement $\mathcal{P}'$ liquidates under the baseline pivot-control rule. Let $C_p \equiv C_{\text{pivot}}(\mathcal{P})$. Since $\mathcal{P}$ continues under that rule, and the payoff calculation above applies to the pivot class under $\mathcal{P}$, its strictly positive liquidation payoff implies that it strictly prefers extension without consent transfers; equivalently,

$$\mathbb{E}[\min\{C_p, Z(\theta_2, \theta_3)\} \mid \theta_2] > \min\{C_p, V_L(\theta_2)\}.$$

This strict inequality implies $V_L(\theta_2) < C_p$, because if $V_L(\theta_2) \geq C_p$, then the right-hand side would equal C_p while the left-hand side is at most C_p . Hence every cash-uncovered class junior to the pivot has zero liquidation payoff under absolute priority: for such a class k ,

$$\Pi_k^L(\theta_2) = (\min\{C_k, V_L(\theta_2)\} - C_{k-1})^+ = 0,$$

and therefore accepts by the first clause of (122). The pivot class accepts by the strict inequality above, and more senior cash-covered classes do not vote. Since the proposal with no consent transfers satisfies (119), all voting classes under $\mathcal{P}$ accept continuation. If (125) holds for $\mathcal{P}'$, the first part of the proposition, applied to $\mathcal{P}'$, implies that $\mathcal{P}'$ still liquidates even when bounded consent transfers are allowed. Hence θ_2 remains in the fragmentation region. $\square$

Online Appendix

to “The Curse of Seniority: Debt Priority Design and Inefficient Liquidation”

Jingxiong (Tony) Hu Chao Ying

OA.1 Strict-Objection Robustness of Priority Refinement

The main text uses the weak-objection convention in Assumption 2. This section records two robustness results under the alternative convention in which a cash-uncovered class objects only when its liquidation payoff is *strictly* greater than its expected continuation payoff.

Fix a forced-default state θ_2 and a finite partition

$$\mathcal{P} = \{0 = C_0 < C_1 < \cdots < C_K = D_1\}.$$

For class i , define its liquidation and continuation payoffs by

$$\begin{aligned}\pi_i^L(\mathcal{P}; \theta_2) &\equiv \min\{C_i, V_L(\theta_2)\} - \min\{C_{i-1}, V_L(\theta_2)\}, \\ \pi_i^C(\mathcal{P}; \theta_2) &\equiv \mathbb{E}[\min\{C_i, Z(\theta_2, \theta_3)\} - \min\{C_{i-1}, Z(\theta_2, \theta_3)\} \mid \theta_2].\end{aligned}$$

Under the strict-objection convention, a cash-uncovered class objects when $\pi_i^L(\mathcal{P}; \theta_2) > \pi_i^C(\mathcal{P}; \theta_2)$, and the firm liquidates if at least one class objects.

Proposition 11 (Priority refinement under strict objections). *Fix total face D_1 and a forced-default state θ_2 .*

- (i) *If $\mathcal{P}'$ refines $\mathcal{P}$ and the firm liquidates under $\mathcal{P}$ under the strict-objection convention, then it also liquidates under $\mathcal{P}'$.*
- (ii) *Define the conditional downside continuation pool by*

$$Z^-(\theta_2) \equiv \text{ess inf}(Z(\theta_2, \theta_3) \mid \theta_2).$$

Suppose

$$Z^-(\theta_2) = G(\theta_2), \tag{127}$$

which is equivalent to requiring that, for every $\varepsilon > 0$,

$$\Pr(Z(\theta_2, \theta_3) < G(\theta_2) + \varepsilon \mid \theta_2) > 0.$$

If

$$\text{mesh}(\mathcal{P}) \leq V_L(\theta_2) - G(\theta_2),$$

then the firm liquidates in state θ_2 under the strict-objection convention. Consequently, for any sequence $(\mathcal{P}_n)_{n \geq 1}$ with $\text{mesh}(\mathcal{P}_n) \rightarrow 0$, there exists $N(\theta_2)$ such that the firm liquidates under every $\mathcal{P}_n$ with $n \geq N(\theta_2)$ whenever condition (127) holds.

Proof. Part (i). Suppose class i under $\mathcal{P}$ strictly objects. Let

$$C_{i-1} = \tilde{C}_{m_0} < \tilde{C}_{m_0+1} < \cdots < \tilde{C}_{m_1} = C_i$$

be the cutoffs of $\mathcal{P}'$ that subdivide the interval $[C_{i-1}, C_i]$. Absolute-priority payoffs telescope under either resolution outcome:

$$\pi_i^L(\mathcal{P}; \theta_2) = \sum_{m=m_0+1}^{m_1} \pi_m^L(\mathcal{P}'; \theta_2), \quad \pi_i^C(\mathcal{P}; \theta_2) = \sum_{m=m_0+1}^{m_1} \pi_m^C(\mathcal{P}'; \theta_2).$$

Therefore,

$$0 < \pi_i^L(\mathcal{P}; \theta_2) - \pi_i^C(\mathcal{P}; \theta_2) = \sum_{m=m_0+1}^{m_1} [\pi_m^L(\mathcal{P}'; \theta_2) - \pi_m^C(\mathcal{P}'; \theta_2)].$$

At least one refined class has a strictly positive payoff difference. Every class whose upper cutoff is weakly below $G(\theta_2)$ is fully repaid from contemporaneous cash under both outcomes and has difference zero. Hence a refined class with a strictly positive difference is cash-uncovered and objects. The refined structure therefore liquidates.

Part (ii). Let $C_{j-1} \leq G(\theta_2) < C_j$ denote the first cash-uncovered class. The mesh condition implies

$$C_j \leq G(\theta_2) + \text{mesh}(\mathcal{P}) \leq V_L(\theta_2),$$

so this class is fully repaid under liquidation and receives $C_j - C_{j-1} > 0$. Since $Z(\theta_2, \theta_3) \geq G(\theta_2) \geq$

C_{j-1} almost surely, its expected continuation payoff is

$$\mathbb{E}[\min\{C_j, Z(\theta_2, \theta_3)\} \mid \theta_2] - C_{j-1}.$$

Condition (127) and $C_j > G(\theta_2)$ imply

$$\Pr(Z(\theta_2, \theta_3) < C_j \mid \theta_2) > 0.$$

Hence

$$\mathbb{E}[\min\{C_j, Z(\theta_2, \theta_3)\} \mid \theta_2] < C_j,$$

so the first cash-uncovered class strictly prefers liquidation and objects. The sequence statement follows because $V_L(\theta_2) - G(\theta_2) = L(\theta_2, 2) > 0$ and the mesh converges to zero. $\square$

Part (i) shows that refinement monotonicity does not depend on weak objections: payoff conservation within every class that is subdivided ensures that a strict coarse-class objection survives in at least one refined subtranche. Part (ii) is more demanding. Under strict objections, the fine-partition liquidation limit requires continuation resources to have downside support arbitrarily close to contemporaneous cash, as stated in condition (127).

OA.2 Multiple Financing Dates

We show that the fragmentation-dilution trade-off also applies to a more general setting with n pre-maturity financing dates at which the firm either issues senior debt that is senior to every debt issued at later dates or junior debt without covenant.

Fix an integer $n \geq 3$. The firm has n pre-maturity financing dates $t = 0, 1, \dots, n-1$ and a common maturity date $t = n$. At each date $t = i \in \{0, \dots, n-1\}$, the firm must finance an exogenous outlay $I_i > 0$ with debt due at $t = n$. The firm may issue more than required and distribute the residual as payout.

At maturity $t = n$ with state θ_n , operating cash is $G(\theta_n) \geq 0$, and liquidation at $t = n$ yields proceeds $L(\theta_n, n) > 0$. Define the liquidation pool $V_L(\theta_n) \equiv G(\theta_n) + L(\theta_n, n)$. If creditors extend

maturity to $t = n + 1$, a further state θ_{n+1} is realized and total resources at $t = n + 1$ are

$$Z(\theta_n, \theta_{n+1}) \equiv G(\theta_n) + G(\theta_{n+1}) + L(\theta_{n+1}, n + 1),$$

viewed as a random variable conditional on θ_n . Define the continuation value at $t = n$ by

$$V_{n+1}(\theta_n) \equiv \mathbb{E}[G(\theta_{n+1}) + L(\theta_{n+1}, n + 1) \mid \theta_n],$$

and total repayment capacity

$$V_C^n(\theta_n) \equiv G(\theta_n) + V_{n+1}(\theta_n) = \mathbb{E}[Z(\theta_n, \theta_{n+1}) \mid \theta_n].$$

We maintain the analogue of Assumption 1 at $t = n$:

$$L(\theta_n, n) < V_{n+1}(\theta_n) \quad \text{for all } \theta_n,$$

and define the per-state deadweight loss from liquidation as

$$\Delta(\theta_n) \equiv V_{n+1}(\theta_n) - L(\theta_n, n) > 0.$$

At each financing date $t = i$ the firm may issue either: (i) protected debt that is contractually senior to all debt issued at later dates (senior-at- i); or (ii) unprotected debt that carries no covenant and is pari passu with all other unprotected debt. A protected issuance can, in principle, be split into multiple within-date seniority layers. Proposition 12 gives a fixed-policy dominance result under weak pivot-control and no strategic default at maturity: holding the face vector and continuation issuance plan fixed, within-date splitting weakly increases liquidation (and thus Λ_n) and cannot increase the issuance proceeds of the protected block. We therefore use one protected layer per date as a maintained design class; the proposition does not by itself solve the unrestricted sequential financing problem after all equilibrium prices and future policies are allowed to adjust.

Proposition 12. *Fix a total promised face value D and let $\mathcal{P}$ be any maturity partition of $[0, D]$. Let $\tilde{\mathcal{P}}$ be obtained from $\mathcal{P}$ by splitting some protected issuance made at a single financing date into multiple within-date seniority sublayers, holding fixed all total face values and hence keeping D*

unchanged. Assume weak pivot-control at $t = n$ and no strategic default at maturity (i.e., whenever $V_C^n(\theta_n) \geq D$ the firm repays in full). Then:

(i) More liquidation at fixed promised face: $\tilde{\mathcal{P}}$ is a refinement of $\mathcal{P}$, and

$$\mathbf{1}_{\text{liq}}(\theta_n; \tilde{\mathcal{P}}, D) \geq \mathbf{1}_{\text{liq}}(\theta_n; \mathcal{P}, D) \quad \forall \theta_n,$$

so $\Lambda_n(\tilde{\mathcal{P}}, D) \geq \Lambda_n(\mathcal{P}, D)$.

(ii) Proceeds dominance for the split block: Let $A < B$ be the adjacent cutoffs in $\mathcal{P}$ delimiting the (single) protected block that is split, so that $\tilde{\mathcal{P}}$ differs from $\mathcal{P}$ only by inserting additional cutoffs in (A, B) . For every θ_n , the total expected payoff at $t = n$ of the entire block of claims with priority interval $(A, B]$ is weakly higher under $\mathcal{P}$ than under $\tilde{\mathcal{P}}$. Consequently, under competitive risk-neutral pricing, the issuance proceeds of this block at any financing date are weakly higher under $\mathcal{P}$ than under $\tilde{\mathcal{P}}$ for the same total promised face $B - A$.

Thus, conditional on the fixed face vector and continuation issuance plan, within-date seniority splitting is weakly dominated in liquidation loss and protected-block proceeds.

Part (i) of Proposition 12 isolates an ex post governance cost: within-date refinement makes inefficient liquidation weakly more likely under weak pivot-control for a given total promised face. Part (ii) connects this to ex ante feasibility: because within-date splitting cannot increase the total value (and hence proceeds) of the protected block being issued, splitting cannot relax any proceeds requirement I_i .

Let $D_i^s \geq 0$ and $D_i^u \geq 0$ denote the face values issued at date i in these two categories. Let $\mathcal{S} \equiv \{i : D_i^s > 0\}$ and write its elements in increasing order as $i_1 < \dots < i_K$ (with the convention $K = 0$ if $\mathcal{S} = \emptyset$). Define cumulative protected cutoffs

$$C_k \equiv \sum_{m=1}^k D_{i_m}^s \quad (k = 1, \dots, K), \quad C_0 \equiv 0,$$

total unprotected face $D^u \equiv \sum_{i=0}^{n-1} D_i^u$, and total face due at $t = n$ as $D \equiv C_K + D^u$ (interpreting $C_K = C_0 = 0$ when $K = 0$). The maturity priority structure is summarized by the cutoff set

$$\mathcal{P} \equiv \{0 = C_0 < C_1 < \dots < C_K\} \cup \{D\},$$

where the final cutoff D is redundant if $D^u = 0$ (i.e., if $D = C_K$).

At $t = n$, we assume the firm repays in full whenever repayment is feasible, i.e., whenever $V_C^n(\theta_n) \geq D$; otherwise default is forced. In forced default (so $G(\theta_n) < D$, hence $C_{\text{pivot}}(\mathcal{P})$ is well-defined), the liquidation/extension decision follows the same weak pivot-control rule as in Section 4.1.2: letting $C_{\text{pivot}}(\mathcal{P}) \equiv \min\{C \in \mathcal{P} : C > G(\theta_n)\}$, creditors liquidate at $t = n$ if and only if

$$\min\{C_{\text{pivot}}(\mathcal{P}), V_L(\theta_n)\} \geq \mathbb{E}[\min\{C_{\text{pivot}}(\mathcal{P}), Z(\theta_n, \theta_{n+1})\} \mid \theta_n],$$

with ties broken toward liquidation. Let $\mathbf{1}_{\text{liq}}(\theta_n; \mathcal{P}, D) \in \{0, 1\}$ denote the resulting liquidation indicator (and set it to zero whenever $V_C^n(\theta_n) \geq D$).

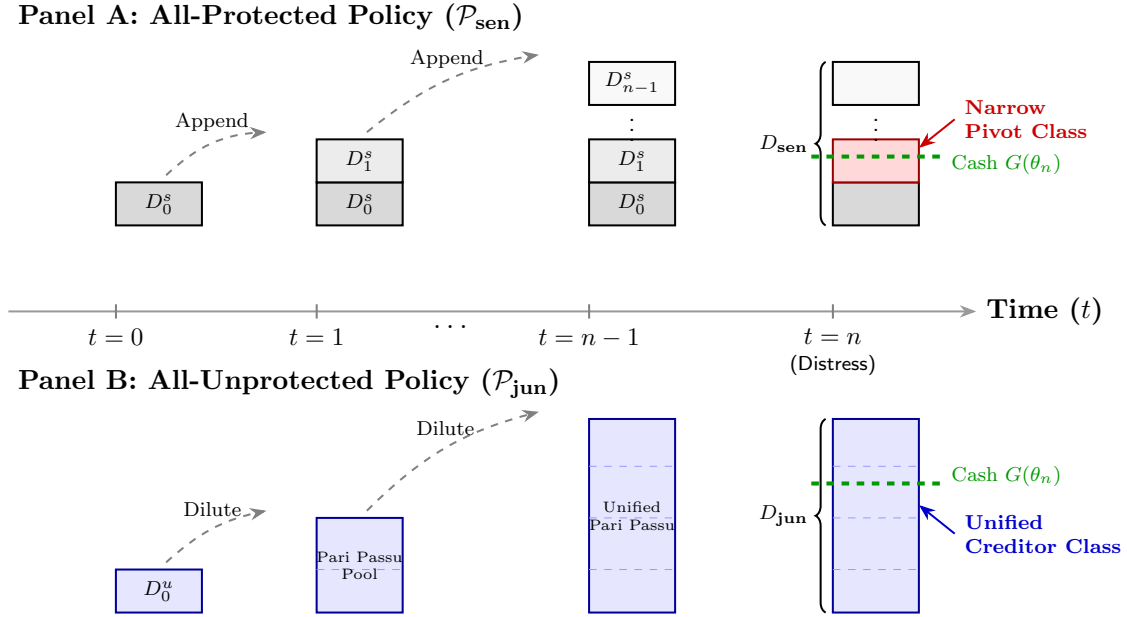


Figure 7: **Dynamic leverage ratchet and priority stacking over multiple financing dates.** The figure illustrates the evolution of debt structure under two polar financing regimes. Panel A depicts an all-protected policy ($\mathcal{P}_{\text{sen}}$), under which successive protected issues generate a layered maturity partition with total face D_{sen} . Panel B depicts an all-unprotected policy ($\mathcal{P}_{\text{jun}}$), under which all issues enter a common pari passu pool with total face D_{jun} . The green dashed line indicates realized cash $G(\theta_n)$. The figure illustrates the fragmentation and dilution forces isolated in Proposition 13; it does not assert that the two polar policies always select liquidation and continuation, respectively, or that either policy is equilibrium-optimal.

For any maturity partition $\mathcal{P}$ and total face D , define expected deadweight loss

$$\Lambda_n(\mathcal{P}, D) := \mathbb{E}[\Delta(\theta_n) \mathbf{1}_{\text{liq}}(\theta_n; \mathcal{P}, D)]. \quad (128)$$

For one-layer debt (no seniority), define $\Lambda_n^{\text{one}}(D) := \Lambda_n(\{0, D\}, D)$.

Proposition 13. *Let $(\mathcal{P}_{\text{sen}}, D_{\text{sen}})$ denote the maturity partition and total face value induced by an all-protected (layered) policy, and let $(\mathcal{P}_{\text{jun}}, D_{\text{jun}})$ denote those induced by an all-unprotected (one-layer) policy. Then*

$$\begin{aligned} & \Lambda_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) - \Lambda_n(\mathcal{P}_{\text{jun}}, D_{\text{jun}}) \\ &= \underbrace{\left(\Lambda_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) - \Lambda_n^{\text{one}}(D_{\text{sen}}) \right)}_{\text{fragmentation wedge at fixed } D_{\text{sen}}} - \underbrace{\left(\Lambda_n^{\text{one}}(D_{\text{jun}}) - \Lambda_n^{\text{one}}(D_{\text{sen}}) \right)}_{\text{leverage ratchet wedge under one-layer debt}}. \end{aligned} \quad (129)$$

Moreover, the fragmentation wedge is weakly nonnegative whenever $\mathcal{P}_{\text{sen}}$ is a refinement of $\{0, D_{\text{sen}}\}$ (in particular, for any nontrivial layered structure under weak pivot-control), and it is strictly positive whenever

$$\Pr(\mathbf{1}_{\text{liq}}(\theta_n; \mathcal{P}_{\text{sen}}, D_{\text{sen}}) > \mathbf{1}_{\text{liq}}(\theta_n; \{0, D_{\text{sen}}\}, D_{\text{sen}})) > 0.$$

If in addition $\Lambda_n^{\text{one}}(\cdot)$ is weakly increasing and $D_{\text{jun}} \geq D_{\text{sen}}$, then the leverage ratchet wedge is weakly nonnegative as well.

Corollary 5 (Total-loss decomposition). *Let $\Psi_n(D)$ denote the maturity- n analogue of the distress cost in Assumption 3, and define $\mathcal{T}_n(\mathcal{P}, D) \equiv \Lambda_n(\mathcal{P}, D) + \Psi_n(D)$. Then*

$$\begin{aligned} \mathcal{T}_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) - \mathcal{T}_n(\mathcal{P}_{\text{jun}}, D_{\text{jun}}) &= \left(\Lambda_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) - \Lambda_n^{\text{one}}(D_{\text{sen}}) \right) \\ &\quad - \left(\Lambda_n^{\text{one}}(D_{\text{jun}}) - \Lambda_n^{\text{one}}(D_{\text{sen}}) \right) - \left(\Psi_n(D_{\text{jun}}) - \Psi_n(D_{\text{sen}}) \right). \end{aligned} \quad (130)$$

The fragmentation wedge is identical to that in equation (129) and is independent of the distress-cost specification; the cost enters only through total face. If $D_{\text{jun}} \geq D_{\text{sen}}$, then $\Psi_n(D_{\text{jun}}) - \Psi_n(D_{\text{sen}}) \geq 0$. Because this bracket enters the decomposition with a minus sign, the distress-cost component weakly favors the protected policy whenever that policy induces weakly lower total face.

Proposition 13 and Corollary 5 decompose the difference between two polar financing policies; they do not characterize the unrestricted sequential equilibrium. The first bracket in equation (129) is the fixed-face fragmentation wedge, weakly nonnegative whenever the protected policy induces a refinement of one-layer debt at the same total face, and independent of the distress-cost specification. The second bracket is the leverage-ratchet component under one-layer debt; it is weakly

nonnegative under the additional conditions stated in the proposition, namely $D_{\text{jun}} \geq D_{\text{sen}}$ and monotonicity of one-layer liquidation loss over the relevant face domain. When total face differs across the two policies, Corollary 5 adds a distress-cost component: when $D_{\text{jun}} \geq D_{\text{sen}}$, this component weakly favors the protected policy, because the unprotected policy incurs the larger cost of total face.

Figure 7 illustrates the two forces. Protected issuance can insulate earlier claims from later pari passu dilution while generating a finer terminal priority partition; unprotected issuance preserves a single pari passu class while leaving inherited claims exposed to dilution by subsequent borrowing. The figure is schematic: it depicts the two polar mechanisms and does not assert that either polar policy is equilibrium-optimal, that the protected policy always induces lower total face, or that the unprotected policy continues in every default state.

The fixed-face refinement results, the conditional coarsening result (Corollary 4), and the decomposition above do not solve the unrestricted multi-date architecture-design problem, which would require backward induction with the inherited priority partition and face vector as state variables, date-by-date issuance best responses, and competitive prices determined by the continuation equilibrium. Accordingly, this subsection isolates the two polar financing forces rather than characterizing the unrestricted sequential equilibrium. Its multi-date contribution is to establish the fixed-face governance cost of priority refinement, to show that unnecessary subdivision of a protected block is dominated by coarsening, and to decompose the fragmentation and leverage-ratchet components. The paper’s positive ex ante design results are obtained in the completed two-date environment: Corollary 1 provides the conditional implementation result for safe seniority, and the numerical partial-seniority plateau in Figure 4 reports the corresponding numerical design locus.

OA.2.1 Proofs

Proof of Proposition 12

Proof. Fix D and partitions $\mathcal{P}$ and $\tilde{\mathcal{P}}$ as in Proposition 12.

Part (i). By construction, $\tilde{\mathcal{P}}$ is a refinement of $\mathcal{P}$. Proposition 7(i) therefore implies that, for every forced-default state θ_n ,

$$\mathbf{1}_{\text{liq}}(\theta_n; \tilde{\mathcal{P}}, D) \geq \mathbf{1}_{\text{liq}}(\theta_n; \mathcal{P}, D).$$

Under our convention that $\mathbf{1}_{\text{liq}}(\theta_n; \cdot, D) = 0$ whenever repayment is feasible, and because repayment feasibility depends only on D , the same inequality holds for all θ_n . Multiplying by $\Delta(\theta_n) > 0$ and taking expectations yields $\Lambda_n(\tilde{\mathcal{P}}, D) \geq \Lambda_n(\mathcal{P}, D)$.

Part (ii). Let $A < B$ be the adjacent cutoffs in $\mathcal{P}$ delimiting the protected block that is split, so that $\tilde{\mathcal{P}}$ differs from $\mathcal{P}$ only by inserting additional cutoffs in (A, B) . For $Q \in \{\mathcal{P}, \tilde{\mathcal{P}}\}$, define the time- n value of the entire block $(A, B]$ in state θ_n by

$$\Pi_{A,B}(\theta_n; Q, D) := \begin{cases} B - A, & \text{if } V_C^n(\theta_n) \geq D, \\ \mathbf{1}_{\text{liq}}(\theta_n; Q, D) \left(\min\{B, V_L(\theta_n)\} - \min\{A, V_L(\theta_n)\} \right) \\ + (1 - \mathbf{1}_{\text{liq}}(\theta_n; Q, D)) \mathbb{E} \left[\min\{B, Z(\theta_n, \theta_{n+1})\} - \min\{A, Z(\theta_n, \theta_{n+1})\} \mid \theta_n \right], & \text{if } V_C^n(\theta_n) < D. \end{cases}$$

We claim that $\Pi_{A,B}(\theta_n; \mathcal{P}, D) \geq \Pi_{A,B}(\theta_n; \tilde{\mathcal{P}}, D)$ for all θ_n .

Fix any θ_n . If $V_C^n(\theta_n) \geq D$, then full repayment occurs under both partitions (no strategic default), so $\Pi_{A,B}(\theta_n; \mathcal{P}, D) = \Pi_{A,B}(\theta_n; \tilde{\mathcal{P}}, D) = B - A$.

Now suppose $V_C^n(\theta_n) < D$ (forced default), and write $g := G(\theta_n)$. If $g \notin [A, B)$, then inserting additional cutoffs strictly inside (A, B) does not change the pivot cap: $C_{\text{pivot}}(\mathcal{P}) = C_{\text{pivot}}(\tilde{\mathcal{P}})$. Hence weak pivot-control makes the same liquidation/extension decision under $\mathcal{P}$ and $\tilde{\mathcal{P}}$, so $\Pi_{A,B}(\theta_n; \mathcal{P}, D) = \Pi_{A,B}(\theta_n; \tilde{\mathcal{P}}, D)$.

Finally, suppose $g \in [A, B)$. Since A and B are adjacent in $\mathcal{P}$, we have $C_{\text{pivot}}(\mathcal{P}) = B$. Under $\tilde{\mathcal{P}}$, let $\tilde{C} := C_{\text{pivot}}(\tilde{\mathcal{P}}) = \min\{C \in \tilde{\mathcal{P}} : C > g\}$, so $\tilde{C} \in (g, B]$. Define the block's liquidation payoff and conditional continuation payoff by

$$\begin{aligned} \pi_L(\theta_n) &:= \min\{B, V_L(\theta_n)\} - \min\{A, V_L(\theta_n)\}, \\ \pi_C(\theta_n) &:= \mathbb{E} \left[\min\{B, Z(\theta_n, \theta_{n+1})\} - \min\{A, Z(\theta_n, \theta_{n+1})\} \mid \theta_n \right]. \end{aligned}$$

Since $g \geq A$ and $V_L(\theta_n) = g + L(\theta_n, n) > g$, we have $\min\{A, V_L(\theta_n)\} = A$. Also $Z(\theta_n, \theta_{n+1}) \geq g \geq A$ almost surely, so $\min\{A, Z(\theta_n, \theta_{n+1})\} = A$ almost surely and hence $\mathbb{E}[\min\{A, Z(\theta_n, \theta_{n+1})\} \mid \theta_n] = A$. Therefore, under $\mathcal{P}$ the weak pivot-control liquidation condition at the pivot cap B ,

$$\min\{B, V_L(\theta_n)\} \geq \mathbb{E} \left[\min\{B, Z(\theta_n, \theta_{n+1})\} \mid \theta_n \right],$$

is equivalent to $\pi_L(\theta_n) \geq \pi_C(\theta_n)$. Thus, in state θ_n the induced payoff of the whole block $(A, B]$ under $\mathcal{P}$ is

$$\Pi_{A,B}(\theta_n; \mathcal{P}, D) = \max\{\pi_L(\theta_n), \pi_C(\theta_n)\}.$$

Under $\tilde{\mathcal{P}}$, weak pivot-control chooses either liquidation or extension based on the smaller pivot cap $\tilde{C}$, but the block payoff under liquidation is still $\pi_L(\theta_n)$ and the block payoff under extension is still $\pi_C(\theta_n)$. Hence

$$\Pi_{A,B}(\theta_n; \tilde{\mathcal{P}}, D) \in \{\pi_L(\theta_n), \pi_C(\theta_n)\},$$

so $\Pi_{A,B}(\theta_n; \tilde{\mathcal{P}}, D) \leq \max\{\pi_L(\theta_n), \pi_C(\theta_n)\} = \Pi_{A,B}(\theta_n; \mathcal{P}, D)$.

This proves $\Pi_{A,B}(\theta_n; \mathcal{P}, D) \geq \Pi_{A,B}(\theta_n; \tilde{\mathcal{P}}, D)$ for all θ_n . By monotonicity of conditional expectation, for any financing date $t = i$ and any public information at $t = i$, the competitive risk-neutral market value (and hence issuance proceeds) of the entire block $(A, B]$ is weakly higher under $\mathcal{P}$ than under $\tilde{\mathcal{P}}$ for the same promised face $B - A$. $\square$

Proof of Proposition 13 and Corollary 5

Proof. Under the all-unprotected (one-layer) policy, the maturity partition is $\mathcal{P}_{\text{jun}} = \{0, D_{\text{jun}}\}$, hence by definition $\Lambda_n(\mathcal{P}_{\text{jun}}, D_{\text{jun}}) = \Lambda_n^{\text{one}}(D_{\text{jun}})$. Therefore,

$$\Lambda_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) - \Lambda_n(\mathcal{P}_{\text{jun}}, D_{\text{jun}}) = \Lambda_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) - \Lambda_n^{\text{one}}(D_{\text{jun}}).$$

Adding and subtracting $\Lambda_n^{\text{one}}(D_{\text{sen}})$ yields

$$\begin{aligned} \Lambda_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) - \Lambda_n^{\text{one}}(D_{\text{jun}}) &= \left[\Lambda_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) - \Lambda_n^{\text{one}}(D_{\text{sen}}) \right] \\ &\quad - \left[\Lambda_n^{\text{one}}(D_{\text{jun}}) - \Lambda_n^{\text{one}}(D_{\text{sen}}) \right]. \end{aligned}$$

which is exactly (129).

For the sign of the fragmentation wedge, suppose $\mathcal{P}_{\text{sen}}$ is a refinement of $\{0, D_{\text{sen}}\}$. (The proof of Proposition 7(i) uses only the weak pivot-control rule, so it applies verbatim at $t = n$.) Proposition 7(i) implies that, state by state in forced default at $t = n$,

$$\mathbf{1}_{\text{liq}}(\theta_n; \mathcal{P}_{\text{sen}}, D_{\text{sen}}) \geq \mathbf{1}_{\text{liq}}(\theta_n; \{0, D_{\text{sen}}\}, D_{\text{sen}}).$$

Because $\mathbf{1}_{\text{liq}}(\theta_n; \cdot, D_{\text{sen}})$ is defined to be zero whenever $V_C^n(\theta_n) \geq D_{\text{sen}}$, and repayment feasibility depends only on D_{sen} (not on the partition), the same inequality holds for all θ_n . Since $\Delta(\theta_n) > 0$ pointwise, multiplying by $\Delta(\theta_n)$ and taking expectations yields $\Lambda_n(\mathcal{P}_{\text{sen}}, D_{\text{sen}}) \geq \Lambda_n^{\text{one}}(D_{\text{sen}})$. Moreover, the inequality is strict whenever

$$\Pr(\mathbf{1}_{\text{liq}}(\theta_n; \mathcal{P}_{\text{sen}}, D_{\text{sen}}) > \mathbf{1}_{\text{liq}}(\theta_n; \{0, D_{\text{sen}}\}, D_{\text{sen}})) > 0.$$

Finally, if $\Lambda_n^{\text{one}}(\cdot)$ is weakly increasing and $D_{\text{jun}} \geq D_{\text{sen}}$, then $\Lambda_n^{\text{one}}(D_{\text{jun}}) \geq \Lambda_n^{\text{one}}(D_{\text{sen}})$, so the leverage ratchet wedge $\Lambda_n^{\text{one}}(D_{\text{jun}}) - \Lambda_n^{\text{one}}(D_{\text{sen}})$ is weakly nonnegative.

For the total-loss decomposition, use

$$\mathcal{T}_n(\mathcal{P}, D) = \Lambda_n(\mathcal{P}, D) + \Psi_n(D).$$

Subtract the all-unprotected total loss from the all-protected total loss, then add and subtract $\Lambda_n^{\text{one}}(D_{\text{sen}})$ and $\Psi_n(D_{\text{sen}})$. This gives equation (130). The first bracket is the fixed-face fragmentation wedge. The second is the one-layer liquidation component of the leverage-ratchet wedge. The final bracket records the change in the private real distress cost induced by the change in total face. $\square$

OA.2.2 General Fragmentation and Divisible-Asset Liquidation

This subsection isolates a general implication of creditor fragmentation that does not depend on a particular priority structure. The results apply whenever all creditor classes have objection rights over the same resolution choice and the allocation of creditor payoffs is exhaustive.

Fix a forced-default state and consider first a binary choice between liquidation, L , and continuation, C . Let $U_i^r \geq 0$ denote the payoff of creditor class $i \in \{1, \dots, n\}$ under outcome $r \in \{L, C\}$, and let U_{one}^r denote the payoff of a unified creditor class with the same total face value. A fragmentation is *payoff conserving* if

$$U_{\text{one}}^r = \sum_{i=1}^n U_i^r, \quad r \in \{L, C\}. \quad (131)$$

As in Assumption 2, class i objects to continuation if and only if $U_i^L > 0$ and $U_i^L \geq U_i^C$, and

liquidation occurs if at least one class objects. The same convention applies to the unified class.

Lemma 13 (Payoff-conserving creditor fragmentation). *Under any payoff-conserving fragmentation, if the unified creditor class objects to continuation, then at least one fragmented creditor class objects to continuation. Consequently, replacing a unified class with multiple classes cannot eliminate liquidation when all classes have objection rights over the same resolution choice.*

The inclusion can be strict. In particular, the fragmented structure liquidates while the unified structure continues whenever

$$U_{\text{one}}^L < U_{\text{one}}^C$$

and there exists a class i such that

$$U_i^L > 0 \quad \text{and} \quad U_i^L \geq U_i^C.$$

Proof. Suppose that the unified class objects, so $U_{\text{one}}^L > 0$ and $U_{\text{one}}^L \geq U_{\text{one}}^C$. If no fragmented class objected, then $U_i^L < U_i^C$ for every class with $U_i^L > 0$, while $U_i^L = 0 \leq U_i^C$ for every remaining class. Because $U_{\text{one}}^L = \sum_i U_i^L > 0$, at least one class belongs to the first group. Summing across classes would therefore give

$$U_{\text{one}}^L = \sum_{i=1}^n U_i^L < \sum_{i=1}^n U_i^C = U_{\text{one}}^C,$$

a contradiction. The strictness statement follows immediately from the objection rule. $\square$

We next extend the argument to divisible assets. Let $q \in [0, 1]$ denote the fraction of firm assets liquidated at $t = 2$. Asset payoffs are proportional, so the resources available to creditors when fraction q is liquidated and the remainder continues are

$$X(q) = G_2 + qL_2 + (1 - q)(G_3 + L_3). \tag{132}$$

Thus, $q = 0$ represents full continuation and $q = 1$ represents full liquidation. All expectations below are conditional on the date-2 state. For a unified class with total face D , define

$$U_{\text{one}}(q; D) = \mathbb{E}[\min\{D, X(q)\}].$$

Consider any fragmentation into classes with payoffs $U_i(q) \geq 0$ satisfying

$$U_{\text{one}}(q; D) = \sum_{i=1}^n U_i(q) \quad \text{for every } q \in [0, 1]. \quad (133)$$

Payoffs are path independent: $U_i(q)$ depends only on the final liquidation fraction, rather than on the sequence through which that fraction is reached. All classes have common incremental objection rights. Starting from a liquidation fraction q , class i can require an increase to $q' > q$ whenever $U_i(q') > 0$ and $U_i(q') \geq U_i(q)$. A liquidation fraction is stable if no class can require such an increase.

Proposition 14 (Fragmentation with divisible assets). *Suppose that $U_{\text{one}}(1; D) > 0$ and full liquidation maximizes the unified class's payoff:*

$$1 \in \arg \max_{q \in [0, 1]} U_{\text{one}}(q; D).$$

Then, under any payoff-conserving fragmentation satisfying equation (133), full liquidation is the only stable liquidation level.

Proof. Because $X(q)$ is affine in q and $x \mapsto \min\{D, x\}$ is concave, $U_{\text{one}}(q; D)$ is concave in q . Fix $0 \leq q < q' \leq 1$ and write $q' = (1 - \lambda)q + \lambda$ for some $\lambda \in (0, 1]$. Concavity and the optimality of $q = 1$ imply

$$U_{\text{one}}(q'; D) \geq (1 - \lambda)U_{\text{one}}(q; D) + \lambda U_{\text{one}}(1; D) \geq U_{\text{one}}(q; D).$$

Moreover, concavity between $q = 0$ and $q = 1$, together with $U_{\text{one}}(1; D) > 0$, implies $U_{\text{one}}(q'; D) > 0$. Payoff conservation and Lemma 13, applied to the higher-liquidation outcome q' and the lower-liquidation outcome q , therefore imply that some class i satisfies

$$U_i(q') > 0 \quad \text{and} \quad U_i(q') \geq U_i(q).$$

That class objects to stopping at q and can require the additional liquidation. Hence no $q < 1$ is stable, and full liquidation is the only stable liquidation level. $\square$

The proposition does not require the same class to favor every successive increase in liquidation. As liquidation proceeds, the identity of the objecting class may change; payoff conservation

ensures that at least one class supports each additional increment. The result does require common incremental objection rights and path-independent payoffs. If objection instead required a strict payoff improvement, full liquidation would need to be the unique maximizer of $U_{\text{one}}(q; D)$.